

Does adding temp * julian date interaction improve model fit for results?

Objectives:

1. Add interaction between julian date and max temp to model for each response: growth, survival, cort, provisioning
2. Determine whether that interaction improves the fit of the model
3. Determine whether adding that interaction changes the results (probably will need to make a table)
4. Describe the direction of interaction for each response

Growth

```
prior_model <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat + age_s  
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),  
    ~ scale(.x)[,1],  
    .names = "{.col}_scaled"),  
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)  
  
g_lintemp <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat + age_s  
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),  
    ~ scale(.x)[,1],  
    .names = "{.col}_scaled"),  
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)  
  
g_lintemp_addmax <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat + age_s  
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),  
    ~ scale(.x)[,1],  
    .names = "{.col}_scaled"),  
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)  
  
g_lintemp_addmin <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + age_s  
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),  
    ~ scale(.x)[,1],  
    .names = "{.col}_scaled"),  
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)  
  
g_lintemp_noint <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled + meanmintempI_scaled + habitat + age_s  
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),  
    ~ scale(.x)[,1],  
    .names = "{.col}_scaled"),  
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)  
  
c1 <- anova(g_lintemp,g_lintemp_addmin,g_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",
```

WEBL

```
## refitting model(s) with ML (instead of REML)
```

```

c2 <- anova(g_lintemp,g_lintemp_addmax,g_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction", "single interaction", "both interacting"))

## refitting model(s) with ML (instead of REML)
(int_tab_growth_webl <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 4)),
    P = `Pr(>Chisq)` %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

## Warning: `as.tibble()` was deprecated in tibble 2.0.0.
## i Please use `as_tibble()` instead.
## i The signature and semantics have changed, see `?as_tibble`.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

Model	AIC	Chisq	P
Max temp			
no interaction	794.26	NA	NA
single interaction	772.23	28.04	<0.001
both interacting	754.48	23.75	<0.001
Min temp			
no interaction	794.26	NA	NA
single interaction	789.06	11.20	0.0107
both interacting	754.48	40.59	<0.001

```

summary(g_lintemp)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled *
##   habitat + age_scaled + meanmaxtempI_scaled * juliandate_scaled +
##   year_fct + (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
##   mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate,
##     brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##     meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
##
## REML criterion at convergence: 756.8
##
## Scaled residuals:
##   Min       1Q   Median       3Q      Max
## -2.8745 -0.5725  0.0013  0.6288  2.6105
##

```

```

## Random effects:
##   Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.4867  0.6976
## Residual              0.1929  0.4392
## Number of obs: 438, groups: attempt_id, 85
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)      1.763042   0.197818  54.097387   8.912
## meanmaxtempI_scaled      0.914179   0.184623 327.266126   4.952
## habitatOrchard          0.047645   0.215058  51.771999   0.222
## habitatGrassland        0.029385   0.265560  59.641235   0.111
## habitatRow crop       -0.273612   0.237549  47.864111  -1.152
## meanmintempI_scaled     -0.900892   0.205663 202.804917  -4.380
## age_scaled            -0.559397   0.031812 420.785344 -17.585
## juliandate_scaled      -0.161913   0.097565  71.993854  -1.660
## year_fct2022           0.005082   0.195659  51.774858   0.026
## year_fct2023           0.048130   0.229296  54.316569   0.210
## meanmaxtempI_scaled:habitatOrchard -0.507216   0.234034 330.697411  -2.167
## meanmaxtempI_scaled:habitatGrassland -1.105991   0.220279 332.023227  -5.021
## meanmaxtempI_scaled:habitatRow crop -1.529207   0.245209 277.044610  -6.236
## habitatOrchard:meanmintempI_scaled  0.999487   0.257304 192.457635   3.884
## habitatGrassland:meanmintempI_scaled  1.470341   0.297500 267.611469   4.942
## habitatRow crop:meanmintempI_scaled  1.111948   0.265576 137.150598   4.187
## meanmaxtempI_scaled:juliandate_scaled -0.164348   0.051470 334.740947  -3.193
##
##              Pr(>|t|)
## (Intercept)      3.37e-12 ***
## meanmaxtempI_scaled      1.18e-06 ***
## habitatOrchard          0.825541
## habitatGrassland        0.912262
## habitatRow crop       0.255118
## meanmintempI_scaled     1.90e-05 ***
## age_scaled            < 2e-16 ***
## juliandate_scaled      0.101359
## year_fct2022           0.979378
## year_fct2023           0.834527
## meanmaxtempI_scaled:habitatOrchard  0.030927 *
## meanmaxtempI_scaled:habitatGrassland 8.41e-07 ***
## meanmaxtempI_scaled:habitatRow crop  1.67e-09 ***
## habitatOrchard:meanmintempI_scaled  0.000141 ***
## habitatGrassland:meanmintempI_scaled 1.36e-06 ***
## habitatRow crop:meanmintempI_scaled  5.03e-05 ***
## meanmaxtempI_scaled:juliandate_scaled 0.001541 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE) or
##       vcov(x)           if you need it
check_collinearity(g_lintemp_noint)

## # Check for Multicollinearity
##

```

```
## Low Correlation
##
##
##          Term   VIF   VIF 95% CI Increased SE Tolerance
##          meanmaxtempI_scaled 2.30 [2.01, 2.67]      1.52      0.44
##          meanmintempI_scaled 2.60 [2.26, 3.02]      1.61      0.39
##          habitat 1.32 [1.20, 1.51]      1.15      0.76
##          age_scaled 1.11 [1.04, 1.30]      1.05      0.90
##          juliandate_scaled 1.85 [1.64, 2.14]      1.36      0.54
##          year_fct 1.28 [1.17, 1.47]      1.13      0.78
## meanmaxtempI_scaled:juliandate_scaled 1.13 [1.05, 1.31]      1.06      0.89
## Tolerance 95% CI
## [0.37, 0.50]
## [0.33, 0.44]
## [0.66, 0.83]
## [0.77, 0.96]
## [0.47, 0.61]
## [0.68, 0.85]
## [0.76, 0.95]
```

```
anova(g_lintemp,prior_model)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>% ...
## Models:
## prior_model: gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat + age_scaled + ...
## g_lintemp: gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat + age_scaled + ...
##          npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model   18 762.11 835.59 -363.05   726.11
## g_lintemp     19 754.48 832.04 -358.24   716.48 9.6316  1  0.001913 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
g_lintemp_webl <- g_lintemp
```

For WEBL growth, adding the interaction improves fit.

Do the results change?

```
weblgrowthtrendmax_prior <- emtrends(prior_model,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "t")
mutate(Model = "No temp * juliandate int")

weblgrowthtrendmax <- emtrends(g_lintemp,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test()
mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "t")
mutate(Model = "interaction")

(trend_comp <- weblgrowthtrendmax_prior %>% rbind(weblgrowthtrendmax) %>% gt())
```

Comparing temp trend and significance across land covers without and with temp * julian date interaction

Habitat	Max temp trend	SE	Df	T-ratio	P	Model
Forest	0.878	0.187	348	4.706	<0.001	No temp * juliandate int
Orchard	0.458	0.148	395	3.100	0.002	No temp * juliandate int
Grassland	-0.108	0.119	422	-0.908	0.364	No temp * juliandate int
Row crop	-0.642	0.160	257	-4.009	<0.001	No temp * juliandate int
Forest	0.914	0.186	359	4.915	<0.001	interaction
Orchard	0.407	0.148	400	2.752	0.006	interaction
Grassland	-0.192	0.121	417	-1.584	0.114	interaction
Row crop	-0.615	0.161	265	-3.813	<0.001	interaction

No direction or significance change, or really large coefficient change, between the two models.

```
samp <- g_lintemp@frame %>% group_by(habitat) %>% summarize(count = n())

dat_text_webl <- data.frame(
  label = paste("N =", samp$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_webl = dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

mean_temp_webl <- mean(data_webl %>% pull(meanmaxtempI))
sd_temp_webl <- sd(data_webl %>% pull(meanmaxtempI))

temp_trans_webl <- trans_new("temp_trans_webl",
  transform = function(x){(x * sd_temp_webl) + mean_temp_webl},
  inverse = function(x){x})

(fig2_webl <- predict_response(prior_model, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = FALSE) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Growth (g/day)") +
```

```

scale_fill_viridis(discrete = TRUE) +
scale_color_viridis(discrete = TRUE) +
scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "solid", "Grassland" = "dotted", "Row
theme(text = element_text(size = 16)) +
labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_webl,
                    breaks = c((20-mean_temp_webl)/sd_temp_webl,
                               (25-mean_temp_webl)/sd_temp_webl,
                               (30-mean_temp_webl)/sd_temp_webl,
                               (35-mean_temp_webl)/sd_temp_webl,
                               (40-mean_temp_webl)/sd_temp_webl),
                    # breaks = c(20,30,40,50),
                    # labels = c("20", "30", "40", "50"),
                    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
                    #          (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
                    # limits = c(18,5)
                    ) +
#ylim(-5,5) +
geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust =
theme(legend.position = "none")
)

```

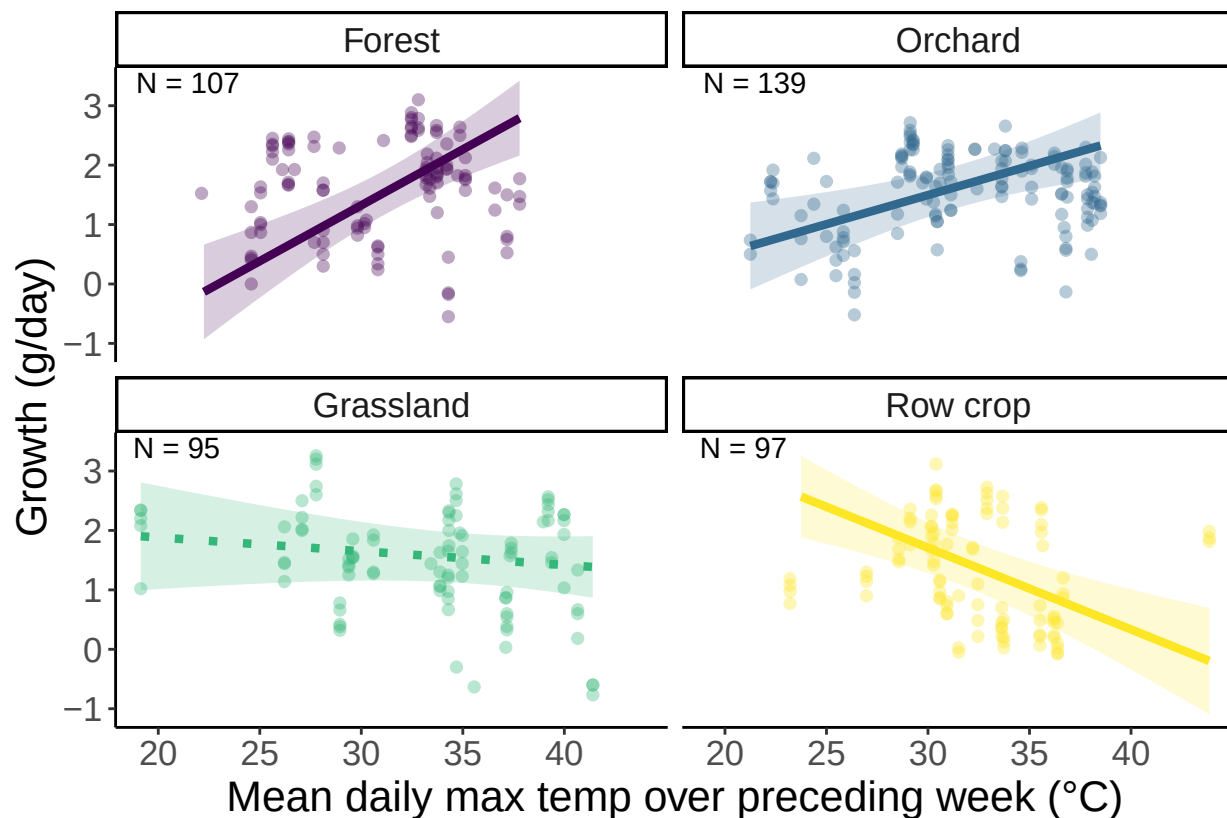
Plot of growth without interaction

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



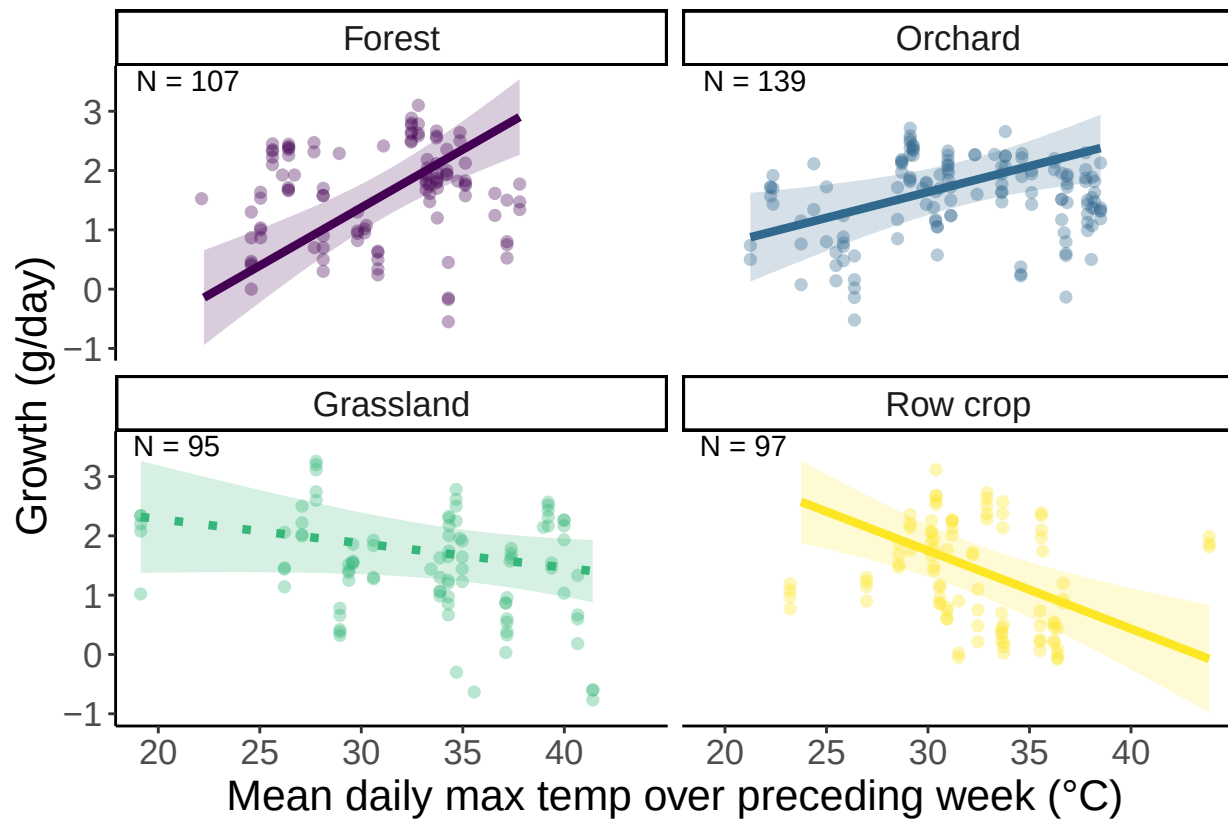
```
(fig2_webl <- predict_response(g_lintemp, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = "none",
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Growth (g/day)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "solid", "Grassland" = "dotted", "Row crop" = "dotted")) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_webl,
    breaks = c((20 - mean_temp_webl) / sd_temp_webl,
      (25 - mean_temp_webl) / sd_temp_webl,
      (30 - mean_temp_webl) / sd_temp_webl,
      (35 - mean_temp_webl) / sd_temp_webl,
      (40 - mean_temp_webl) / sd_temp_webl),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18, 5)
  ) +
```

```
#ylim(-5,5) +
geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1)
theme(legend.position = "none")
)
```

Plot of growth with interaction

```
## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
(plott <- predict_response(g_lintemp, terms = c("meanmaxtempI_scaled [all]", "juliandate_scaled"), bias_correction = "none",
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
# aes(linetype = .data[["group"]]) +
theme_classic() +
# facet_wrap(~ group, ncol = 2) +
xlab("Mean daily max temp over preceding week (°C)") +
ylab("Growth (g/day)") +
# scale_fill_viridis(discrete = TRUE) +
# scale_color_viridis(discrete = TRUE) +
# scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "solid", "Grassland" = "dotted", "Row crop" = "solid")) +
theme(text = element_text(size = 16)) +
# labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_webl,
```

```

breaks = c((20-mean_temp_webl)/sd_temp_webl,
            (25-mean_temp_webl)/sd_temp_webl,
            (30-mean_temp_webl)/sd_temp_webl,
            (35-mean_temp_webl)/sd_temp_webl,
            (40-mean_temp_webl)/sd_temp_webl),
# breaks = c(20,30,40,50),
# labels = c("20","30","40","50"),
# limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
#            (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
# limits = c(18,5)

) ##
#ylim(-5,5) +
# geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust
# theme(legend.position = "none")
)

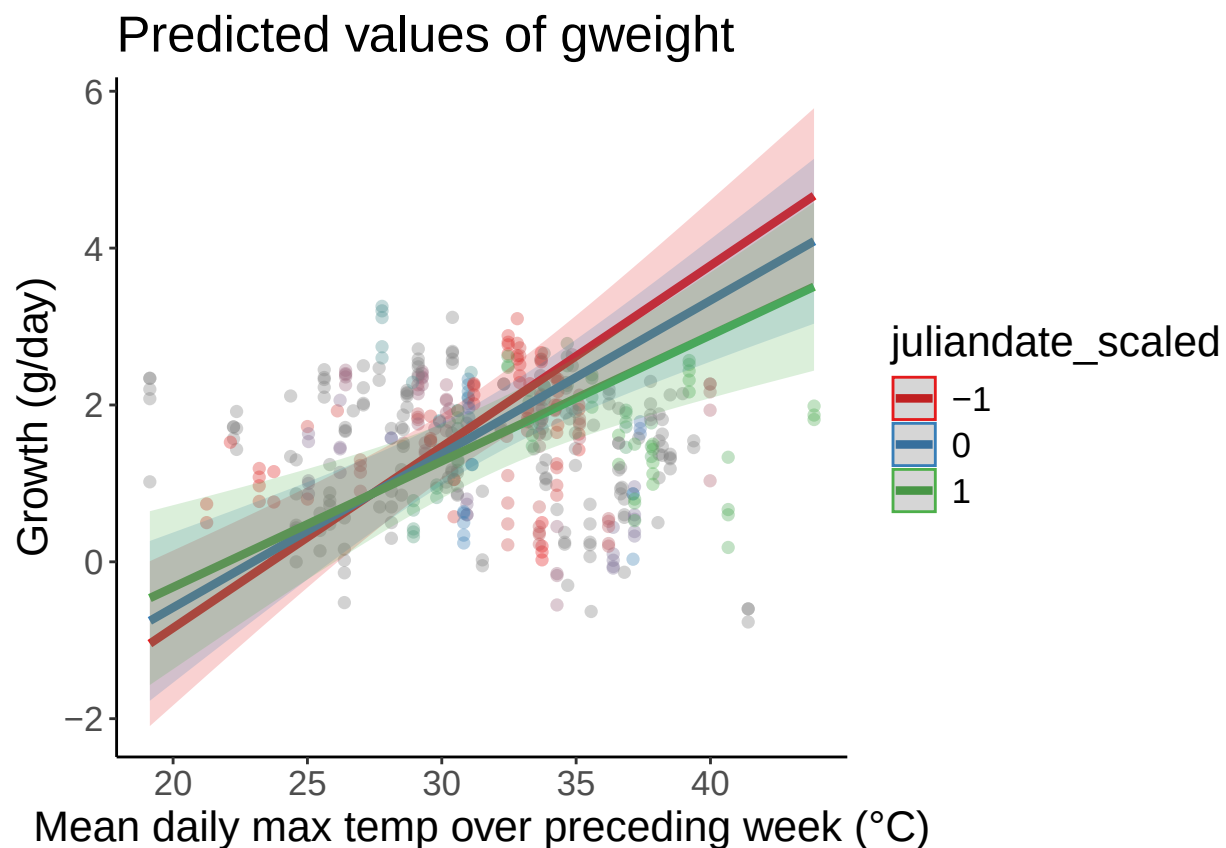
```

Visualize temp * juliandate interaction

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Conclusion: for WEBL growth, the interaction improves fit, but does not change growth results. In addition, the direction of interaction is opposite that expected: later in the season, hotter temps are worse for growth than in earlier season.

```

prior_model <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + age_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

g_lintemp <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat + age_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

g_lintemp_addmax <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat + age_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

g_lintemp_addmin <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + age_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

g_lintemp_noint <- lmerTest::lmer(gweight ~ meanmaxtempI_scaled + meanmintempI_scaled + habitat + age_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

c1 <- anova(g_lintemp,g_lintemp_addmin,g_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",

```

TRES

```

## refitting model(s) with ML (instead of REML)
c2 <- anova(g_lintemp,g_lintemp_addmax,g_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",
## refitting model(s) with ML (instead of REML)
(int_tab_growth_tres <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 4)),
    P = `Pr(>Chisq)` ) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	513.03	NA	NA

single interaction	478.45	40.58	<0.001
both interacting	479.69	4.76	0.1904
Min temp			
no interaction	513.03	NA	NA
single interaction	506.20	12.83	0.005
both interacting	479.69	32.51	<0.001

```
check_collinearity(g_lintemp_noint)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##
##              Term  VIF    VIF 95% CI Increased SE Tolerance
## meanmaxtempI_scaled 2.04 [1.77, 2.40]          1.43      0.49
## meanmintempI_scaled 2.03 [1.76, 2.39]          1.42      0.49
##          habitat 1.46 [1.30, 1.70]          1.21      0.69
##          age_scaled 1.15 [1.06, 1.36]          1.07      0.87
##      juliandate_scaled 1.41 [1.26, 1.64]          1.19      0.71
##          year_fct 1.22 [1.12, 1.43]          1.11      0.82
## meanmaxtempI_scaled:juliandate_scaled 1.17 [1.08, 1.38]          1.08      0.85
## Tolerance 95% CI
##      [0.42, 0.57]
##      [0.42, 0.57]
##      [0.59, 0.77]
##      [0.74, 0.94]
##      [0.61, 0.79]
##      [0.70, 0.90]
##      [0.72, 0.93]
```

```
summary(g_lintemp_noint)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: gweight ~ meanmaxtempI_scaled + meanmintempI_scaled + habitat +
##          age_scaled + meanmaxtempI_scaled * juliandate_scaled + year_fct +
##          (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), meanmaxtempI <
## 45, !is.na(meanmintempI)) %>% mutate(across(c(gweight, meanmaxtempI,
## meanmintempI, juliandate, brood_size, age), ~scale(.x)[,
## 1], .names = "{.col}_scaled"), meanmaxtempI_scaled_sq = meanmaxtempI_scaled *
## meanmaxtempI_scaled)
##
## REML criterion at convergence: 521.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.77827 -0.53100  0.06156  0.49655  2.84200
##
## Random effects:
## Groups      Name                Variance Std.Dev.
```

```

## attempt_id (Intercept) 0.1815 0.4260
## Residual 0.1774 0.4211
## Number of obs: 347, groups: attempt_id, 66
##
## Fixed effects:
##
## Estimate Std. Error df t value
## (Intercept) 1.15264 0.20429 54.51447 5.642
## meanmaxtempI_scaled -0.07990 0.06002 163.05828 -1.331
## meanmintempI_scaled -0.06395 0.07426 67.91584 -0.861
## habitatOrchard 0.44707 0.24486 51.60549 1.826
## habitatGrassland 0.04514 0.20986 56.02929 0.215
## habitatRow crop 0.20571 0.21127 53.10778 0.974
## age_scaled -0.26283 0.03470 295.04507 -7.575
## juliandate_scaled -0.02811 0.06836 63.70594 -0.411
## year_fct2022 0.05051 0.13821 49.78715 0.365
## year_fct2023 0.24889 0.16521 51.42231 1.506
## meanmaxtempI_scaled:juliandate_scaled -0.02782 0.04725 230.11645 -0.589
##
## Pr(>|t|)
## (Intercept) 6.19e-07 ***
## meanmaxtempI_scaled 0.1850
## meanmintempI_scaled 0.3922
## habitatOrchard 0.0737 .
## habitatGrassland 0.8305
## habitatRow crop 0.3346
## age_scaled 4.64e-13 ***
## juliandate_scaled 0.6823
## year_fct2022 0.7163
## year_fct2023 0.1381
## meanmaxtempI_scaled:juliandate_scaled 0.5566
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) mnmXI_ mnmnI_ hbttOr hbttGr hbttRc ag_scl jlndt_ y_2022
## mnmxtmpI_sc 0.240
## mnmntmpI_sc -0.197 -0.574
## hbttOrchrd -0.628 -0.029 0.025
## hbttGrsslnd -0.840 -0.367 0.299 0.586
## hbttRwcrp -0.758 -0.192 0.070 0.576 0.743
## age_scaled 0.131 0.241 -0.268 -0.086 -0.136 -0.077
## jlndt_scl 0.073 -0.043 -0.317 0.030 -0.073 -0.064 -0.063
## yer_fct2022 -0.363 0.047 0.017 0.064 0.022 -0.022 -0.155 0.035
## yer_fct2023 -0.383 0.141 -0.017 0.032 0.121 0.051 0.017 -0.241 0.436
## mnmxtmpI:_ -0.195 -0.301 0.150 0.062 0.141 0.180 -0.040 -0.145 -0.088
## y_2023
## mnmxtmpI_sc
## mnmntmpI_sc
## hbttOrchrd
## hbttGrsslnd
## hbttRwcrp
## age_scaled
## jlndt_scl
## yer_fct2022
## yer_fct2023

```

```
## mnmxtmpI:_ -0.016
g_lintemp_addmin_tres <- g_lintemp_addmin

(anova(g_lintemp_addmin,prior_model))

## refitting model(s) with ML (instead of REML)

## Data: dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), meanmaxtempI < ...
## Models:
## prior_model: gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + age_scaled + juliandate
## g_lintemp_addmin: gweight ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + age_scaled + meanm
##
##          npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model      15 476.59 534.33 -223.29   446.59
## g_lintemp_addmin  16 478.45 540.04 -223.23   446.45 0.1375  1    0.7107
```

For TRES growth, same model is significant. Interaction is not significant. Adding the interaction does not improve fit.

Do the results change?

```
tresgrowthtrendmax_prior <- emtrends(prior_model, specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend", Df = "df", `T-ratio` = "t")
  mutate(Model = "No temp * juliandate int")

tresgrowthtrendmax <- emtrends(g_lintemp_addmin, specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend", Df = "df", `T-ratio` = "t")
  mutate(Model = "interaction")

(trend_comp <- tresgrowthtrendmax_prior %>% rbind(tresgrowthtrendmax) %>% gt())
```

Comparing temp trend and significance across land covers without and with temp * julian date interaction

Habitat	Max temp trend	SE	Df	T-ratio	P	Model
Forest	0.289	0.150	84	1.931	0.057	No temp * juliandate int
Orchard	0.188	0.270	125	0.695	0.488	No temp * juliandate int
Grassland	0.205	0.081	281	2.534	0.012	No temp * juliandate int
Row crop	-0.432	0.077	291	-5.622	<0.001	No temp * juliandate int
Forest	0.294	0.151	84	1.947	0.055	interaction
Orchard	0.194	0.271	127	0.714	0.476	interaction
Grassland	0.198	0.085	290	2.335	0.02	interaction
Row crop	-0.440	0.082	300	-5.366	<0.001	interaction

No direction or significance change, or really large coefficient change, between the two models.

```

samp <- g_lintemp_addmin@frame %>% group_by(habitat) %>% summarize(count = n())

dat_text_tres <- data.frame(
  label = paste("N =", samp$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_tres = dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), meanmaxtempI < 45, !is.na(meanmintempI),
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

mean_temp_tres <- mean(data_tres %>% pull(meanmaxtempI))
sd_temp_tres <- sd(data_tres %>% pull(meanmaxtempI))

temp_trans_tres <- trans_new("temp_trans",
  transform = function(x){(x * sd_temp_tres) + mean_temp_tres},
  inverse = function(x){x})

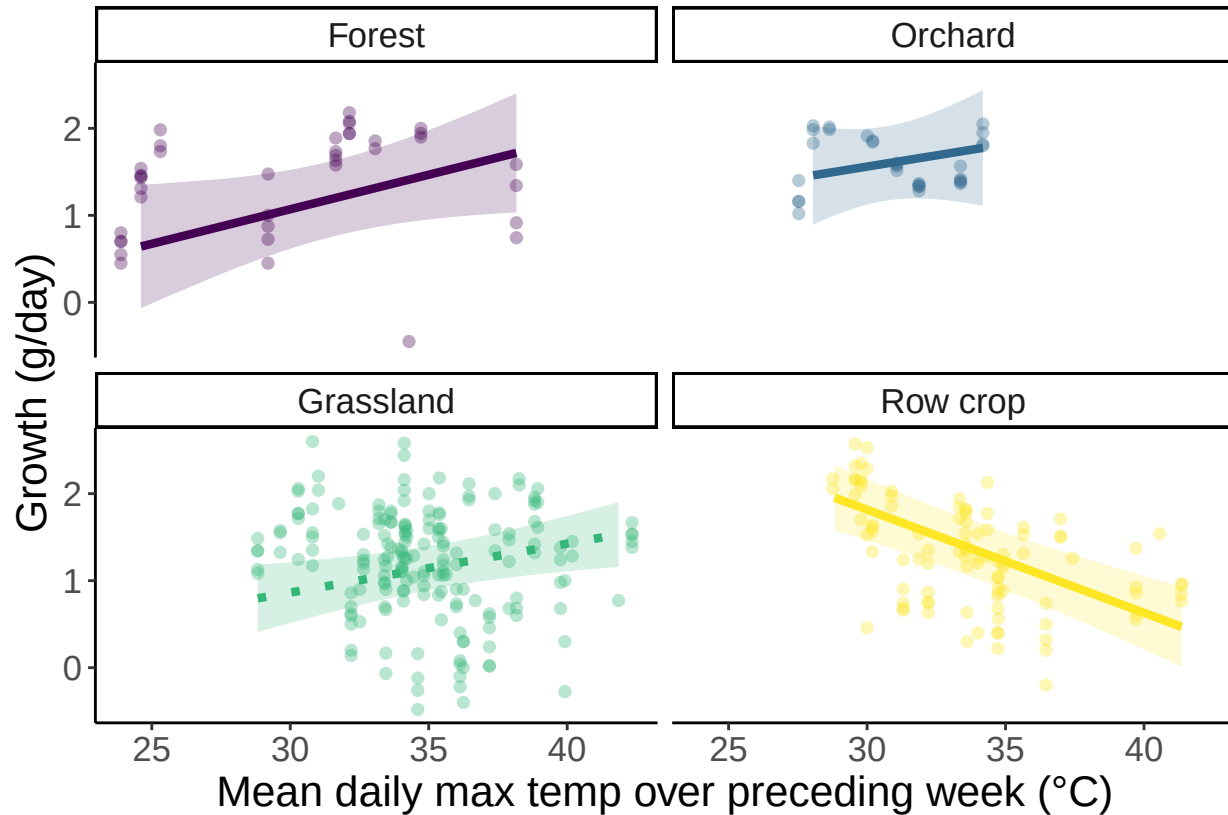
(fig2_tres <- predict_response(prior_model, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = TRUE) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Growth (g/day)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "solid", "Grassland" = "dotted", "Row crop" = "dotted")) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_tres,
    breaks = c((20-mean_temp_tres)/sd_temp_tres,
      (25-mean_temp_tres)/sd_temp_tres,
      (30-mean_temp_tres)/sd_temp_tres,
      (35-mean_temp_tres)/sd_temp_tres,
      (40-mean_temp_tres)/sd_temp_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18,5)
  ) +
  #ylim(-5,5) +
  #geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1) +
  theme(legend.position = "none")
)

```

Plot of growth without interaction

```
## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
(fig2_tres <- predict_response(g_lintemp_admin, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_c
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Growth (g/day)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "solid", "Grassland" = "dotted", "Row
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_tres,
    breaks = c((20-mean_temp_tres)/sd_temp_tres,
               (25-mean_temp_tres)/sd_temp_tres,
               (30-mean_temp_tres)/sd_temp_tres,
               (35-mean_temp_tres)/sd_temp_tres,
               (40-mean_temp_tres)/sd_temp_tres),
```

```

# breaks = c(20,30,40,50),
# labels = c("20", "30", "40", "50"),
# limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
#           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
# limits = c(18,5)
) +
#ylim(-5,5) +
geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust = 1)
theme(legend.position = "none")
)

```

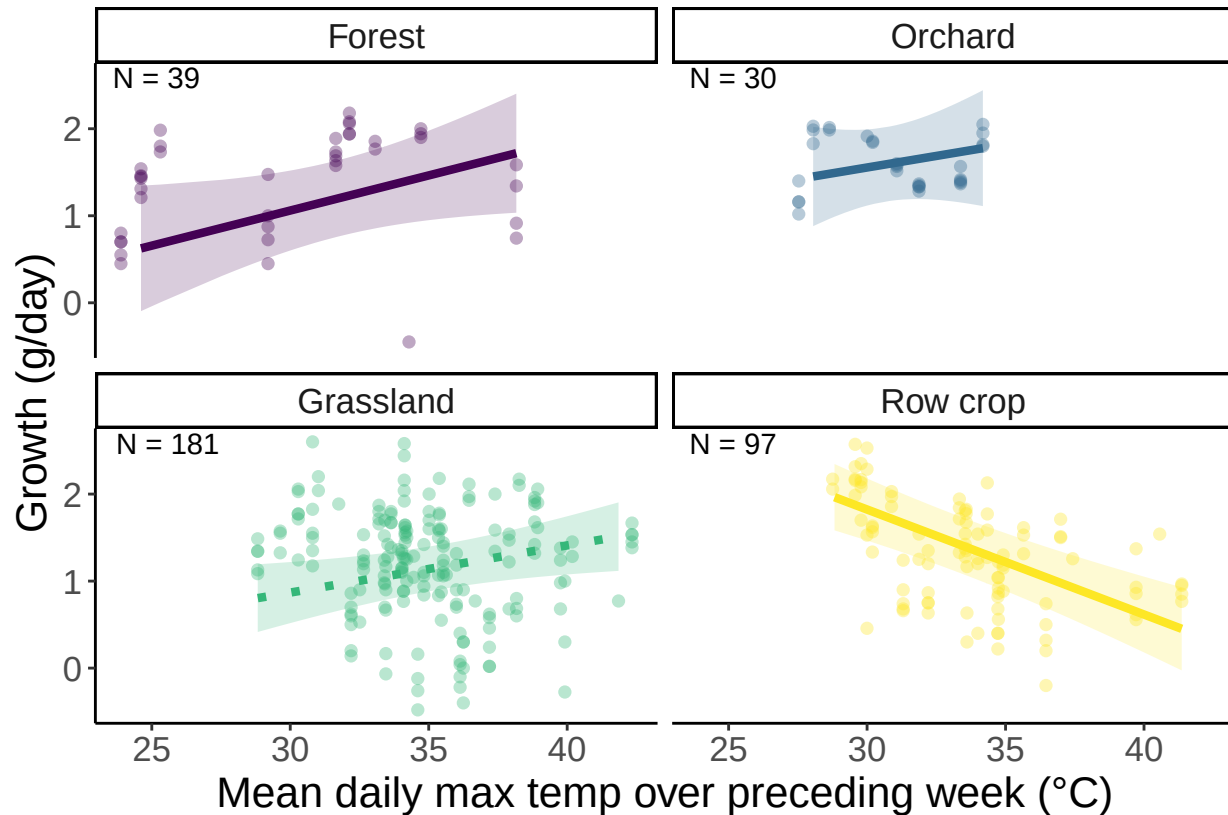
Plot of growth with interaction

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(plott <- predict_response(g_lintemp_addmin,terms = c("meanmaxtempI_scaled [all]","juliandate_scaled"),
plot(line_size = 1.5,alpha = .2,show_data = TRUE,limit_range = TRUE) +
# aes(linetype = .data[["group"]]) +
theme_classic() +
# facet_wrap(~ group, ncol = 2) +
xlab("Mean daily max temp over preceding week (°C)") +
ylab("Growth (g/day)") +

```

```

# scale_fill_viridis(discrete = TRUE) +
# scale_color_viridis(discrete = TRUE) +
# scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "solid", "Grassland" = "dotted", "R
theme(text = element_text(size = 16)) +
# labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_tres,
  breaks = c((20-mean_temp_tres)/sd_temp_tres,
    (25-mean_temp_tres)/sd_temp_tres,
    (30-mean_temp_tres)/sd_temp_tres,
    (35-mean_temp_tres)/sd_temp_tres,
    (40-mean_temp_tres)/sd_temp_tres),
  # breaks = c(20,30,40,50),
  # labels = c("20", "30", "40", "50"),
  # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
  # (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
  # limits = c(18,5)
) #+
#ylim(-5,5) +
# geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust
# theme(legend.position = "none")
)

```

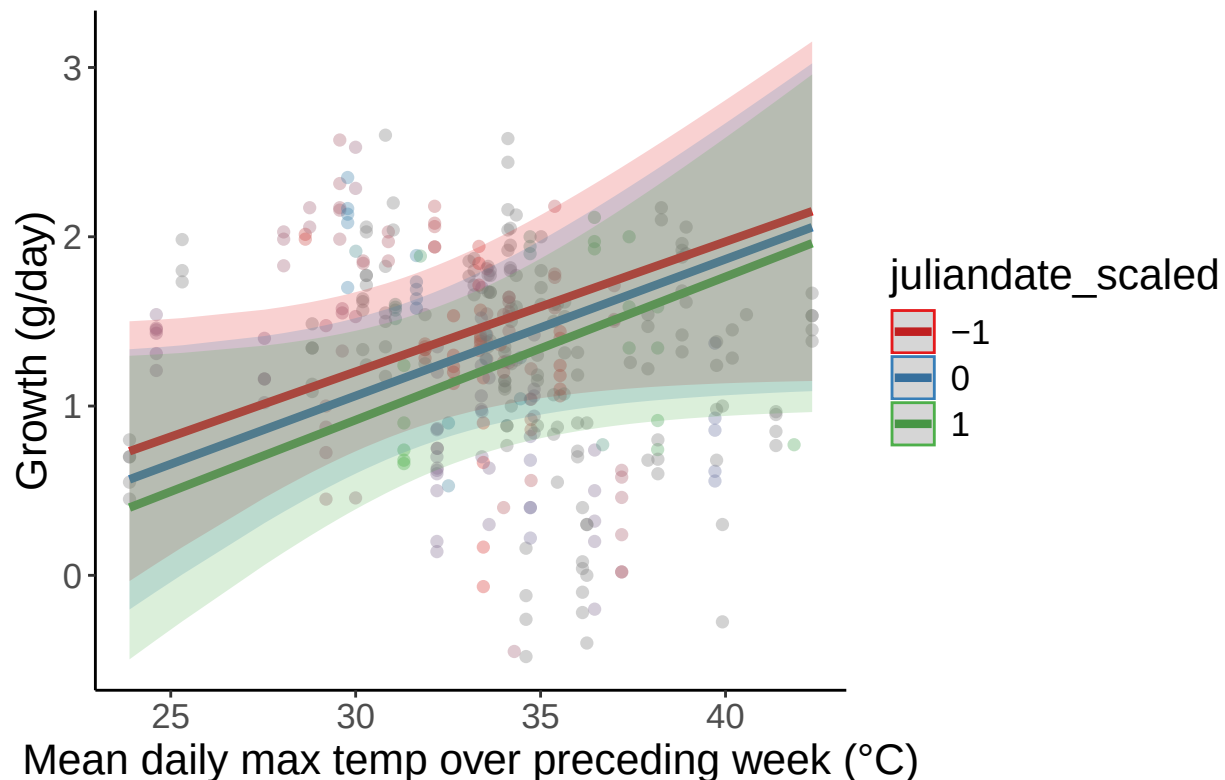
Visualize temp * juliandate interaction

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```

Predicted values of gweight



Conclusion: for TRES growth, the interaction neither improves fit nor changes the results. There does not appear to be an interaction between juliandate and max temperature for TRES growth.

Survival

```
prior_model <- glm(cbind(nest_fledged,clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled * habitat + m
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "WEBL",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
    mutate(across(c(meanmaxt_nestpd,meanmint_nestpd,juliandate_hatch),
      ~ scale(.x)[,1],
      .names = "{.col}_scaled")
    )
)

s_nestpd_WEBL <- glm(cbind(nest_fledged,clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled * habitat + m
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "WEBL",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
    mutate(across(c(meanmaxt_nestpd,meanmint_nestpd,juliandate_hatch),
      ~ scale(.x)[,1],
      .names = "{.col}_scaled")
    )
)

s_nestpd_WEBL_addmax <- glm(cbind(nest_fledged,clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled + m
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "WEBL",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
    mutate(across(c(meanmaxt_nestpd,meanmint_nestpd,juliandate_hatch),
      ~ scale(.x)[,1],
      .names = "{.col}_scaled")
    )
)

s_nestpd_WEBL_addmin <- glm(cbind(nest_fledged,clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled * h
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "WEBL",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
    mutate(across(c(meanmaxt_nestpd,meanmint_nestpd,juliandate_hatch),
      ~ scale(.x)[,1],
      .names = "{.col}_scaled")
    )
)
```

```

s_nestpd_WEBL_noint <- glm(cbind(nest_fledged,clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled + me
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
                        Species == "WEBL",
                        !is.na(nest_fledged),
                        !is.na(clutch_size)) %>%
    mutate(across(c(meanmaxt_nestpd,meanmint_nestpd,juliandate_hatch),
                  ~ scale(.x)[,1],
                  .names = "{.col}_scaled")
    )
)

c1 <- anova(s_nestpd_WEBL,s_nestpd_WEBL_addmin,s_nestpd_WEBL_noint,test="Chisq") %>% tibble() %>%
  rename(Chisq = Deviance) %>%
  mutate(AIC = c(AIC(s_nestpd_WEBL),AIC(s_nestpd_WEBL_addmin),AIC(s_nestpd_WEBL_noint)),.before = Df) %>%
  mutate(Model = c("no interaction","single interaction","both interacting"),.before = Df)

c2 <- anova(s_nestpd_WEBL,s_nestpd_WEBL_addmax,s_nestpd_WEBL_noint,test="Chisq") %>% tibble() %>%
  rename(Chisq = Deviance) %>%
  mutate(AIC = c(AIC(s_nestpd_WEBL),
                AIC(s_nestpd_WEBL_addmax),
                AIC(s_nestpd_WEBL_noint)),.before = Df) %>%
  mutate(Model = c("no interaction","single interaction","both interacting"),.before = Df)

(int_tab_survival_nestpd_webl <- bind_rows(c1,c2) %>%
  as_tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 4)),
         P = `Pr(>Chi)`)) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
         across(c(Chisq), ~ round(.x, digits = 2)),
         P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

WEBL

Model	AIC	Chisq	P
Max temp			
no interaction	319.0342	NA	NA
single interaction	320.0632	-7.03	0.071
both interacting	314.9549	-0.89	0.8274
Min temp			
no interaction	319.0342	NA	NA
single interaction	319.7139	-6.68	0.0828
both interacting	314.9549	-1.24	0.7432

```

# gtsave(int_tab_survival_nestpd_webl,"figures/int_tab_survival_nestpd_webl.html")

```

#Conclusion: For nest attempt overall, land cover interacts with either max or min temp but not both to

```
summary(s_nestpd_WEBL_addmin)
```

```
##
## Call:
## glm(formula = cbind(nest_fledged, clutch_size - nest_fledged) ~
##     meanmaxt_nestpd_scaled * habitat + meanmint_nestpd_scaled +
##     meanmaxt_nestpd_scaled * juliandate_hatch_scaled + year_fct +
##     site, family = binomial(link = "logit"), data = dplyr::filter(s,
##     Species == "WEBL", !is.na(nest_fledged), !is.na(clutch_size)) %>%
##     mutate(across(c(meanmaxt_nestpd, meanmint_nestpd, juliandate_hatch),
##     ~scale(.x)[, 1], .names = "{.col}_scaled")))
##
## Coefficients:
##                                     Estimate Std. Error z value
## (Intercept)                      0.000244   0.433466   0.001
## meanmaxt_nestpd_scaled            -0.645886   0.337151  -1.916
## habitatOrchard                    1.700254   0.441699   3.849
## habitatGrassland                  2.610218   0.522510   4.996
## habitatRow crop                   1.512034   0.409349   3.694
## meanmint_nestpd_scaled             0.452489   0.263654   1.716
## juliandate_hatch_scaled           -0.016409   0.274881  -0.060
## year_fct2022                     -0.239180   0.415544  -0.576
## year_fct2023                     -1.130950   0.474165  -2.385
## sitePG                           0.435061   0.357239   1.218
## siteRR                           0.334349   0.381945   0.875
## siteWI                           -0.708859   0.563649  -1.258
## meanmaxt_nestpd_scaled:habitatOrchard  0.082738   0.309438   0.267
## meanmaxt_nestpd_scaled:habitatGrassland  0.340118   0.404784   0.840
## meanmaxt_nestpd_scaled:habitatRow crop -0.034385   0.440675  -0.078
## meanmaxt_nestpd_scaled:juliandate_hatch_scaled -0.185452   0.161162  -1.151
##                                     Pr(>|z|)
## (Intercept)                      0.999551
## meanmaxt_nestpd_scaled            0.055401 .
## habitatOrchard                    0.000118 ***
## habitatGrassland                  5.87e-07 ***
## habitatRow crop                   0.000221 ***
## meanmint_nestpd_scaled             0.086122 .
## juliandate_hatch_scaled           0.952399
## year_fct2022                     0.564897
## year_fct2023                     0.017073 *
## sitePG                           0.223283
## siteRR                           0.381365
## siteWI                           0.208527
## meanmaxt_nestpd_scaled:habitatOrchard  0.789174
## meanmaxt_nestpd_scaled:habitatGrassland  0.400771
## meanmaxt_nestpd_scaled:habitatRow crop  0.937806
## meanmaxt_nestpd_scaled:juliandate_hatch_scaled 0.249849
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 282.86 on 83 degrees of freedom
## Residual deviance: 206.66 on 68 degrees of freedom
## AIC: 320.06
##
## Number of Fisher Scoring iterations: 4
anova(s_nestpd_WEBL_addmin, prior_model)

## Analysis of Deviance Table
##
## Model 1: cbind(nest_fledged, clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled *
## habitat + meanmint_nestpd_scaled + meanmaxt_nestpd_scaled *
## juliandate_hatch_scaled + year_fct + site
## Model 2: cbind(nest_fledged, clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled *
## habitat + meanmint_nestpd_scaled + juliandate_hatch_scaled +
## year_fct + site
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1 68 206.66
## 2 69 207.98 -1 -1.3199 0.2506
```

Interaction does not improve model fit for survival.

Do the results change?

```
(webbsurvival_nestpd_trendmax_prior_model <- emtrends(prior_model, specs = pairwise ~ habitat, var = c("meanmaxt_nestpd_scaled", "meanmint_nestpd_scaled"),
mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
df = round(df),
p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
rename(Habitat = "habitat", `Max temp trend` = "meanmaxt_nestpd_scaled.trend", Df = "df", `T-ratio` =
gt()))
```

Comparing temp trend and significance across land covers without and with temp * julian date interaction

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.586	0.330	Inf	-1.775	0.076
Orchard	-0.586	0.348	Inf	-1.683	0.092
Grassland	-0.370	0.356	Inf	-1.040	0.298
Row crop	-0.747	0.439	Inf	-1.703	0.088

```
(webbsurvival_nestpd_trendmax <- emtrends(s_nestpd_WEBL_addmin, specs = pairwise ~ habitat, var = c("meanmaxt_nestpd_scaled", "meanmint_nestpd_scaled"),
mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
df = round(df),
p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
rename(Habitat = "habitat", `Max temp trend` = "meanmaxt_nestpd_scaled.trend", Df = "df", `T-ratio` =
gt()))
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.646	0.337	Inf	-1.916	0.055

Orchard	-0.563	0.348	Inf	-1.618	0.106
Grassland	-0.306	0.364	Inf	-0.839	0.401
Row crop	-0.680	0.442	Inf	-1.538	0.124

Forest trend becomes slightly more significant with the addition of the interaction

```
data_webl = s_nestpd_WEBL_addmin$data

mean_temp_webl <- mean(data_webl %>% pull(meanmaxt_nestpd),na.rm = TRUE)
sd_temp_webl <- sd(data_webl %>% pull(meanmaxt_nestpd),na.rm = TRUE)

temp_trans_webl <- trans_new("temp_trans_webl",
                             transform = function(x){(x * sd_temp_webl) + mean_temp_webl},
                             inverse = function(x){x})

samp_webl <- data_webl %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("../figures/ss_webl.html")

dat_text_webl <- data.frame(
  label = paste("N =",samp_webl$count),
  group = factor(samp_webl$habitat)
)

(fig3_webl <- predict_response(prior_model,terms = c("meanmaxt_nestpd_scaled [all]","habitat")) %>%
  plot(line_size = 1.5,alpha = .2,show_data = TRUE,limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Predicted percent of eggs fledging") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_webl,
                     breaks = c((20-mean_temp_webl)/sd_temp_webl,
                                (25-mean_temp_webl)/sd_temp_webl,
                                (30-mean_temp_webl)/sd_temp_webl,
                                (35-mean_temp_webl)/sd_temp_webl,
                                (40-mean_temp_webl)/sd_temp_webl#,
                                #(45-mean_temp)/sd_temp
                                ),
                     # breaks = c(20,30,40,50),
                     # labels = c("20","30","40","50"),
                     # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
                     #
                     # (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
                     # limits = c(18,5)
                     ) +
  # ylim(0,100) +
```

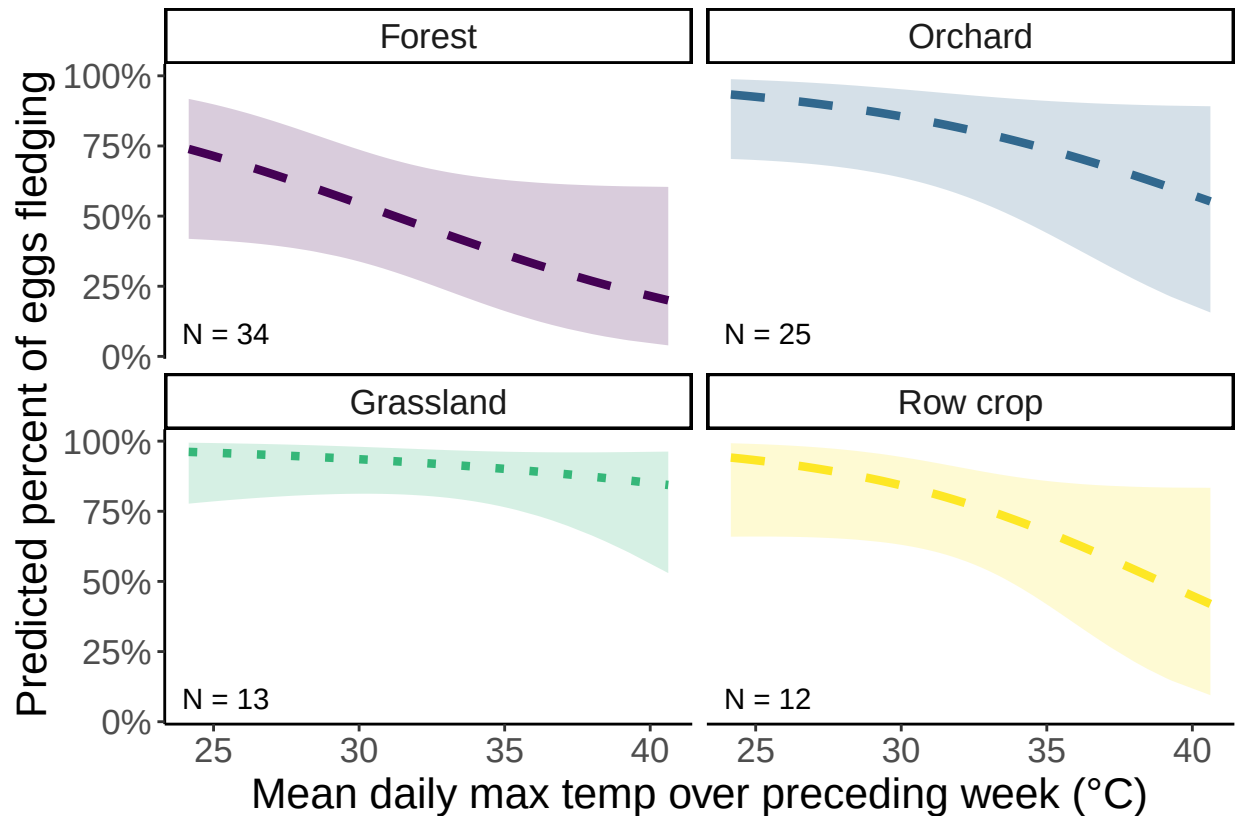
```
scale_linetype_manual(values = c("Forest" = "dashed", "Orchard" = "dashed", "Grassland" = "dotted", "Row crop" = "dotted"),
  geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = -Inf, label = label), hjust = -.2, vjust = .5,
  theme(legend.position = "none")
)
```

Plot of survival without interaction

Raw data not available.

Scale for colour is already present.

Adding another scale for colour, which will replace the existing scale.



```
(fig3_webl <- predict_response(s_nestpd_WEBL_admin, terms = c("meanmaxt_nestpd_scaled [all]", "habitat")) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Predicted percent of eggs fledging") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_webl,
    breaks = c((20 - mean_temp_webl) / sd_temp_webl,
      (25 - mean_temp_webl) / sd_temp_webl,
```

```

      (30-mean_temp_webl)/sd_temp_webl,
      (35-mean_temp_webl)/sd_temp_webl,
      (40-mean_temp_webl)/sd_temp_webl#,
      #(45-mean_temp)/sd_temp
    ),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) +
  # ylim(0,100) +
  scale_linetype_manual(values = c("Forest" = "dashed", "Orchard" = "dotted", "Grassland" = "dotted", "Row
  geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = -Inf, label = label), hjust = -.2, vjust = .
  theme(legend.position = "none")
)

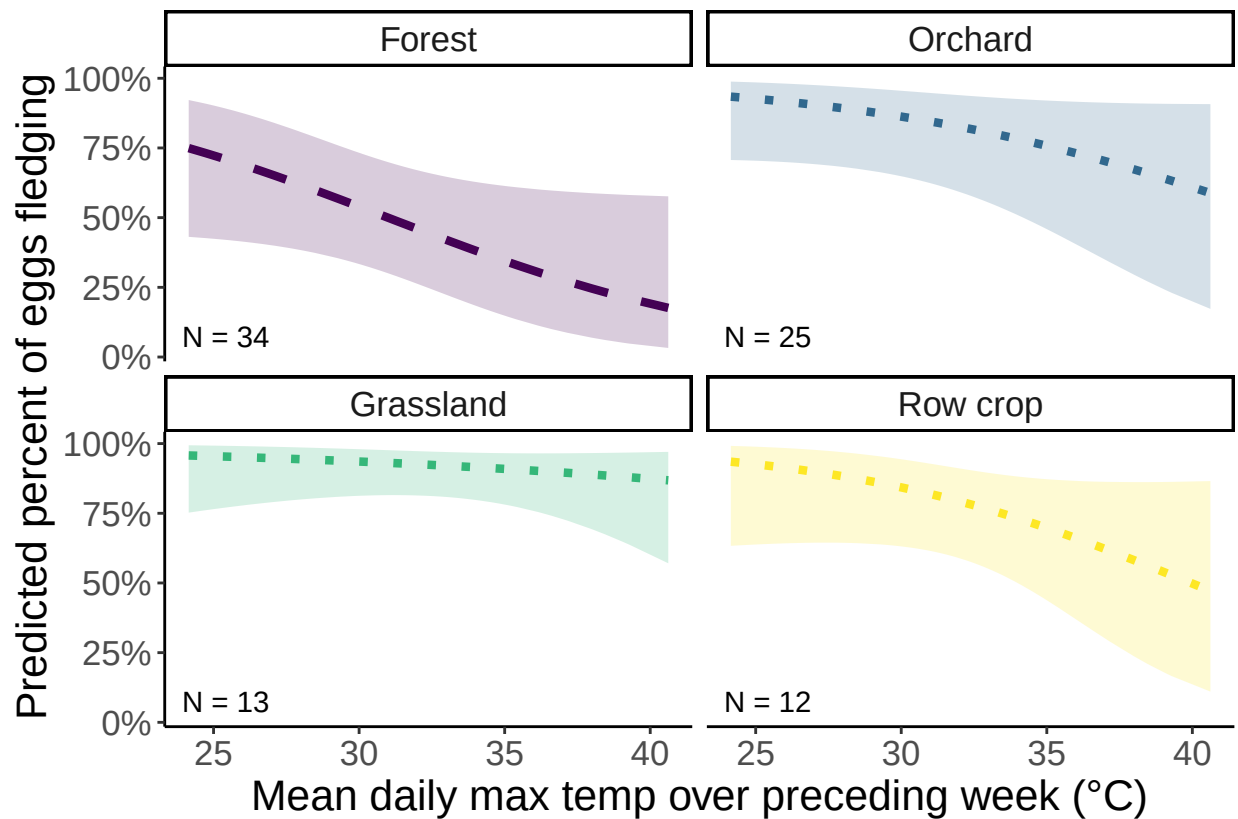
```

With interaction

Raw data not available.

Scale for colour is already present.

Adding another scale for colour, which will replace the existing scale.



```

(plott <- predict_response(s_nestpd_WEBL_addmin, terms = c("meanmaxt_nestpd_scaled [all]", "juliandate_ha
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +

```

```

# aes(linetype = .data[["group"]]) +
theme_classic() +
# facet_wrap(~ group, ncol = 2) +
xlab("Mean daily max temp over preceding week (\u00b0C)") +
ylab("Survival") +
# scale_fill_viridis(discrete = TRUE) +
# scale_color_viridis(discrete = TRUE) +
# scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "solid", "Grassland" = "dotted", "R
theme(text = element_text(size = 16)) +
# labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_tres,
                    breaks = c((20-mean_temp_tres)/sd_temp_tres,
                               (25-mean_temp_tres)/sd_temp_tres,
                               (30-mean_temp_tres)/sd_temp_tres,
                               (35-mean_temp_tres)/sd_temp_tres,
                               (40-mean_temp_tres)/sd_temp_tres),
                    # breaks = c(20,30,40,50),
                    # labels = c("20", "30", "40", "50"),
                    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
                    #          (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
                    # limits = c(18,5)
) #+
#ylim(-5,5) +
# geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust
# theme(legend.position = "none")
)

```

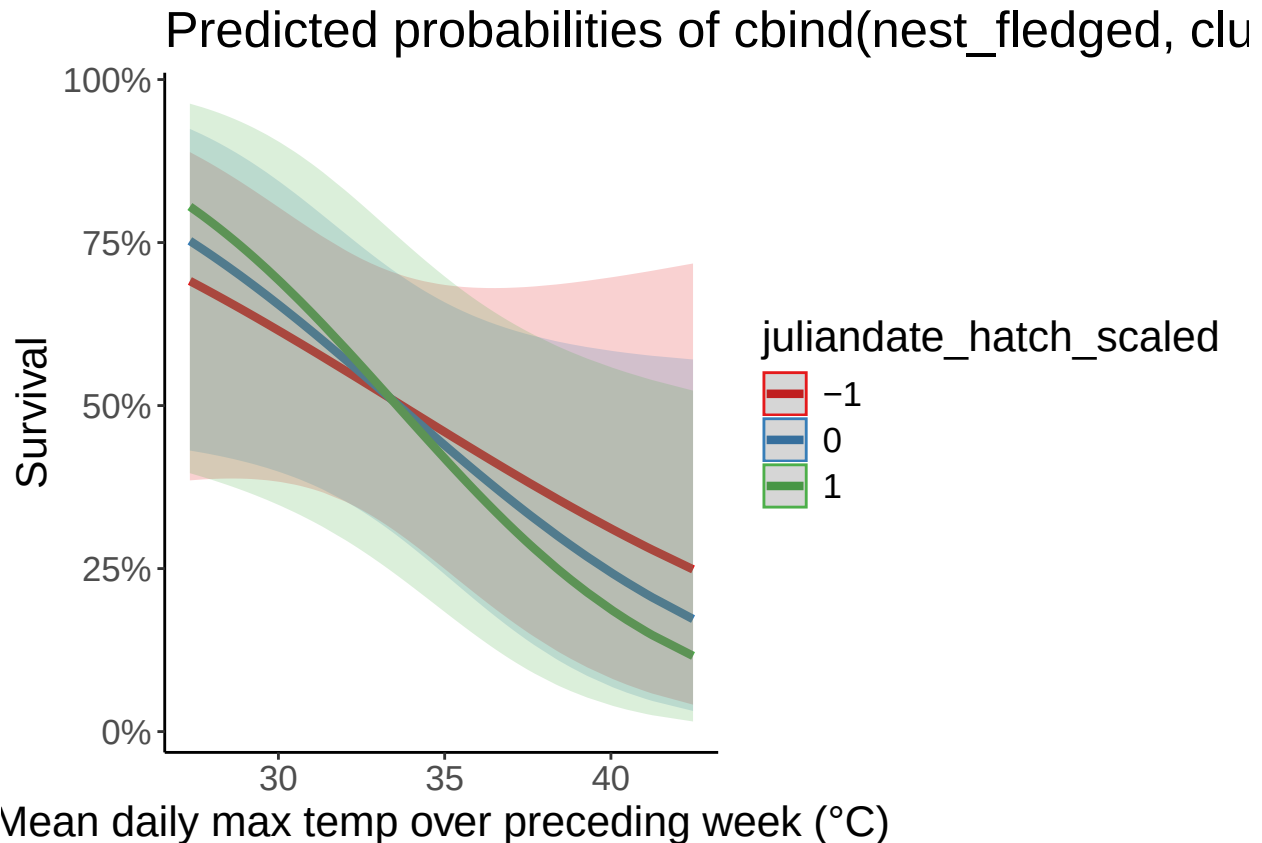
Visualize direction of interaction

```

## Bias-correction is currently only supported for mixed or gee models. No
##   bias-correction is applied.

## Raw data not available.

```



Conclusion: for WEBL survival, the interaction is not significant nor does it improve fit. Adding the interaction slightly changes p-values, which eliminates marginal significance in orchard and grassland, leaving only Forest with a marginally significant negative temp trend. It does not change the magnitude or direction of the effect of temperature.

```
prior_model <- glm(cbind(nest_fledged, clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled + meanmint_nestpd_scaled,
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "TRES",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
  mutate(across(c(meanmaxt_nestpd, meanmint_nestpd, juliandate_hatch),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"))
)

s_nestpd_TRES <- glm(cbind(nest_fledged, clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled * habitat,
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "TRES",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
  mutate(across(c(meanmaxt_nestpd, meanmint_nestpd, juliandate_hatch),
```

```

        ~ scale(.x)[,1],
        .names = "{.col}_scaled"
      )
    )

s_nestpd_TRES_addmax <- glm(cbind(nest_fledged,clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled + m
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "TRES",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
  mutate(across(c(meanmaxt_nestpd,meanmint_nestpd,juliandate_hatch),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"
  )
)

s_nestpd_TRES_addmin <- glm(cbind(nest_fledged,clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled * h
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "TRES",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
  mutate(across(c(meanmaxt_nestpd,meanmint_nestpd,juliandate_hatch),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"
  )
)

s_nestpd_TRES_noint <- glm(cbind(nest_fledged,clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled + me
  family = binomial(link = "logit"),
  data = dplyr::filter(s,
    Species == "TRES",
    !is.na(nest_fledged),
    !is.na(clutch_size)) %>%
  mutate(across(c(meanmaxt_nestpd,meanmint_nestpd,juliandate_hatch),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"
  )
)

c1 <- anova(s_nestpd_TRES,s_nestpd_TRES_addmin,s_nestpd_TRES_noint,test="Chisq") %>% tibble() %>%
  rename(Chisq = Deviance) %>%
  mutate(AIC = c(AIC(s_nestpd_TRES),AIC(s_nestpd_TRES_addmin),AIC(s_nestpd_TRES_noint)),.before = Df) %>%
  mutate(Model = c("no interaction","single interaction","both interacting"),.before = Df)

c2 <- anova(s_nestpd_TRES,s_nestpd_TRES_addmax,s_nestpd_TRES_noint,test="Chisq") %>% tibble() %>%
  rename(Chisq = Deviance) %>%
  mutate(AIC = c(AIC(s_nestpd_TRES),
    AIC(s_nestpd_TRES_addmax),
    AIC(s_nestpd_TRES_noint)),.before = Df) %>%
  mutate(Model = c("no interaction","single interaction","both interacting"),.before = Df)

(int_tab_survival_nestpd_tres <- bind_rows(c1,c2) %>%

```

```

as_tibble() %>%
mutate(across(where(is.numeric),~round(.x,digits = 4)),
      P = `Pr(>Chi)` ) %>%
dplyr::select(Model,AIC,Chisq,P) %>%
mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
      across(c(Chisq), ~ round(.x, digits = 2)),
      P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
group_by(max_or_min) %>% gt()

```

TRES

Model	AIC	Chisq	P
Max temp			
no interaction	238.4467	NA	NA
single interaction	233.4696	-1.02	0.7957
both interacting	230.7602	-3.29	0.349
Min temp			
no interaction	238.4467	NA	NA
single interaction	234.8721	-2.43	0.4889
both interacting	230.7602	-1.89	0.596

```
# gtsave(int_tab_survival_nestpd_tres,"figures/int_tab_survival_nestpd_tres.html")
```

#Conclusion: For nest attempt overall, land cover interacts with either max or min temp but not both to.

```
summary(s_nestpd_TRES_noint)
```

```

##
## Call:
## glm(formula = cbind(nest_fledged, clutch_size - nest_fledged) ~
##      meanmaxt_nestpd_scaled + meanmint_nestpd_scaled + habitat +
##      meanmaxt_nestpd_scaled * juliandate_hatch_scaled + year_fct +
##      site, family = binomial(link = "logit"), data = dplyr::filter(s,
##      Species == "TRES", !is.na(nest_fledged), !is.na(clutch_size)) %>%
##      mutate(across(c(meanmaxt_nestpd, meanmint_nestpd, juliandate_hatch),
##      ~scale(.x)[, 1], .names = "{.col}_scaled")))
##
## Coefficients:
##
##              Estimate Std. Error z value
## (Intercept)      -0.16378    0.36465  -0.449
## meanmaxt_nestpd_scaled    0.06783    0.21491   0.316
## meanmint_nestpd_scaled    0.52303    0.28376   1.843
## habitatOrchard         0.82033    0.76985   1.066
## habitatGrassland        1.61011    0.49921   3.225
## habitatRow crop         1.39421    0.42445   3.285
## juliandate_hatch_scaled  -0.81095    0.28047  -2.891

```

```
## year_fct2022          0.34104      0.32428      1.052
## year_fct2023          1.01083      0.45742      2.210
## sitePG                -0.22248      0.39844     -0.558
## siteRR                -0.63973      0.40734     -1.570
## siteWI                -1.84477      0.92834     -1.987
## meanmaxt_nestpd_scaled:juliandate_hatch_scaled  0.52889      0.20394      2.593
##                      Pr(>|z|)
## (Intercept)           0.65332
## meanmaxt_nestpd_scaled 0.75230
## meanmint_nestpd_scaled 0.06530 .
## habitatOrchard        0.28662
## habitatGrassland      0.00126 **
## habitatRow crop       0.00102 **
## juliandate_hatch_scaled 0.00384 **
## year_fct2022          0.29295
## year_fct2023          0.02712 *
## sitePG                0.57659
## siteRR                0.11630
## siteWI                0.04690 *
## meanmaxt_nestpd_scaled:juliandate_hatch_scaled 0.00950 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 185.30  on 67  degrees of freedom
## Residual deviance: 135.27  on 55  degrees of freedom
## AIC: 230.76
##
## Number of Fisher Scoring iterations: 5
```

```
anova(s_nestpd_TRES_noint, prior_model)
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(nest_fledged, clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled +
##   meanmint_nestpd_scaled + habitat + meanmaxt_nestpd_scaled *
##   juliandate_hatch_scaled + year_fct + site
## Model 2: cbind(nest_fledged, clutch_size - nest_fledged) ~ meanmaxt_nestpd_scaled +
##   meanmint_nestpd_scaled + habitat + juliandate_hatch_scaled +
##   year_fct + site
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         55      135.28
## 2         56      142.62 -1   -7.3425 0.006734 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interaction improves model fit for survival - same model comes out of the comparison

Do the results change?

```
(tressurvival_nestpd_trendmax_prior_model <- emtrends(prior_model, specs = pairwise ~ habitat, var = c("year_fct", "site"),
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
  df = round(df),
```

```

p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
rename(Habitat = "habitat", `Max temp trend` = "meanmaxt_nestpd_scaled.trend", Df = "df", `T-ratio` =
gt())

```

Comparing temp trend and significance across land covers without and with temp * julian date interaction

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.006	0.216	Inf	-0.029	0.977
Orchard	-0.006	0.216	Inf	-0.029	0.977
Grassland	-0.006	0.216	Inf	-0.029	0.977
Row crop	-0.006	0.216	Inf	-0.029	0.977

```

(tressurvival_nestpd_trendmax <- emtrends(s_nestpd_TRES_noint, specs = pairwise ~ habitat, var = c("meanmaxt_nestpd_scaled.trend", "Df", "T-ratio"),
mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
df = round(df),
p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
rename(Habitat = "habitat", `Max temp trend` = "meanmaxt_nestpd_scaled.trend", Df = "df", `T-ratio` =
gt())

```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.068	0.215	Inf	0.316	0.752
Orchard	0.068	0.215	Inf	0.316	0.752
Grassland	0.068	0.215	Inf	0.316	0.752
Row crop	0.068	0.215	Inf	0.316	0.752

Basically no change - temp trend becomes slightly more positive with addition of interaction but is still very small in magnitude and not significant.

```

data_tres = s_nestpd_TRES_noint$data

mean_temp_tres <- mean(data_tres %>% pull(meanmaxt_nestpd), na.rm = TRUE)
sd_temp_tres <- sd(data_tres %>% pull(meanmaxt_nestpd), na.rm = TRUE)

temp_trans_tres <- trans_new("temp_trans_tres",
transform = function(x){(x * sd_temp_tres) + mean_temp_tres},
inverse = function(x){x})

samp_tres <- data_tres %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("../figures/ss_tres.html")

dat_text_tres <- data.frame(
label = paste("N =", samp_tres$count),
group = factor(samp_tres$habitat)
)

```

```

)

(fig3_tres <- predict_response(prior_model, terms = c("meanmaxt_nestpd_scaled [all]", "habitat")) %>%
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Predicted percent of eggs fledging") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_tres,
    breaks = c((20-mean_temp_tres)/sd_temp_tres,
      (25-mean_temp_tres)/sd_temp_tres,
      (30-mean_temp_tres)/sd_temp_tres,
      (35-mean_temp_tres)/sd_temp_tres,
      (40-mean_temp_tres)/sd_temp_tres#,
      #(45-mean_temp)/sd_temp
    ),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18,5)
  ) +
  # ylim(0,100) +
  scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "dotted", "Ro
  geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = -Inf, label = label), hjust = -.2, vjust = .
  theme(legend.position = "none")
)

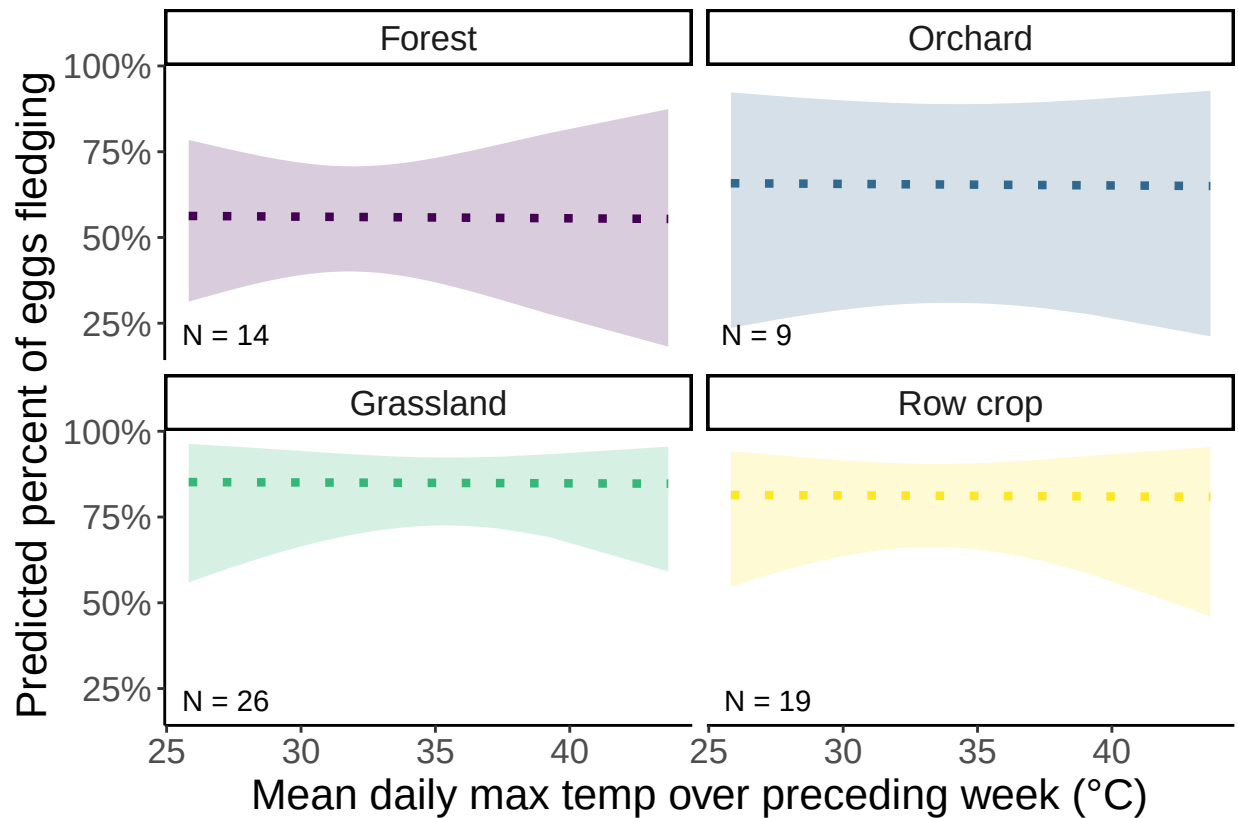
```

Plot of survival without interaction

Raw data not available.

Scale for colour is already present.

Adding another scale for colour, which will replace the existing scale.



```
(fig3_tres <- predict_response(s_nestpd_TRES_noint, terms = c("meanmaxt_nestpd_scaled [all]", "habitat"))
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Predicted percent of eggs fledging") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_tres,
    breaks = c((20-mean_temp_tres)/sd_temp_tres,
      (25-mean_temp_tres)/sd_temp_tres,
      (30-mean_temp_tres)/sd_temp_tres,
      (35-mean_temp_tres)/sd_temp_tres,
      (40-mean_temp_tres)/sd_temp_tres#,
      #(45-mean_temp)/sd_temp
    ),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    #           (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18, 5)
```

```

) +
# ylim(0,100) +
scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "dotted", "Row crop" = "dotted")) +
geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = -Inf, label = label), hjust = -.2, vjust = .5) +
theme(legend.position = "none")
)

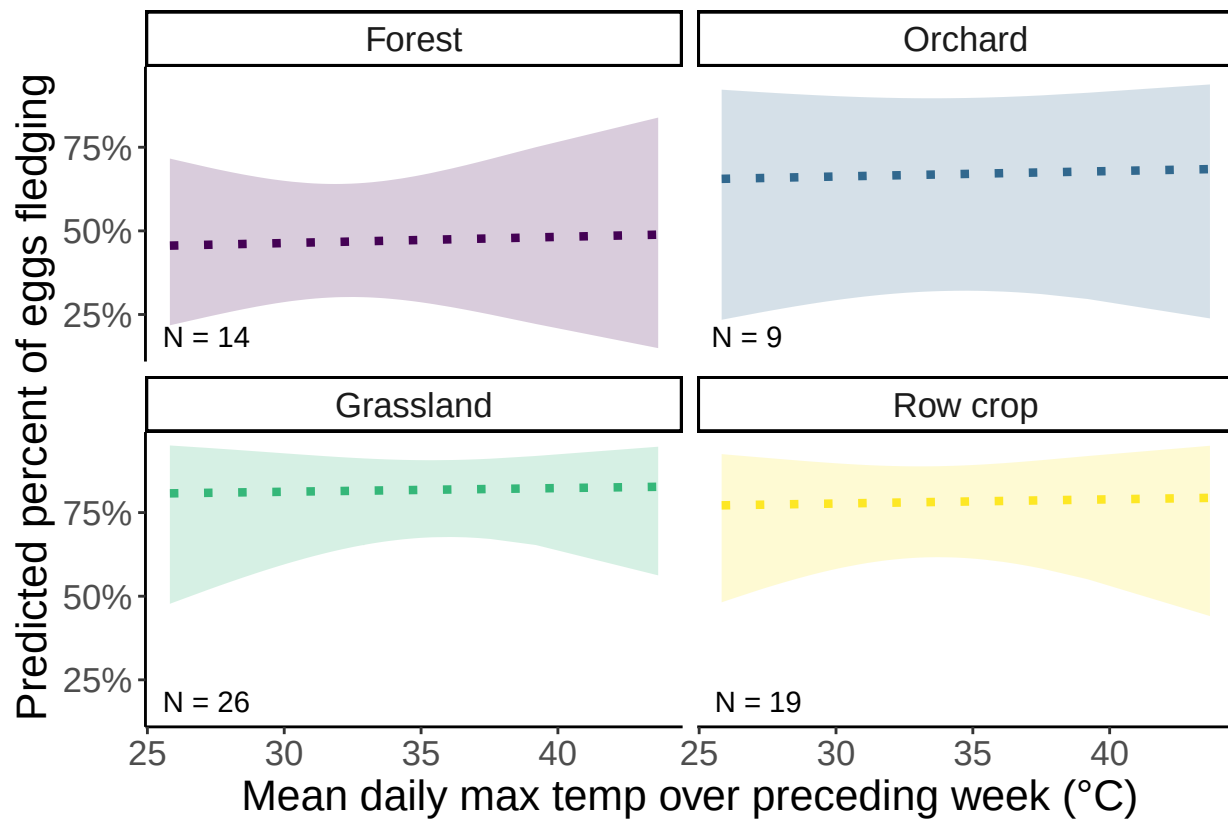
```

With interaction

```
## Raw data not available.
```

```
## Scale for colour is already present.
```

```
## Adding another scale for colour, which will replace the existing scale.
```



```

(plott <- predict_response(s_nestpd_TRES_noint, terms = c("meanmaxt_nestpd_scaled [all]", "juliandate_hatch")) +
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
# aes(linetype = .data[["group"]]) +
theme_classic() +
# facet_wrap(~ group, ncol = 2) +
xlab("Mean daily max temp over preceding week (°C)") +
ylab("Survival") +
# scale_fill_viridis(discrete = TRUE) +
# scale_color_viridis(discrete = TRUE) +
# scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "solid", "Grassland" = "dotted", "Row crop" = "dotted")) +
theme(text = element_text(size = 16)) +
# labs(title = element_blank()) +

```

```

scale_x_continuous(trans = temp_trans_tres,
  breaks = c((20-mean_temp_tres)/sd_temp_tres,
    (25-mean_temp_tres)/sd_temp_tres,
    (30-mean_temp_tres)/sd_temp_tres,
    (35-mean_temp_tres)/sd_temp_tres,
    (40-mean_temp_tres)/sd_temp_tres),
  # breaks = c(20,30,40,50),
  # labels = c("20", "30", "40", "50"),
  # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # limits = c(18,5)
) ##
#ylim(-5,5) +
# geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust
# theme(legend.position = "none")
)

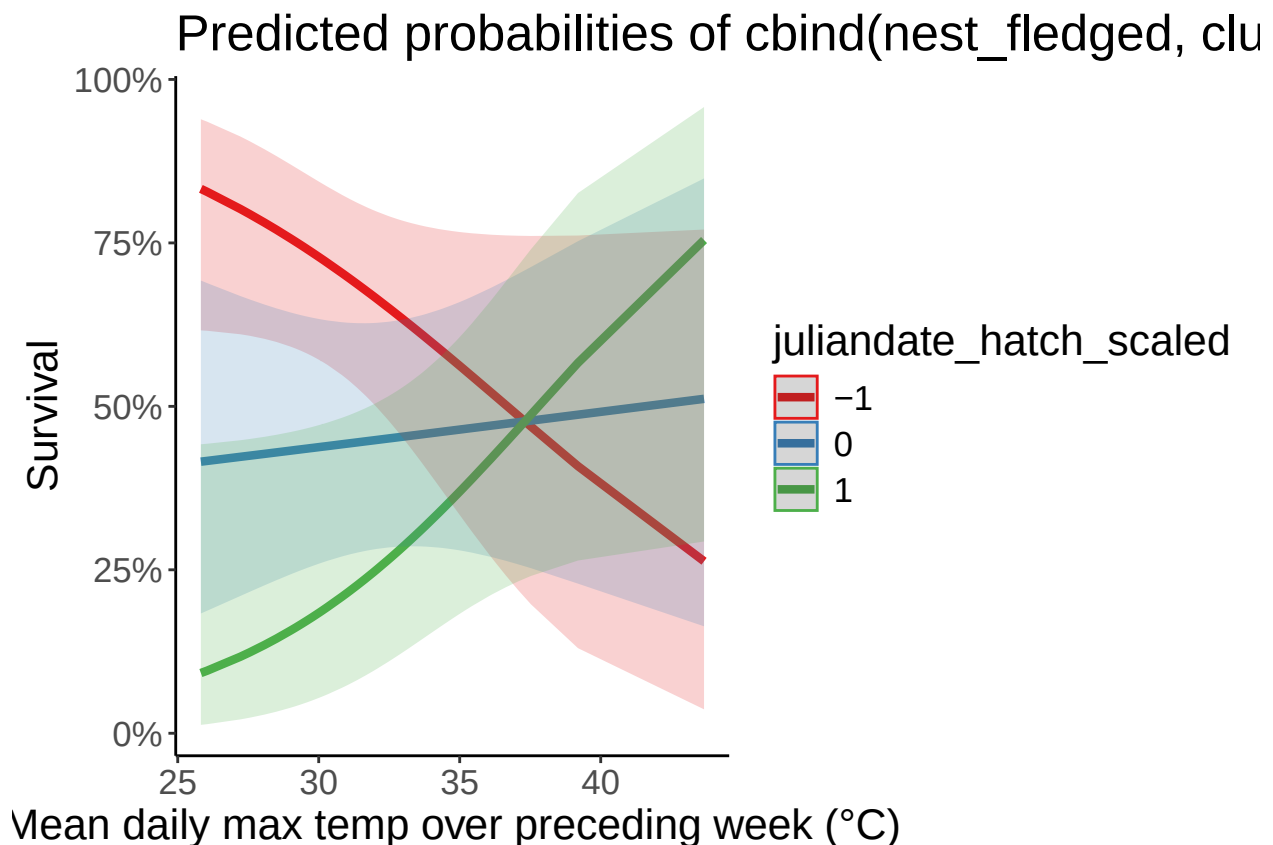
```

Visualize direction of interaction

```

## Bias-correction is currently only supported for mixed or gee models. No
## bias-correction is applied.
## Raw data not available.

```



Conclusion: for TRES survival, the interaction improves fit but does not alter the overall findings re: survival. The interaction between temp and julian date is as expected: Earlier in the season, hot temps decrease survival more than hot temps later in the season.

Baseline cort by prior week temp

```
g <- read_rds("../data/growth_cort_provis_manytempmeasures.rds") %>%
  mutate(year_fct = as.factor(year))

prior_model <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled + habitat + meanmintempI_scaled + habi
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s1_lintemp <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habi
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s1_lintemp_addmax <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s1_lintemp_addmin <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s1_lintemp_noint <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled + meanmintempI_scaled + habitat
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

c1 <- anova(s1_lintemp,s1_lintemp_addmin,s1_lintemp_noint) %>% tibble() %>% mutate(Model = c("no intera
```

WEBL

```
## refitting model(s) with ML (instead of REML)

c2 <- anova(s1_lintemp,s1_lintemp_addmax,s1_lintemp_noint) %>% tibble() %>% mutate(Model = c("no intera

## refitting model(s) with ML (instead of REML)

(int_tab_s1_webl <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
    P = `Pr(>Chisq)` %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())
```

Model	AIC	Chisq	P
Max temp			
no interaction	279.78	NA	NA
single interaction	280.19	5.59	0.133
both interacting	284.07	2.12	0.548
Min temp			
no interaction	279.78	NA	NA
single interaction	285.07	0.71	0.871
both interacting	284.07	7.00	0.072

```
# gtsave(int_tab_s1_webl,"figures/int_tab_s1_webl.html")

s1_webl <- s1_lintemp_noint

anova(s1_lintemp_noint,prior_model)

## refitting model(s) with ML (instead of REML)

## Data: dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>% ...
## Models:
## prior_model: sqrt(cort_s1) ~ meanmaxtempI_scaled + habitat + meanmintempI_scaled + habitat + age_scaled
## s1_lintemp_noint: sqrt(cort_s1) ~ meanmaxtempI_scaled + meanmintempI_scaled + habitat + age_scaled +
##               npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model      11 278.49 313.30 -128.24   256.49
## s1_lintemp_noint  12 279.78 317.76 -127.89   255.78 0.7095  1    0.3996
```

Same conclusion here - noint is the most supported model, and the addition of the temp * julian date interaction doesn't improve fit.

```
(webbs1trendmax_prior_model <- emtrends(prior_model,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "p.value",
         gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.038	0.077	100	0.492	0.624
Orchard	0.038	0.077	100	0.492	0.624
Grassland	0.038	0.077	100	0.492	0.624
Row crop	0.038	0.077	100	0.492	0.624

```
(webbs1trendmax <- emtrends(s1_lintemp_noint,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "p.value",
         gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.04	0.077	97	0.525	0.6
Orchard	0.04	0.077	97	0.525	0.6
Grassland	0.04	0.077	97	0.525	0.6
Row crop	0.04	0.077	97	0.525	0.6

Same results - very slightly positive and not significant. I don't think I'm even going to graph that.

```
samp_s1_webl <- s1_lintemp_noint@frame %>% group_by(habitat) %>% summarize(count = n())
# samp_s1_webl %>% gt() %>% gtsave("figures/ss_webl_s1.html")

dat_text_s1_webl <- data.frame(
  label = paste("N =", samp_s1_webl$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_s1_webl = dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

mean_temp_s1_webl <- mean(data_s1_webl %>% pull(meanmaxtempI))
sd_temp_s1_webl <- sd(data_s1_webl %>% pull(meanmaxtempI))

temp_trans_s1_webl <- trans_new("temp_trans_s1_webl",
  transform = function(x){(x * sd_temp_s1_webl) + mean_temp_s1_webl},
  inverse = function(x){x})

(figs2_webl <- ggpredict(s1_lintemp_noint, terms = c("meanmaxtempI_scaled [all]", "juliandate_scaled"), bi
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Baseline corticosterone (ng/mL)") +
  #scale_fill_viridis(discrete = TRUE) +
  #scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s1_webl,
    breaks = c((20-mean_temp_s1_webl)/sd_temp_s1_webl,
      (25-mean_temp_s1_webl)/sd_temp_s1_webl,
```

```

      (30-mean_temp_s1_webl)/sd_temp_s1_webl,
      (35-mean_temp_s1_webl)/sd_temp_s1_webl,
      (40-mean_temp_s1_webl)/sd_temp_s1_webl),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) ##
  # scale_linetype_manual(values = c("dotted","dotted","dotted","dotted")) +
  #ylim(-5,5) +
  #geom_text(data = dat_text_s1_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vju
  #theme(legend.position = "none")
)

```

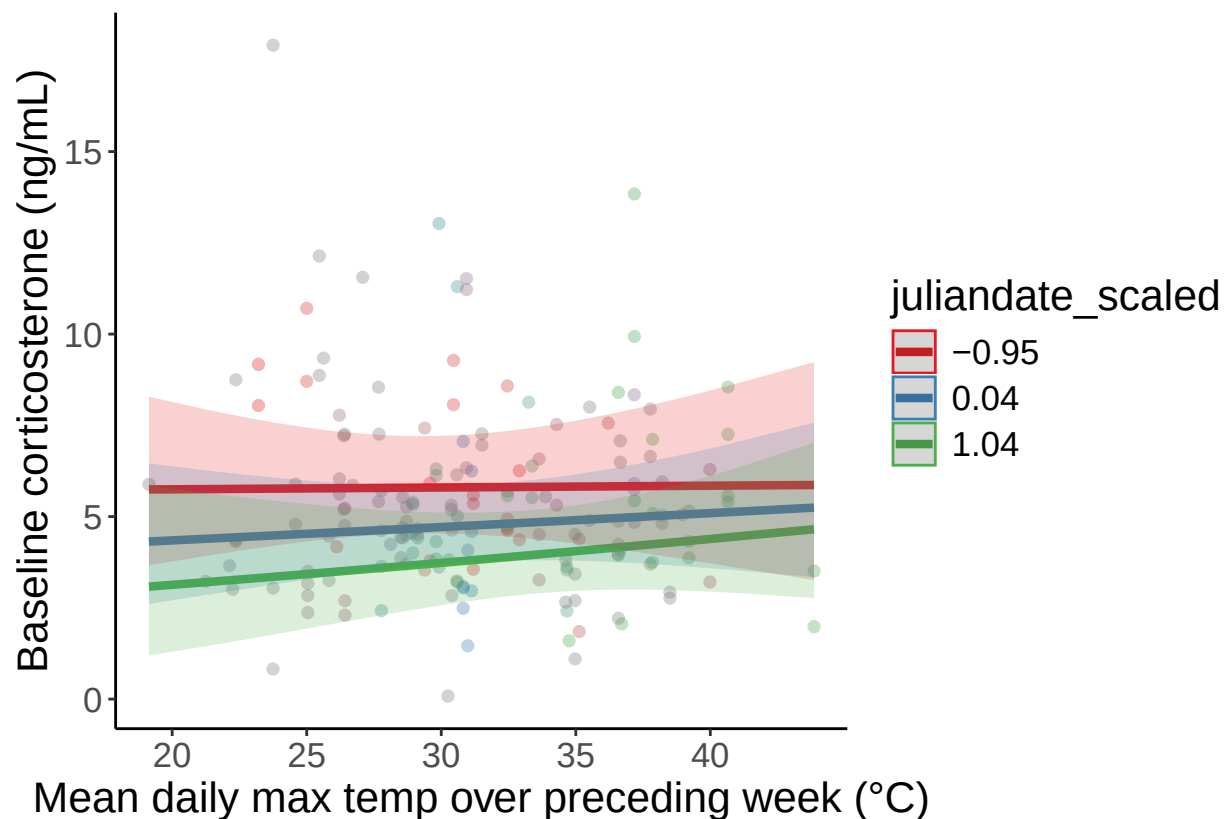
Visualization of the temp * julian date interaction

```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Looks like baseline cort is higher later in the season, but there isn't much of an interaction.

Conclusion: juliandate * temp interaction neither adds to the fit nor changes the results.

```

prior_model <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat + age,
                             mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size),
                                             ~ scale(.x)[,1],
                                             .names = "{.col}_scaled"),
                             meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

s1_lintemp <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat,
                             mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size),
                                             ~ scale(.x)[,1],
                                             .names = "{.col}_scaled"),
                             meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

s1_lintemp_addmax <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat,
                                   mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size),
                                                   ~ scale(.x)[,1],
                                                   .names = "{.col}_scaled"),
                                   meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

s1_lintemp_addmin <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat,
                                   mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size),
                                                   ~ scale(.x)[,1],
                                                   .names = "{.col}_scaled"),
                                   meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

s1_lintemp_noint <- lmerTest::lmer(sqrt(cort_s1) ~ meanmaxtempI_scaled + meanmintempI_scaled + habitat,
                                   mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size),
                                                   ~ scale(.x)[,1],
                                                   .names = "{.col}_scaled"),
                                   meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

c1 <- anova(s1_lintemp,s1_lintemp_addmin,s1_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",

```

TRES

```

## refitting model(s) with ML (instead of REML)
c2 <- anova(s1_lintemp,s1_lintemp_addmax,s1_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",
## refitting model(s) with ML (instead of REML)
(int_tab_s1_tres <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
         P = `Pr(>Chisq)` ) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
         across(c(AIC,Chisq), ~ round(.x, digits = 2)),
         P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	197.59	NA	NA

single interaction	203.28	0.31	0.958
both interacting	193.99	15.29	0.002
Min temp			
no interaction	197.59	NA	NA
single interaction	191.64	11.95	0.008
both interacting	193.99	3.65	0.302

```
#gtsave(int_tab_s1_tres,"figures/int_tab_s1_tres.html")

s1_tres <- s1_lintemp_addmax

anova(s1_lintemp_addmax,prior_model)

## refitting model(s) with ML (instead of REML)
## Data: dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>% ...
## Models:
## prior_model: sqrt(cort_s1) ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat + age_scaled + julia
## s1_lintemp_addmax: sqrt(cort_s1) ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat + age_scaled + julia
##               npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model      14 190.32 227.48 -81.161   162.32
## s1_lintemp_addmax 15 191.64 231.45 -80.821   161.64 0.6794  1    0.4098

Same model results, and temp * julian date does not add to model fit

(tress1trendmax_prior <- emtrends(prior_model,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "p.value")
gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.155	0.119	32	1.309	0.2
Orchard	0.155	0.119	32	1.309	0.2
Grassland	0.155	0.119	32	1.309	0.2
Row crop	0.155	0.119	32	1.309	0.2

```
(tress1trendmax <- emtrends(s1_lintemp_addmax,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "p.value")
gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.182	0.125	26	1.458	0.157
Orchard	0.182	0.125	26	1.458	0.157

Grassland	0.182	0.125	26	1.458	0.157
Row crop	0.182	0.125	26	1.458	0.157

No difference between models.

```
samp_s1_tres <- s1_lintemp_addmax@frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_tres_s1.html")

dat_text_s1_tres <- data.frame(
  label = paste("N =", samp_s1_tres$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_s1_tres = dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

mean_temp_s1_tres <- mean(data_s1_tres %>% pull(meanmaxtempI))
sd_temp_s1_tres <- sd(data_s1_tres %>% pull(meanmaxtempI))

temp_trans_s1_tres <- trans_new("temp_trans_s1_tres",
  transform = function(x){x * sd_temp_s1_tres} + mean_temp_s1_tres,
  inverse = function(x){x})

(figs2_tres <- ggpredict(prior_model, terms = c("meanmaxtempI_scaled [all]"), bias_correction = TRUE) %>%
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Baseline corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s1_tres,
    breaks = c((20-mean_temp_s1_tres)/sd_temp_s1_tres,
      (25-mean_temp_s1_tres)/sd_temp_s1_tres,
      (30-mean_temp_s1_tres)/sd_temp_s1_tres,
      (35-mean_temp_s1_tres)/sd_temp_s1_tres,
      (40-mean_temp_s1_tres)/sd_temp_s1_tres),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18, 5)
  ) +
```

```

# scale_linetype_manual(values = c("dotted","dotted","dotted","dotted")) +
# ylim(-5,5) +
# geom_text(data = dat_text_s1_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vj
theme(legend.position = "none")
)

```

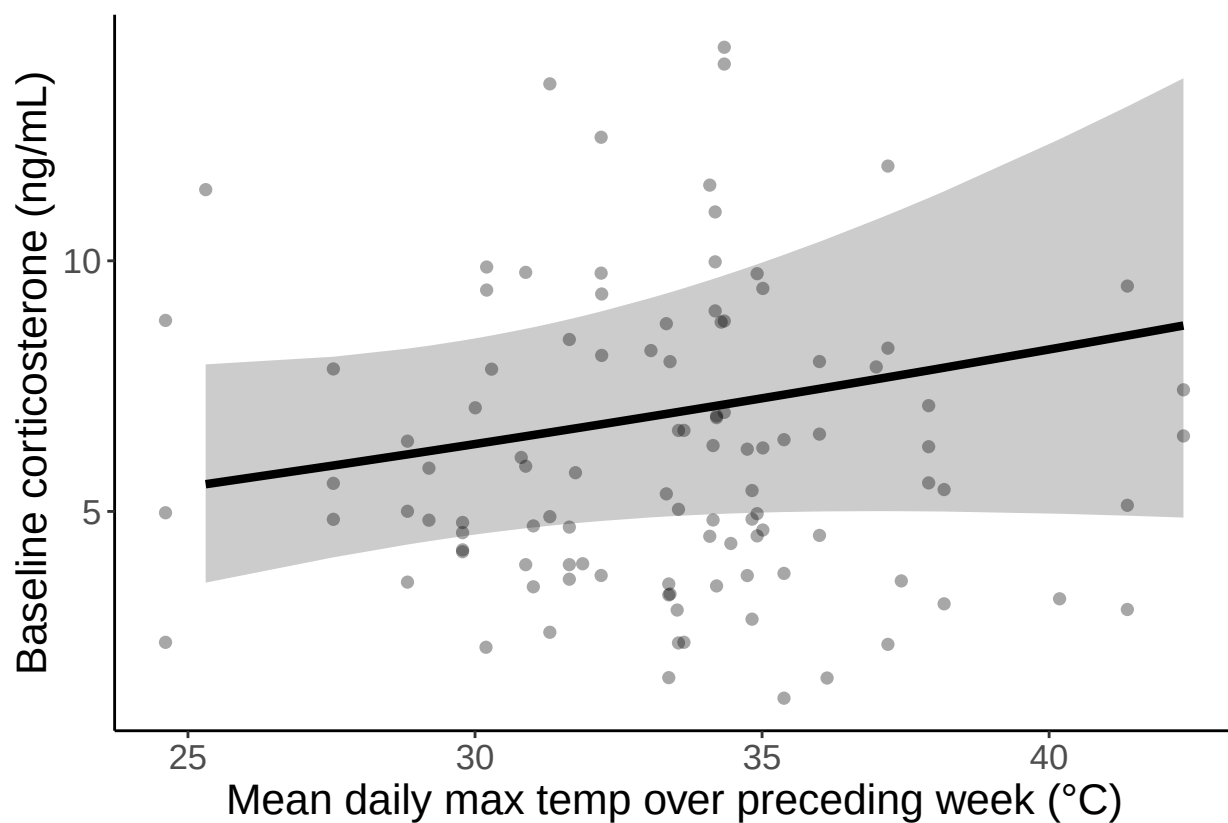
Without interaction

```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



```

samp_s1_tres <- s1_lintemp_addmax@frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_tres_s1.html")

dat_text_s1_tres <- data.frame(
  label = paste("N =", samp_s1_tres$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_s1_tres = dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),

```

```

      ~ scale(.x)[,1],
      .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

mean_temp_s1_tres <- mean(data_s1_tres %>% pull(meanmaxtempI))
sd_temp_s1_tres <- sd(data_s1_tres %>% pull(meanmaxtempI))

temp_trans_s1_tres <- trans_new("temp_trans_s1_tres",
                                transform = function(x){(x * sd_temp_s1_tres) + mean_temp_s1_tres},
                                inverse = function(x){x})

(figs2_tres <- ggpredict(s1_lintemp_addmax, terms = c("meanmaxtempI_scaled [all]"), bias_correction = TRUE) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Baseline corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s1_tres,
                     breaks = c((20-mean_temp_s1_tres)/sd_temp_s1_tres,
                                (25-mean_temp_s1_tres)/sd_temp_s1_tres,
                                (30-mean_temp_s1_tres)/sd_temp_s1_tres,
                                (35-mean_temp_s1_tres)/sd_temp_s1_tres,
                                (40-mean_temp_s1_tres)/sd_temp_s1_tres),
                     # breaks = c(20,30,40,50),
                     # labels = c("20", "30", "40", "50"),
                     # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
                     #           (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
                     # limits = c(18,5)
  ) +
  # scale_linetype_manual(values = c("dotted", "dotted", "dotted", "dotted")) +
  # ylim(-5,5) +
  # geom_text(data = dat_text_s1_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1)
  theme(legend.position = "none")
)

```

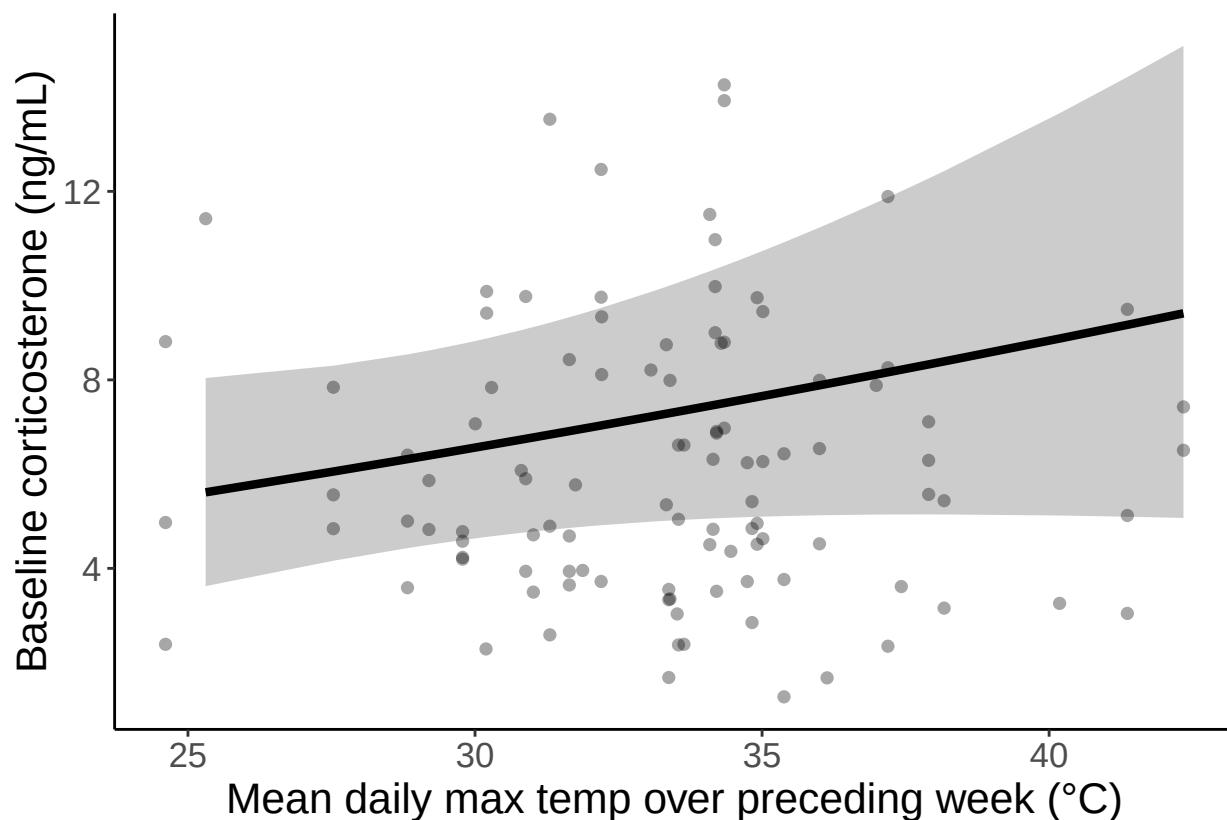
With interaction

```

## Model has sqrt-transformed response. Back-transforming predictions to
##   original response scale. Standard errors are still on the transformed
##   scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
##   random variation to the location of data points and avoid overplotting.

```



```
(figs2_tres <- ggpredict(s1_lintemp_addmax, terms = c("meanmaxtempI_scaled [all]", "juliandate_scaled"), b
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Baseline corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s1_tres,
    breaks = c((20-mean_temp_s1_tres)/sd_temp_s1_tres,
      (25-mean_temp_s1_tres)/sd_temp_s1_tres,
      (30-mean_temp_s1_tres)/sd_temp_s1_tres,
      (35-mean_temp_s1_tres)/sd_temp_s1_tres,
      (40-mean_temp_s1_tres)/sd_temp_s1_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    #           (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18,5)
  ) ##
  # scale_linetype_manual(values = c("dotted", "dotted", "dotted", "dotted")) +
```

```

#ylim(-5,5) +
# geom_text(data = dat_text_s1_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vj
#theme(legend.position = "none")
)

```

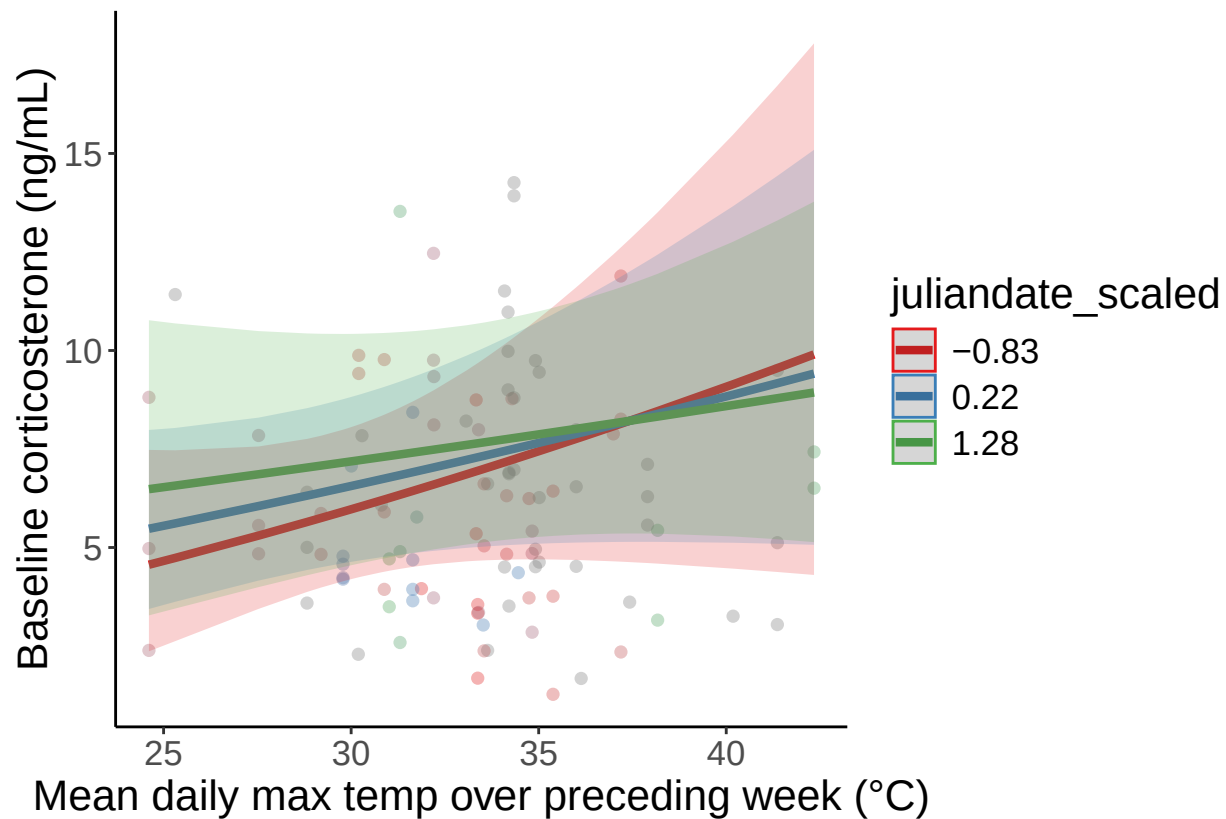
Visualize interaction

```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Interaction is not significant, does not add to fit, does not change results.

Abs difference cort by prior week temp

```

prior_model <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled
mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
~ scale(.x)[,1],
.names = "{.col}_scaled"),
meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

```

WEBL

```

## Warning in sqrt(abs_change_cort): NaNs produced

```

```

## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

abs_lintemp <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled +
                             mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
                                             ~ scale(.x)[,1],
                                             .names = "{.col}_scaled"),
                             meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

abs_lintemp_addmax <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled + meanmintempI_scaled +
                                     mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
                                                     ~ scale(.x)[,1],
                                                     .names = "{.col}_scaled"),
                                     meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

abs_lintemp_addmin <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled +
                                     mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
                                                     ~ scale(.x)[,1],
                                                     .names = "{.col}_scaled"),
                                     meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

abs_lintemp_noint <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled + meanmintempI_scaled +
                                     mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
                                                     ~ scale(.x)[,1],
                                                     .names = "{.col}_scaled"),
                                     meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

```

```

c1 <- anova(abs_lintemp,abs_lintemp_addmin,abs_lintemp_noint) %>% tibble() %>% mutate(Model = c("no int
## refitting model(s) with ML (instead of REML)
c2 <- anova(abs_lintemp,abs_lintemp_addmax,abs_lintemp_noint) %>% tibble() %>% mutate(Model = c("no int
## refitting model(s) with ML (instead of REML)
(int_tab_abs_webl <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
    P = `Pr(>Chisq)` ) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	558.54	NA	NA
single interaction	564.35	0.20	0.978
both interacting	557.08	13.27	0.004
Min temp			
no interaction	558.54	NA	NA
single interaction	559.36	5.18	0.159
both interacting	557.08	8.28	0.041

```

#gtsave(int_tab_abs_webl,"figures/int_tab_abs_webl.html")

```

```

check_collinearity(abs_lintemp_noint)

```

```

## # Check for Multicollinearity

```

```

##

```

```

## Low Correlation

```

```

##

```

```

##               Term  VIF    VIF 95% CI Increased SE Tolerance
## meanmaxtempI_scaled 2.09 [1.70, 2.69]          1.45      0.48
## meanmintempI_scaled 2.73 [2.18, 3.55]          1.65      0.37
##          habitat 1.29 [1.13, 1.66]          1.14      0.78
##          age_scaled 1.07 [1.00, 1.88]          1.03      0.94
##      juliandate_scaled 2.28 [1.84, 2.94]          1.51      0.44
##          year_fct 1.17 [1.05, 1.55]          1.08      0.86
## meanmaxtempI_scaled:juliandate_scaled 1.09 [1.01, 1.63]          1.05      0.91

```

```

## Tolerance 95% CI

```

```

## [0.37, 0.59]

```

```

## [0.28, 0.46]

```

```

## [0.60, 0.89]

```

```

## [0.53, 1.00]

```

```

## [0.34, 0.54]

```

```
##      [0.64, 0.95]
##      [0.61, 0.99]
abs_webl <- abs_lintemp
summary(abs_lintemp)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled *
##      habitat + age_scaled + meanmaxtempI_scaled * juliandate_scaled +
##      year_fct + (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
##      mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate,
##      brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##      meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
##
## REML criterion at convergence: 522
##
## Scaled residuals:
##      Min      1Q   Median      3Q      Max
## -2.00443 -0.64056  0.06119  0.59631  1.77399
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 1.918      1.385
## Residual              1.611      1.269
## Number of obs: 142, groups: attempt_id, 49
##
## Fixed effects:
##
##              Estimate Std. Error    df t value
## (Intercept)      3.65050    0.54914 30.00645    6.648
## meanmaxtempI_scaled -0.33290    0.59170 77.27571   -0.563
## habitatOrchard      0.49749    0.65865 32.62781    0.755
## habitatGrassland     1.47048    0.85051 42.22069    1.729
## habitatRow crop      0.50632    0.72691 29.65188    0.697
## meanmintempI_scaled  0.43783    0.62223 62.84732    0.704
## age_scaled          0.13572    0.15853 125.69432    0.856
## juliandate_scaled   -0.74829    0.35733 41.67364   -2.094
## year_fct2023        -0.07093    0.56928 34.89660   -0.125
## meanmaxtempI_scaled:habitatOrchard  0.57321    0.75595 82.59512    0.758
## meanmaxtempI_scaled:habitatGrassland -1.34009    0.77837 97.42958   -1.722
## meanmaxtempI_scaled:habitatRow crop -0.30841    0.81055 69.19441   -0.380
## habitatOrchard:meanmintempI_scaled -0.41148    0.77304 57.73853   -0.532
## habitatGrassland:meanmintempI_scaled  2.91116    1.04984 100.75820    2.773
## habitatRow crop:meanmintempI_scaled  0.34028    0.78588 49.88824    0.433
## meanmaxtempI_scaled:juliandate_scaled  0.06086    0.20467 83.11525    0.297
##
##              Pr(>|t|)
## (Intercept)      2.31e-07 ***
## meanmaxtempI_scaled  0.57532
## habitatOrchard      0.45548
## habitatGrassland     0.09113 .
## habitatRow crop      0.49152
## meanmintempI_scaled  0.48425
```

```
## age_scaled 0.39358
## juliandate_scaled 0.04237 *
## year_fct2023 0.90155
## meanmaxtempI_scaled:habitatOrchard 0.45045
## meanmaxtempI_scaled:habitatGrassland 0.08830 .
## meanmaxtempI_scaled:habitatRow crop 0.70474
## habitatOrchard:meanmintempI_scaled 0.59657
## habitatGrassland:meanmintempI_scaled 0.00662 **
## habitatRow crop:meanmintempI_scaled 0.66688
## meanmaxtempI_scaled:juliandate_scaled 0.76693
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 16 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)         if you need it
anova(abs_lintemp,prior_model)

## refitting model(s) with ML (instead of REML)

## Data: dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>% ...
## Models:
## prior_model: sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat +
## abs_lintemp: sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat +
##          npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model   17 555.26 605.51 -260.63   521.26
## abs_lintemp   18 557.08 610.28 -260.54   521.08 0.1852  1    0.6669
```

Does not add to fit, is not significant in model

```
(weblabstrendmax_prior <- emtrends(prior_model,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% t
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "
  gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.333	0.602	83	-0.554	0.581
Orchard	0.224	0.519	113	0.432	0.667
Grassland	-1.693	0.523	126	-3.240	0.002
Row crop	-0.608	0.548	79	-1.109	0.271

```
(weblabstrendmax <- emtrends(abs_lintemp,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test() %
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "
  gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
---------	----------------	----	----	---------	---

Forest	-0.330	0.604	83	-0.547	0.586
Orchard	0.243	0.524	109	0.463	0.644
Grassland	-1.670	0.530	126	-3.153	0.002
Row crop	-0.639	0.560	73	-1.140	0.258

No change to coefficients or significance

```
samp <- abs_lintemp@frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_webl_abs.html")

dat_text_webl <- data.frame(
  label = paste("N =", samp$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_webl = dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

mean_temp_webl <- mean(data_webl %>% pull(meanmaxtempI))
sd_temp_webl <- sd(data_webl %>% pull(meanmaxtempI))

temp_trans_webl <- trans_new("temp_trans_webl",
  transform = function(x){(x * sd_temp_webl) + mean_temp_webl},
  inverse = function(x){x})

(fig4_webl <- ggpredict(prior_model, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = TRUE) %>%
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Stress-induced - Baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "WEBL cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_webl,
    breaks = c((20-mean_temp_webl)/sd_temp_webl,
      (25-mean_temp_webl)/sd_temp_webl,
      (30-mean_temp_webl)/sd_temp_webl,
      (35-mean_temp_webl)/sd_temp_webl,
      (40-mean_temp_webl)/sd_temp_webl),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18, 5))
```

```

) +
scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "solid", "Row crop" = "solid"),
ylim(0,60) +
geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1.5) +
theme(legend.position = "none")
)

```

```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

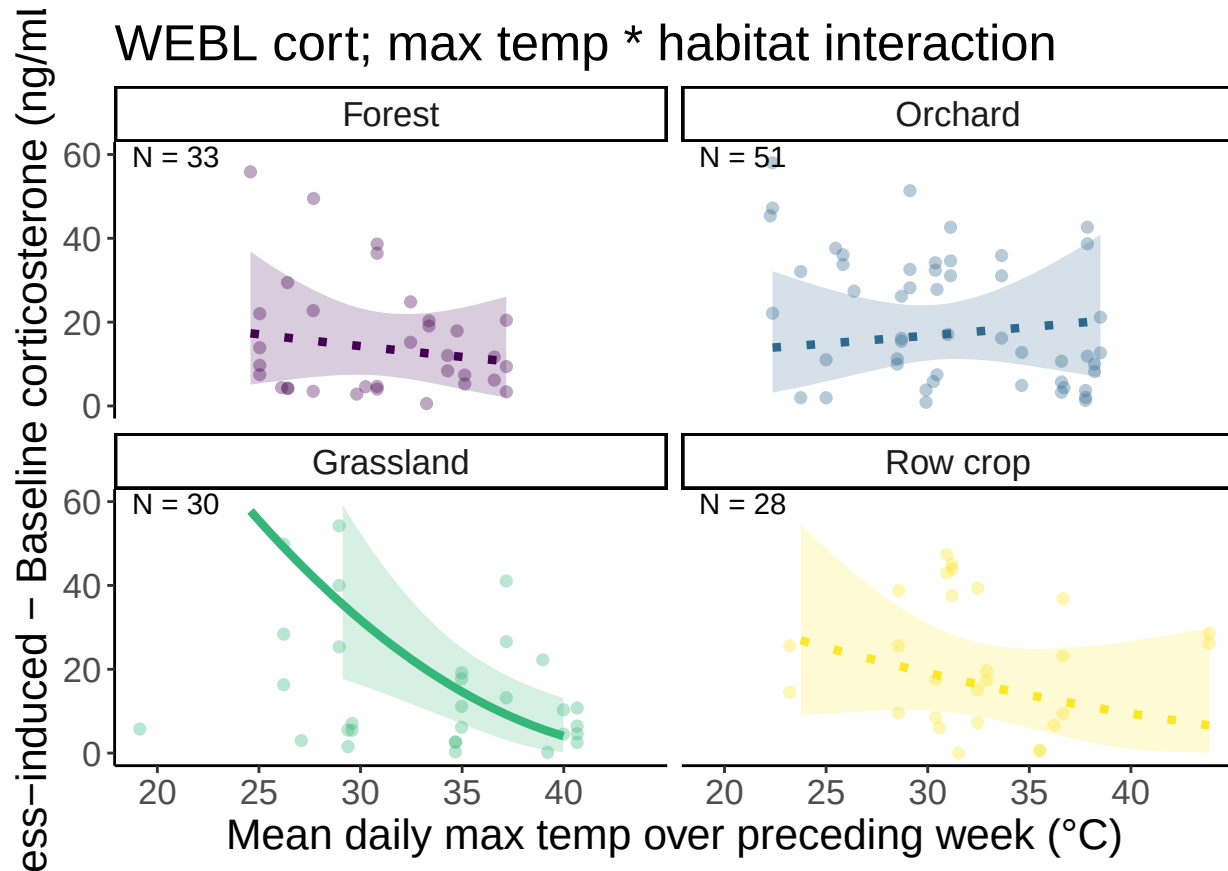
## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_line()`).

```



```

(fig4_webl <- ggpredict(abs_lintemp, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = "none",
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
aes(linetype = .data[["group"]])) +

```

```

theme_classic() +
facet_wrap(~ group, ncol = 2) +
xlab("Mean daily max temp over preceding week (\u00b0C)") +
ylab("Stress-induced - Baseline corticosterone (ng/mL)") +
scale_fill_viridis(discrete = TRUE) +
scale_color_viridis(discrete = TRUE) +
theme(text = element_text(size = 16)) +
labs(title = "WEBL cort; max temp * habitat interaction") +
scale_x_continuous(trans = temp_trans_webl,
                    breaks = c((20-mean_temp_webl)/sd_temp_webl,
                               (25-mean_temp_webl)/sd_temp_webl,
                               (30-mean_temp_webl)/sd_temp_webl,
                               (35-mean_temp_webl)/sd_temp_webl,
                               (40-mean_temp_webl)/sd_temp_webl),
                    # breaks = c(20,30,40,50),
                    # labels = c("20", "30", "40", "50"),
                    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
                    #          (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
                    # limits = c(18,5)
) +
scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "solid", "Ro
ylim(0,60) +
geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust =
theme(legend.position = "none")
)

```

```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

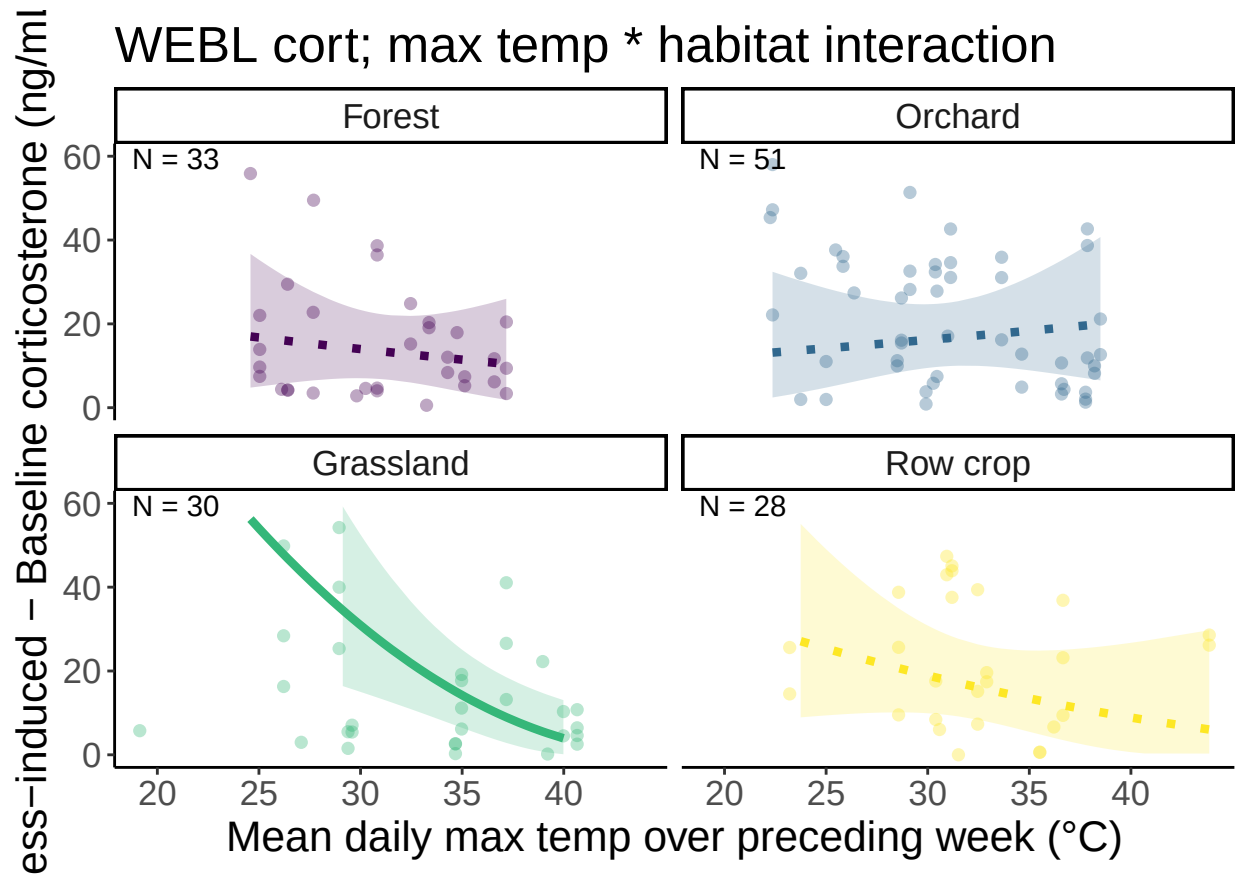
## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_line()`).

```



```
(fig4_webl <- ggpredict(abs_lintemp, terms = c("meanmaxtempI_scaled [all]", "juliandate_scaled"), bias_corr = TRUE,
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Stress-induced - Baseline corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "WEBL cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_webl,
    breaks = c((20-mean_temp_webl)/sd_temp_webl,
      (25-mean_temp_webl)/sd_temp_webl,
      (30-mean_temp_webl)/sd_temp_webl,
      (35-mean_temp_webl)/sd_temp_webl,
      (40-mean_temp_webl)/sd_temp_webl),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18,5)
  ) # +
  # scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "solid", "Row crop" = "solid"))
```

```

# ylim(0,60) +
# geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust
# theme(legend.position = "none")
)

```

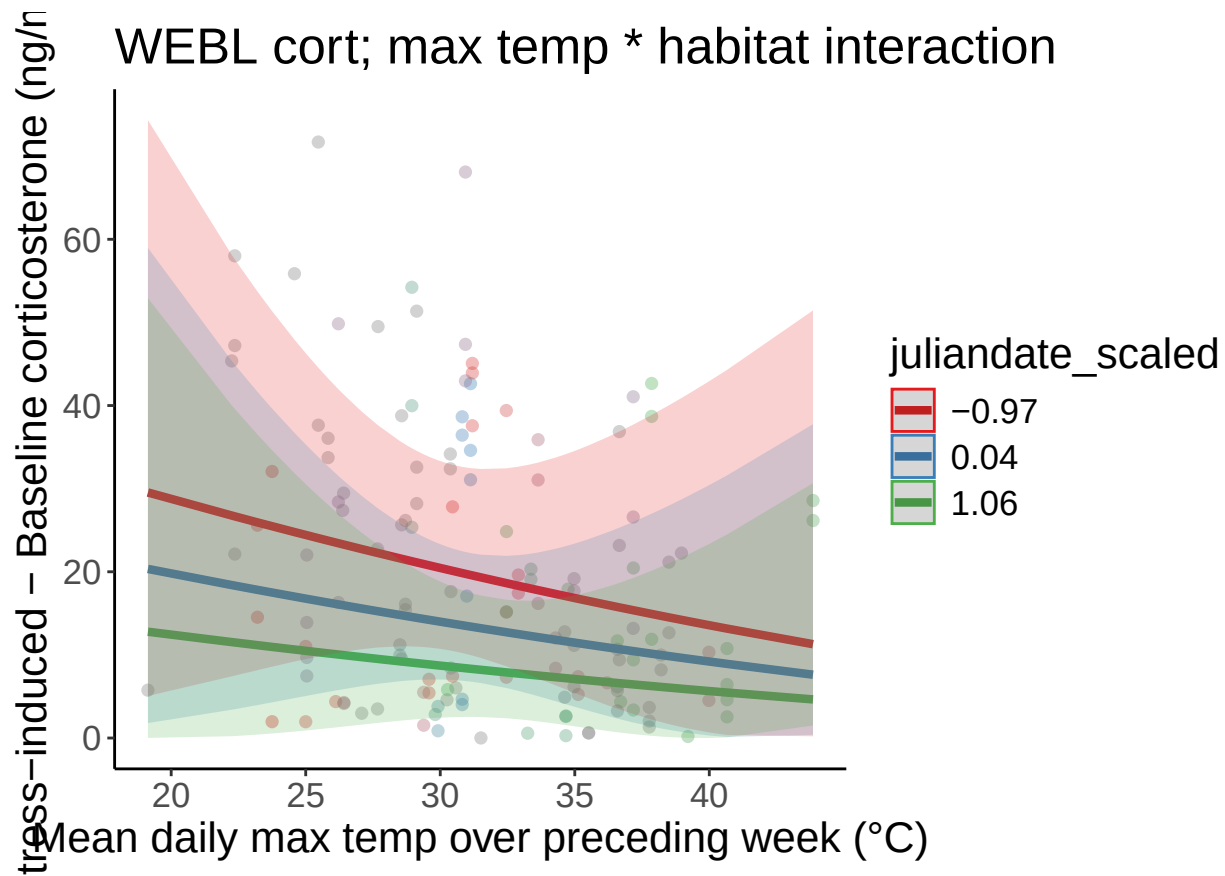
Visualize interaction

```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



No interaction!

```

prior_model <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled
mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size)
~ scale(.x)[,1],
.names = "{.col}_scaled"),
meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

abs_lintemp <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled
mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size)
~ scale(.x)[,1],
.names = "{.col}_scaled"),

```

```

meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

abs_lintemp_addmax <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled + meanmintempI_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

prior_model_addmin <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

abs_lintemp_addmin <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

abs_lintemp_noint <- lmerTest::lmer(sqrt(abs_change_cort) ~ meanmaxtempI_scaled + meanmintempI_scaled +
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

c1 <- anova(abs_lintemp,abs_lintemp_addmin,abs_lintemp_noint) %>% tibble() %>% mutate(Model = c("no int", "max temp", "min temp", "both"))

```

TRES

```

## refitting model(s) with ML (instead of REML)
c2 <- anova(abs_lintemp,abs_lintemp_addmax,abs_lintemp_noint) %>% tibble() %>% mutate(Model = c("no int", "max temp", "min temp", "both"))

## refitting model(s) with ML (instead of REML)
(int_tab_abs_tres <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
    P = `Pr(>Chisq)` ) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	348.25	NA	NA
single interaction	344.98	9.27	0.026
both interacting	344.87	6.12	0.106
Min temp			

no interaction	348.25	NA	NA
single interaction	352.86	1.39	0.708
both interacting	344.87	13.99	0.003

```
# gtsave(int_tab_abs_tres,"figures/int_tab_abs_tres.html")

abs_tres <- abs_lintemp

summary(abs_lintemp)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled *
##   habitat + age_scaled + meanmaxtempI_scaled * juliandate_scaled +
##   year_fct + (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
##   mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate,
##     brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##     meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
##
## REML criterion at convergence: 299.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.19919 -0.44841  0.07252  0.45926  2.13480
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.7929   0.8904
## Residual              2.9315   1.7121
## Number of obs: 80, groups: attempt_id, 38
##
## Fixed effects:
##                                     Estimate Std. Error    df t value
## (Intercept)                       5.14205    1.79977 27.02612   2.857
## meanmaxtempI_scaled                -0.01625    1.21203 19.61723  -0.013
## habitatOrchard                    -1.89050    2.02542 25.12134  -0.933
## habitatGrassland                  -0.19533    1.74994 27.21020  -0.112
## habitatRow crop                     0.28297    1.85504 27.65110   0.153
## meanmintempI_scaled                 0.47867    1.20222 26.60680   0.398
## age_scaled                         0.44394    0.39511 28.97532   1.124
## juliandate_scaled                 -0.09183    0.33727 18.35065  -0.272
## year_fct2023                      0.84558    0.67917 18.74338   1.245
## meanmaxtempI_scaled:habitatOrchard -3.78807    1.99799 18.26731  -1.896
## meanmaxtempI_scaled:habitatGrassland 1.96710    1.52975 18.89012   1.286
## meanmaxtempI_scaled:habitatRow crop  0.67682    1.49758 18.33562   0.452
## habitatOrchard:meanmintempI_scaled -3.00167    1.65631 24.25576  -1.812
## habitatGrassland:meanmintempI_scaled -0.85229    1.23591 26.93705  -0.690
## habitatRow crop:meanmintempI_scaled -0.37606    1.51061 24.60833  -0.249
## meanmaxtempI_scaled:juliandate_scaled -0.53143    0.45066 22.32272  -1.179
##                                     Pr(>|t|)
```

```

## (Intercept)                                0.00813 **
## meanmaxtempI_scaled                        0.98944
## habitatOrchard                            0.35950
## habitatGrassland                          0.91194
## habitatRow crop                           0.87987
## meanmintempI_scaled                       0.69369
## age_scaled                               0.27042
## juliandate_scaled                         0.78844
## year_fct2023                             0.22847
## meanmaxtempI_scaled:habitatOrchard        0.07390 .
## meanmaxtempI_scaled:habitatGrassland      0.21402
## meanmaxtempI_scaled:habitatRow crop      0.65661
## habitatOrchard:meanmintempI_scaled        0.08234 .
## habitatGrassland:meanmintempI_scaled      0.49634
## habitatRow crop:meanmintempI_scaled       0.80547
## meanmaxtempI_scaled:juliandate_scaled     0.25072
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 16 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it
anova(abs_lintemp,prior_model)

## refitting model(s) with ML (instead of REML)

## Data: dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>% ...
## Models:
## prior_model: sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat +
## abs_lintemp: sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat +
##           npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model    17 345.00 385.50 -155.50   311.00
## abs_lintemp    18 344.87 387.74 -154.43   308.87 2.1336 1      0.1441
summary(abs_lintemp_addmin)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled +
##   age_scaled + meanmaxtempI_scaled * juliandate_scaled + year_fct +
##   (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
##   mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate,
##     brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##     meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
##
## REML criterion at convergence: 310.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.09654 -0.48543  0.08953  0.46443  2.22980
##

```

```

## Random effects:
##   Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.8858  0.9412
## Residual      2.9298  1.7117
## Number of obs: 80, groups: attempt_id, 38
##
## Fixed effects:
##                                     Estimate Std. Error    df t value
## (Intercept)                      5.99460    1.57700 22.35444   3.801
## meanmaxtempI_scaled                0.35195    1.11446 21.41316   0.316
## habitatOrchard                   -1.71160    1.80530 22.95784  -0.948
## habitatGrassland                 -0.91998    1.55794 22.42793  -0.591
## habitatRow crop                    0.02921    1.53576 20.72112   0.019
## meanmintempI_scaled              -0.43406    0.43055 22.04115  -1.008
## age_scaled                       0.48376    0.39069 31.12896   1.238
## juliandate_scaled                -0.18439    0.33876 20.74336  -0.544
## year_fct2023                      0.50831    0.62180 19.32949   0.817
## meanmaxtempI_scaled:habitatOrchard -2.32710    1.68775 21.64473  -1.379
## meanmaxtempI_scaled:habitatGrassland 1.82192    1.32864 19.51747   1.371
## meanmaxtempI_scaled:habitatRow crop  0.69536    1.39341 19.87978   0.499
## meanmaxtempI_scaled:juliandate_scaled -0.65039    0.44274 25.97160  -1.469
##                                     Pr(>|t|)
## (Intercept)                      0.000957 ***
## meanmaxtempI_scaled                0.755212
## habitatOrchard                    0.352954
## habitatGrassland                  0.560754
## habitatRow crop                    0.985006
## meanmintempI_scaled                0.324317
## age_scaled                        0.224894
## juliandate_scaled                  0.592024
## year_fct2023                      0.423612
## meanmaxtempI_scaled:habitatOrchard  0.182036
## meanmaxtempI_scaled:habitatGrassland 0.185853
## meanmaxtempI_scaled:habitatRow crop  0.623231
## meanmaxtempI_scaled:juliandate_scaled 0.153834
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)      if you need it
anova(abs_lintemp_addmin,prior_model_addmin)

## refitting model(s) with ML (instead of REML)

## Data: dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>% ...
## Models:
## prior_model_addmin: sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + ag
## abs_lintemp_addmin: sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + ag
##
##      npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model_addmin    14 346.03 379.38 -159.01   318.03
## abs_lintemp_addmin    15 344.98 380.71 -157.49   314.98 3.0471  1    0.08088 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

This is the first instance where including the interaction changes which model is the most parsimonious. Here, the interaction with max temp seems obviously more important vs before, the linear temp model was the chosen model. However, the interaction is significant in neither model, and improves fit neither for the model with both min and max interaction with land cover nor for just max interaction with land cover.

```
(tresabstrendmax <- emtrends(prior_model, specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test() %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend", Df = "df", `T-ratio` = "T-ratio",
    gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.231	1.209	22	0.191	0.85
Orchard	-3.633	1.664	17	-2.184	0.043
Grassland	1.445	0.821	41	1.760	0.086
Row crop	0.170	0.779	20	0.218	0.829

```
(tresabstrendmax <- emtrends(prior_model_admin, specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test() %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend", Df = "df", `T-ratio` = "T-ratio",
    gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.746	1.114	25	0.670	0.509
Orchard	-1.544	1.359	23	-1.136	0.268
Grassland	1.645	0.748	44	2.200	0.033
Row crop	0.329	0.680	29	0.484	0.632

```
(tresabstrendmax <- emtrends(abs_lintemp_admin, specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test() %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend", Df = "df", `T-ratio` = "T-ratio",
    gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.157	1.165	21	0.135	0.894
Orchard	-2.170	1.405	21	-1.545	0.137
Grassland	1.979	0.768	32	2.577	0.015
Row crop	0.853	0.756	24	1.127	0.271

There are differences between the final outcome of these two analyses, but those are mostly because of the different structure of the most supported model. The direction of the estimated trends are the same.

```

samp <- abs_lintemp@frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_tres_abs.html")

dat_text_tres <- data.frame(
  label = paste("N =", samp$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

summary(abs_lintemp)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## sqrt(abs_change_cort) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled *
##   habitat + age_scaled + meanmaxtempI_scaled * juliandate_scaled +
##   year_fct + (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
##   mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate,
##     brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##     meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
##
## REML criterion at convergence: 299.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.19919 -0.44841  0.07252  0.45926  2.13480
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.7929  0.8904
## Residual              2.9315  1.7121
## Number of obs: 80, groups: attempt_id, 38
##
## Fixed effects:
##                                     Estimate Std. Error    df t value
## (Intercept)                        5.14205    1.79977 27.02612    2.857
## meanmaxtempI_scaled                 -0.01625    1.21203 19.61723   -0.013
## habitatOrchard                     -1.89050    2.02542 25.12134   -0.933
## habitatGrassland                   -0.19533    1.74994 27.21020   -0.112
## habitatRow crop                     0.28297    1.85504 27.65110    0.153
## meanmintempI_scaled                 0.47867    1.20222 26.60680    0.398
## age_scaled                         0.44394    0.39511 28.97532    1.124
## juliandate_scaled                  -0.09183    0.33727 18.35065   -0.272
## year_fct2023                       0.84558    0.67917 18.74338    1.245
## meanmaxtempI_scaled:habitatOrchard -3.78807    1.99799 18.26731   -1.896
## meanmaxtempI_scaled:habitatGrassland 1.96710    1.52975 18.89012    1.286
## meanmaxtempI_scaled:habitatRow crop  0.67682    1.49758 18.33562    0.452
## habitatOrchard:meanmintempI_scaled -3.00167    1.65631 24.25576   -1.812
## habitatGrassland:meanmintempI_scaled -0.85229    1.23591 26.93705   -0.690
## habitatRow crop:meanmintempI_scaled -0.37606    1.51061 24.60833   -0.249

```

```
## meanmaxtempI_scaled:juliandate_scaled -0.53143    0.45066 22.32272 -1.179
##                                     Pr(>|t|)
## (Intercept)                        0.00813 **
## meanmaxtempI_scaled                0.98944
## habitatOrchard                    0.35950
## habitatGrassland                  0.91194
## habitatRow crop                    0.87987
## meanmintempI_scaled               0.69369
## age_scaled                        0.27042
## juliandate_scaled                 0.78844
## year_fct2023                      0.22847
## meanmaxtempI_scaled:habitatOrchard 0.07390 .
## meanmaxtempI_scaled:habitatGrassland 0.21402
## meanmaxtempI_scaled:habitatRow crop 0.65661
## habitatOrchard:meanmintempI_scaled 0.08234 .
## habitatGrassland:meanmintempI_scaled 0.49634
## habitatRow crop:meanmintempI_scaled 0.80547
## meanmaxtempI_scaled:juliandate_scaled 0.25072
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 16 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it
check_collinearity(abs_lintemp_noint)

## # Check for Multicollinearity
##
## Low Correlation
##
##               Term  VIF    VIF 95% CI Increased SE Tolerance
## meanmaxtempI_scaled 3.14 [2.35, 4.38]          1.77      0.32
## meanmintempI_scaled 1.88 [1.49, 2.57]          1.37      0.53
## habitat              2.05 [1.61, 2.81]          1.43      0.49
## age_scaled           1.28 [1.10, 1.79]          1.13      0.78
## juliandate_scaled    2.02 [1.58, 2.77]          1.42      0.50
## year_fct             1.49 [1.23, 2.03]          1.22      0.67
## meanmaxtempI_scaled:juliandate_scaled 1.32 [1.13, 1.83]          1.15      0.76
## Tolerance 95% CI
## [0.23, 0.42]
## [0.39, 0.67]
## [0.36, 0.62]
## [0.56, 0.91]
## [0.36, 0.63]
## [0.49, 0.81]
## [0.55, 0.89]

data_tres = dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
```

```

mean_temp_tres <- mean(data_tres %>% pull(meanmaxtempI))
sd_temp_tres <- sd(data_tres %>% pull(meanmaxtempI))

temp_trans_tres <- trans_new("temp_trans_tres",
                             transform = function(x){(x * sd_temp_tres) + mean_temp_tres},
                             inverse = function(x){x})

(fig4_tres <- ggpredict(prior_model, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = "none",
                       plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
                       aes(linetype = .data[["group"]]) +
                       theme_classic() +
                       facet_wrap(~ group, ncol = 2) +
                       xlab("Mean daily max temp over preceding week (°C)") +
                       ylab("Stress-induced - baseline corticosterone (ng/mL)") +
                       scale_fill_viridis(discrete = TRUE) +
                       scale_color_viridis(discrete = TRUE) +
                       theme(text = element_text(size = 16)) +
                       labs(title = "TRES cort; max temp * habitat interaction") +
                       scale_x_continuous(trans = temp_trans_tres,
                                           breaks = c((20-mean_temp_tres)/sd_temp_tres,
                                                         (25-mean_temp_tres)/sd_temp_tres,
                                                         (30-mean_temp_tres)/sd_temp_tres,
                                                         (35-mean_temp_tres)/sd_temp_tres,
                                                         (40-mean_temp_tres)/sd_temp_tres),
                                           # breaks = c(20,30,40,50),
                                           # labels = c("20", "30", "40", "50"),
                                           # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
                                           #              (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
                                           # limits = c(18,5)
                       ) +
                       scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "solid", "Grassland" = "dashed", "Row" = "solid"),
                                           #ylim(-5,5) +
                       geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1) +
                       theme(legend.position = "none")
)

```

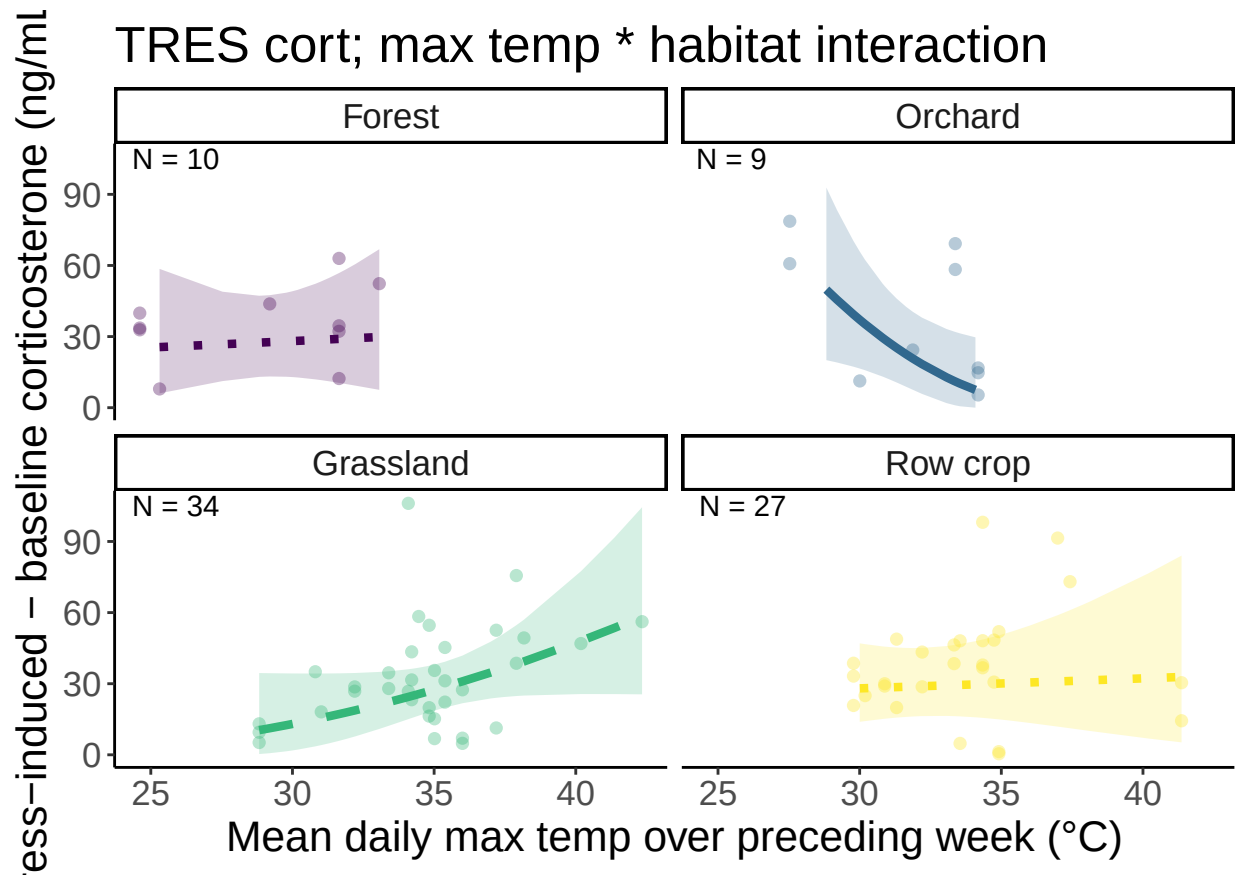
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
(fig4_tres <- ggpredict(abs_lintemp_admin, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_corrected = TRUE,
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Stress-induced - baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "TRES cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_tres,
    breaks = c((20-mean_temp_tres)/sd_temp_tres,
               (25-mean_temp_tres)/sd_temp_tres,
               (30-mean_temp_tres)/sd_temp_tres,
               (35-mean_temp_tres)/sd_temp_tres,
               (40-mean_temp_tres)/sd_temp_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
    #            (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "solid", "Row crop" = "solid")) +
  #ylim(-5,5) +
  geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1.5))
```

```

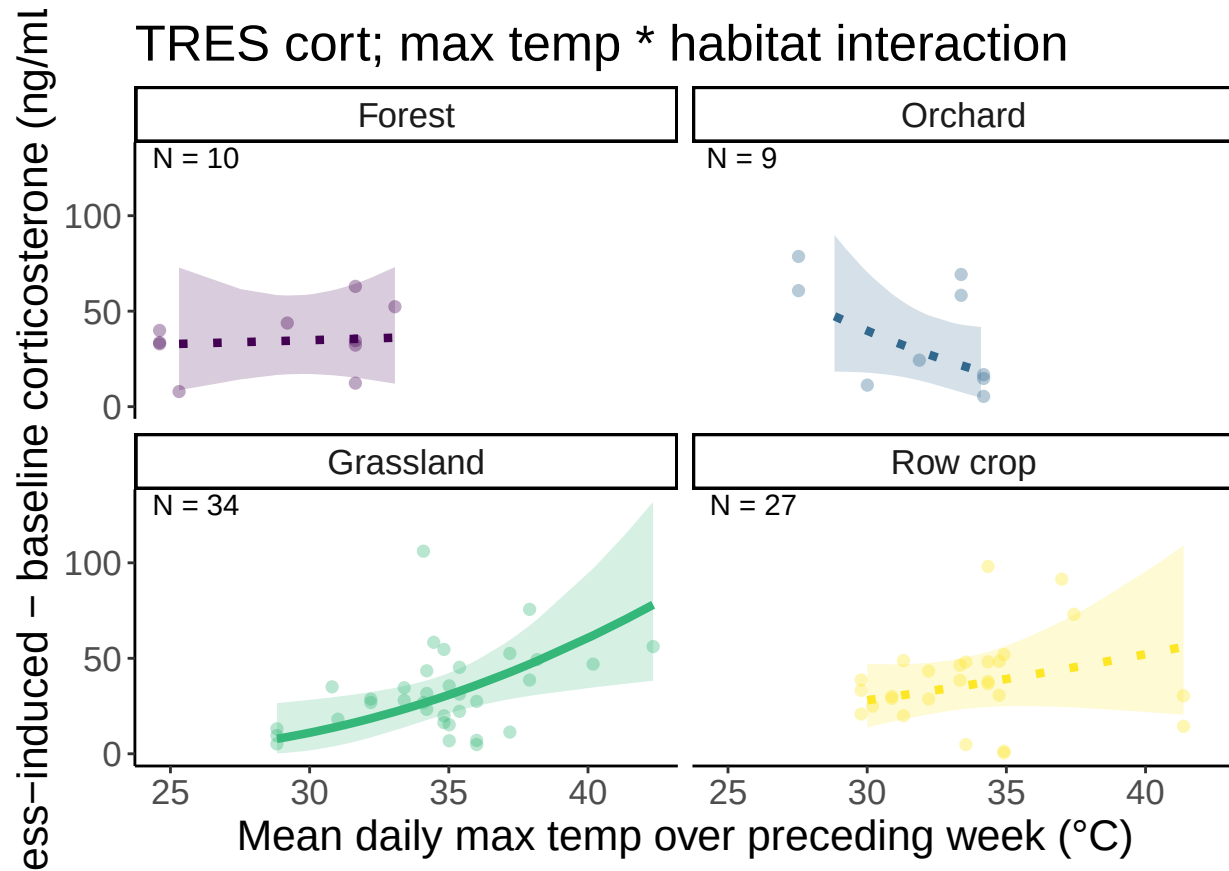
theme(legend.position = "none")
)

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



Honestly the results here seem easier to explain.

```

(fig4_tres <- ggpredict(abs_lintemp_addmin, terms = c("meanmaxtempI_scaled [all]", "juliandate_scaled"), b
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Stress-induced - baseline corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "TRES cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_tres,
    breaks = c((20-mean_temp_tres)/sd_temp_tres,

```

```

      (25-mean_temp_tres)/sd_temp_tres,
      (30-mean_temp_tres)/sd_temp_tres,
      (35-mean_temp_tres)/sd_temp_tres,
      (40-mean_temp_tres)/sd_temp_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) # +
  # scale_linetype_manual(values = c("Forest" = "dotted","Orchard" = "dotted","Grassland" = "solid","
  #ylim(-5,5) +
  # geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust
  # theme(legend.position = "none")
)

```

```

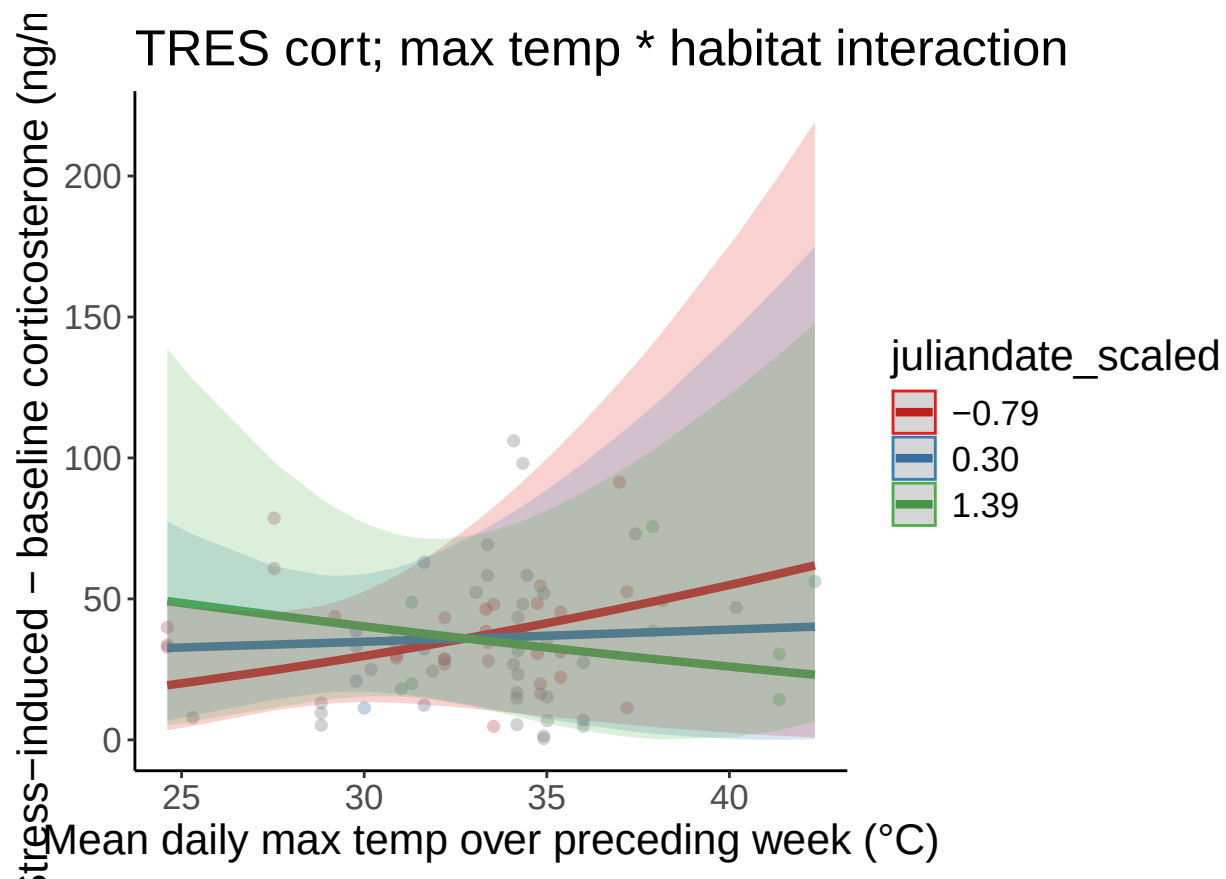
## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Interaction is not significant.

s2 cort by prior week temp

```

prior_model <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habi
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s2_lintemp <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habi
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s2_lintemp_addmax <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s2_lintemp_addmin <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s2_lintemp_noint <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled + meanmintempI_scaled + habitat
  mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood_size,
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

c1 <- anova(s2_lintemp,s2_lintemp_addmin,s2_lintemp_noint) %>% tibble() %>% mutate(Model = c("no intera

```

WEBL

```

## refitting model(s) with ML (instead of REML)
c2 <- anova(s2_lintemp,s2_lintemp_addmax,s2_lintemp_noint) %>% tibble() %>% mutate(Model = c("no intera

## refitting model(s) with ML (instead of REML)
(int_tab_s2_webl <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
    P = `Pr(>Chisq)` %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	592.84	NA	NA

single interaction	597.97	0.87	0.833
both interacting	593.90	10.06	0.018
Min temp			
no interaction	592.84	NA	NA
single interaction	592.99	5.84	0.12
both interacting	593.90	5.09	0.165

```
# gtsave(int_tab_s2_webl, "figures/int_tab_s2_webl.html")

s2_webl <- s2_lintemp

summary(s2_lintemp)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled *
##      habitat + age_scaled + meanmaxtempI_scaled * juliandate_scaled +
##      year_fct + (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
##      mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate,
##      brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##      meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
##
## REML criterion at convergence: 563.7
##
## Scaled residuals:
##      Min      1Q   Median      3Q      Max
## -2.64592 -0.58267 -0.02042  0.51184  2.70175
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 1.594      1.262
## Residual              1.267      1.125
## Number of obs: 162, groups: attempt_id, 53
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)    4.220563   0.483795  32.693862   8.724
## meanmaxtempI_scaled    0.007393   0.520432  88.164464   0.014
## habitatOrchard      0.409246   0.585320  34.688866   0.699
## habitatGrassland    1.380649   0.740916  46.868992   1.863
## habitatRow crop     0.764003   0.628623  31.519275   1.215
## meanmintempI_scaled    0.745083   0.553575  69.603719   1.346
## age_scaled        0.062971   0.127024 144.907848   0.496
## juliandate_scaled   -0.673819   0.297905  44.999509  -2.262
## year_fct2023        0.179402   0.486398  37.022495   0.369
## meanmaxtempI_scaled:habitatOrchard    0.144410   0.666448  92.906813   0.217
## meanmaxtempI_scaled:habitatGrassland -1.067611   0.671934 110.394335  -1.589
## meanmaxtempI_scaled:habitatRow crop  -0.317599   0.695187  78.614991  -0.457
## habitatOrchard:meanmintempI_scaled   -0.708787   0.687253  64.737712  -1.031
```

```
## habitatGrassland:meanmintempI_scaled    1.803171    0.907735 109.273257    1.986
## habitatRow crop:meanmintempI_scaled     -0.422513    0.688256  55.364439   -0.614
## meanmaxtempI_scaled:juliandate_scaled   0.170025    0.174477  97.343425    0.974
##                                          Pr(>|t|)
## (Intercept)                            4.76e-10 ***
## meanmaxtempI_scaled                    0.9887
## habitatOrchard                        0.4891
## habitatGrassland                      0.0687 .
## habitatRow crop                       0.2333
## meanmintempI_scaled                   0.1827
## age_scaled                           0.6208
## juliandate_scaled                     0.0286 *
## year_fct2023                         0.7143
## meanmaxtempI_scaled:habitatOrchard     0.8289
## meanmaxtempI_scaled:habitatGrassland   0.1150
## meanmaxtempI_scaled:habitatRow crop    0.6490
## habitatOrchard:meanmintempI_scaled     0.3062
## habitatGrassland:meanmintempI_scaled   0.0495 *
## habitatRow crop:meanmintempI_scaled    0.5418
## meanmaxtempI_scaled:juliandate_scaled  0.3322
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 16 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)         if you need it
```

```
anova(s2_lintemp,prior_model)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>% ...
## Models:
## prior_model: sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat + age_scaled
## s2_lintemp: sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat + age_scaled
##               npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model    17 593.03 645.52 -279.52   559.03
## s2_lintemp     18 593.90 649.48 -278.95   557.90 1.1301  1    0.2878
```

Same model, interaction is not significant, does not add to fit

```
(webbs2trendmax <- emtrends(prior_model, specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test() %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend", Df = "df", `T-ratio` = "p.value",
         gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.002	0.530	97	0.003	0.997
Orchard	0.099	0.457	129	0.217	0.829
Grassland	-1.125	0.436	146	-2.582	0.011
Row crop	-0.215	0.450	86	-0.479	0.633

```
(webs2trendmax <- emtrends(s2_lintemp, specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test() %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend", Df = "df", `T-ratio` = "T-ratio",
    gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.018	0.530	97	0.034	0.973
Orchard	0.163	0.462	124	0.352	0.725
Grassland	-1.049	0.443	146	-2.369	0.019
Row crop	-0.299	0.459	80	-0.653	0.516

Same direction and significance of trends

```
samp_s2_webl <- s2_lintemp@frame %>% group_by(habitat) %>% summarize(count = n())
# samp_s2_webl %>% gt() %>% gtsave("figures/ss_webl_s2.html")
```

```
dat_text_s2_webl <- data.frame(
  label = paste("N =", samp_s2_webl$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)
```

```
data_s2_webl = dplyr::filter(g, Species == "WEBL", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
```

```
mean_temp_s2_webl <- mean(data_s2_webl %>% pull(meanmaxtempI))
sd_temp_s2_webl <- sd(data_s2_webl %>% pull(meanmaxtempI))
```

```
temp_trans_s2_webl <- trans_new("temp_trans_s2_webl",
  transform = function(x){(x * sd_temp_s2_webl) + mean_temp_s2_webl},
  inverse = function(x){x})
```

```
(fig_webl_s2 <- ggpredict(prior_model, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = FALSE) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
```

```

theme(text = element_text(size = 16)) +
labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_s2_webl,
  breaks = c((20-mean_temp_s2_webl)/sd_temp_s2_webl,
    (25-mean_temp_s2_webl)/sd_temp_s2_webl,
    (30-mean_temp_s2_webl)/sd_temp_s2_webl,
    (35-mean_temp_s2_webl)/sd_temp_s2_webl,
    (40-mean_temp_s2_webl)/sd_temp_s2_webl),
  # breaks = c(20,30,40,50),
  # labels = c("20", "30", "40", "50"),
  # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # limits = c(18,5)
) +
scale_linetype_manual(values = c("dotted","dotted","solid","dotted")) +
#ylim(-5,5) +
geom_text(data = dat_text_s2_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust
theme(legend.position = "none")
)

```

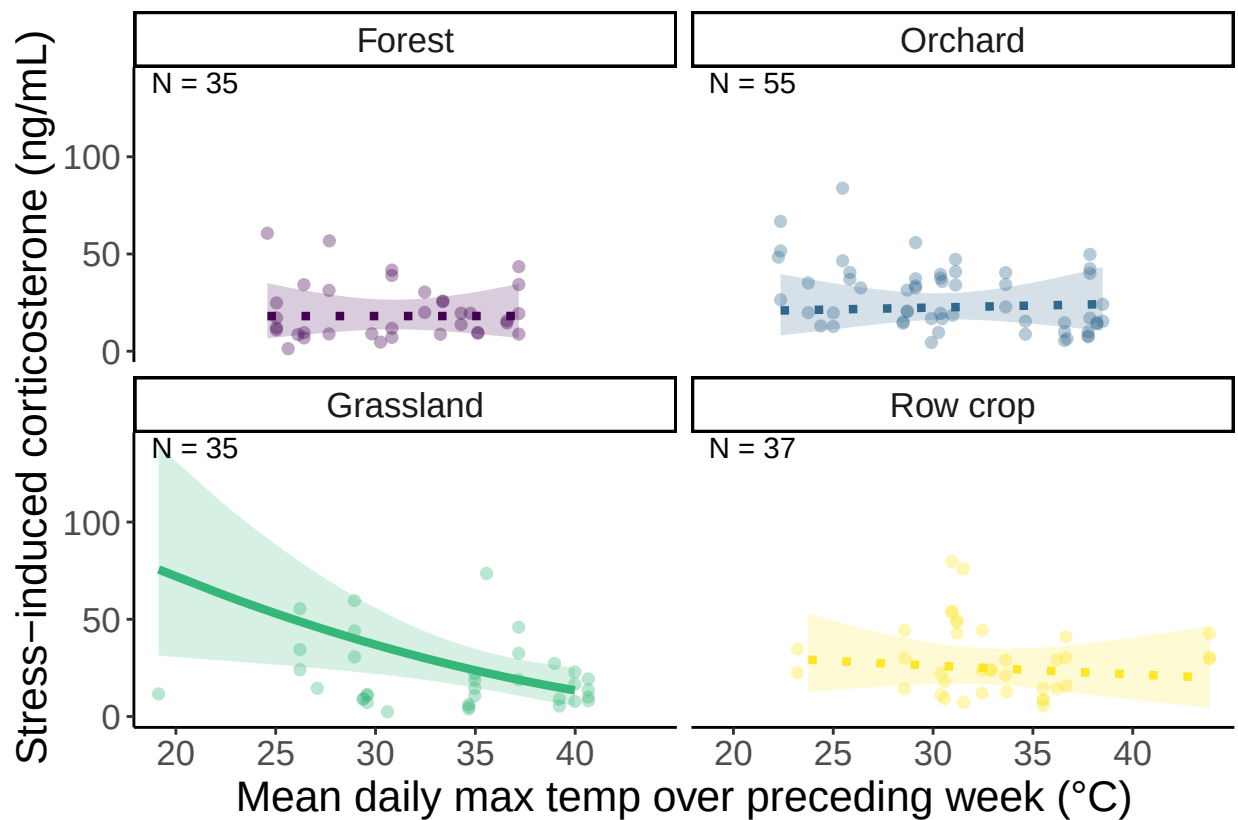
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
(fig_webl_s2 <- ggpredict(s2_lintemp, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = TRUE) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_webl,
    breaks = c((20 - mean_temp_s2_webl) / sd_temp_s2_webl,
               (25 - mean_temp_s2_webl) / sd_temp_s2_webl,
               (30 - mean_temp_s2_webl) / sd_temp_s2_webl,
               (35 - mean_temp_s2_webl) / sd_temp_s2_webl,
               (40 - mean_temp_s2_webl) / sd_temp_s2_webl),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE),
    #            (55 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18, 5)
  ) +
  scale_linetype_manual(values = c("dotted", "dotted", "solid", "dotted")) +
  #ylim(-5, 5) +
  geom_text(data = dat_text_s2_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1.5)
```

```

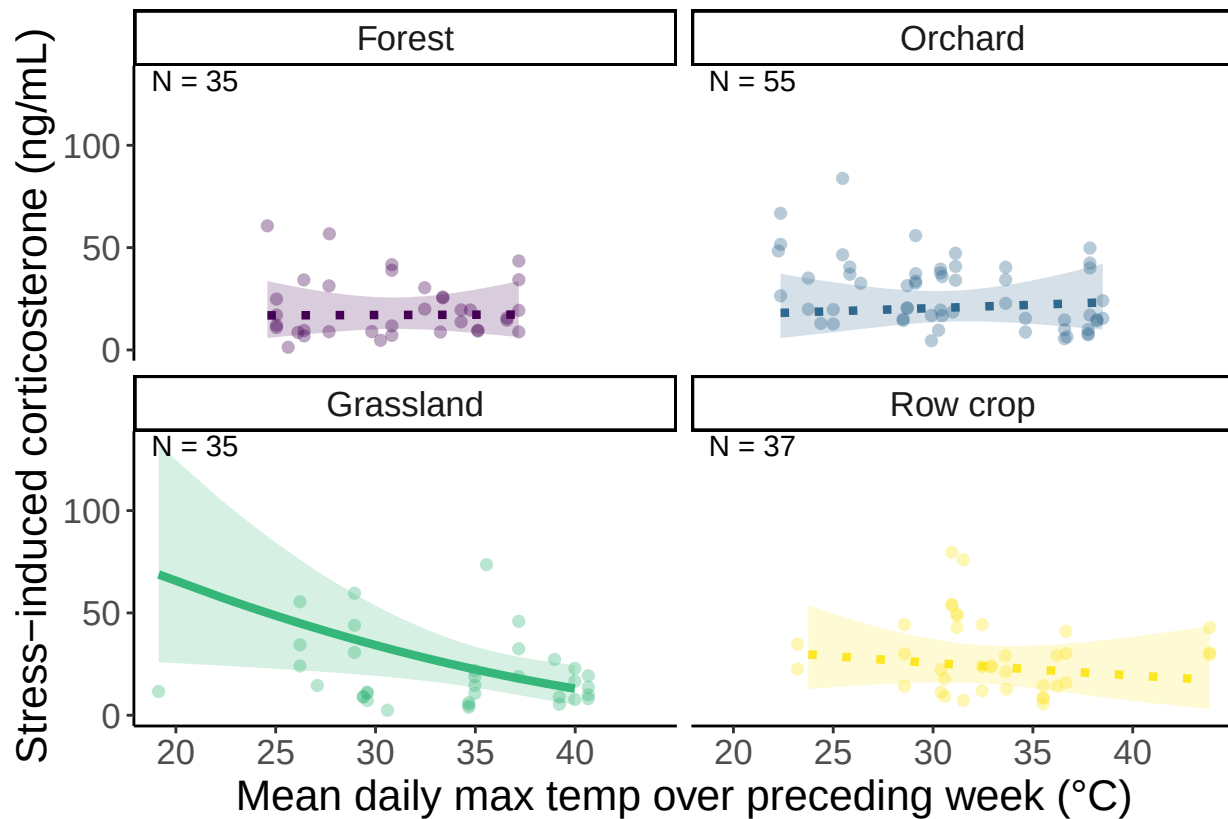
theme(legend.position = "none")
)

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(fig_webl_s2 <- ggpredict(s2_lintemp, terms = c("meanmaxtempI_scaled [all]", "juliandate_scaled"), bias_correction = "none",
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_webl,
    breaks = c((20 - mean_temp_s2_webl) / sd_temp_s2_webl,
      (25 - mean_temp_s2_webl) / sd_temp_s2_webl,

```

```

      (30-mean_temp_s2_webl)/sd_temp_s2_webl,
      (35-mean_temp_s2_webl)/sd_temp_s2_webl,
      (40-mean_temp_s2_webl)/sd_temp_s2_webl),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #             (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) # +
  # scale_linetype_manual(values = c("dotted","dotted","solid","dotted")) +
  # ylim(-5,5) +
  # geom_text(data = dat_text_s2_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vj
  # theme(legend.position = "none")
)

```

```

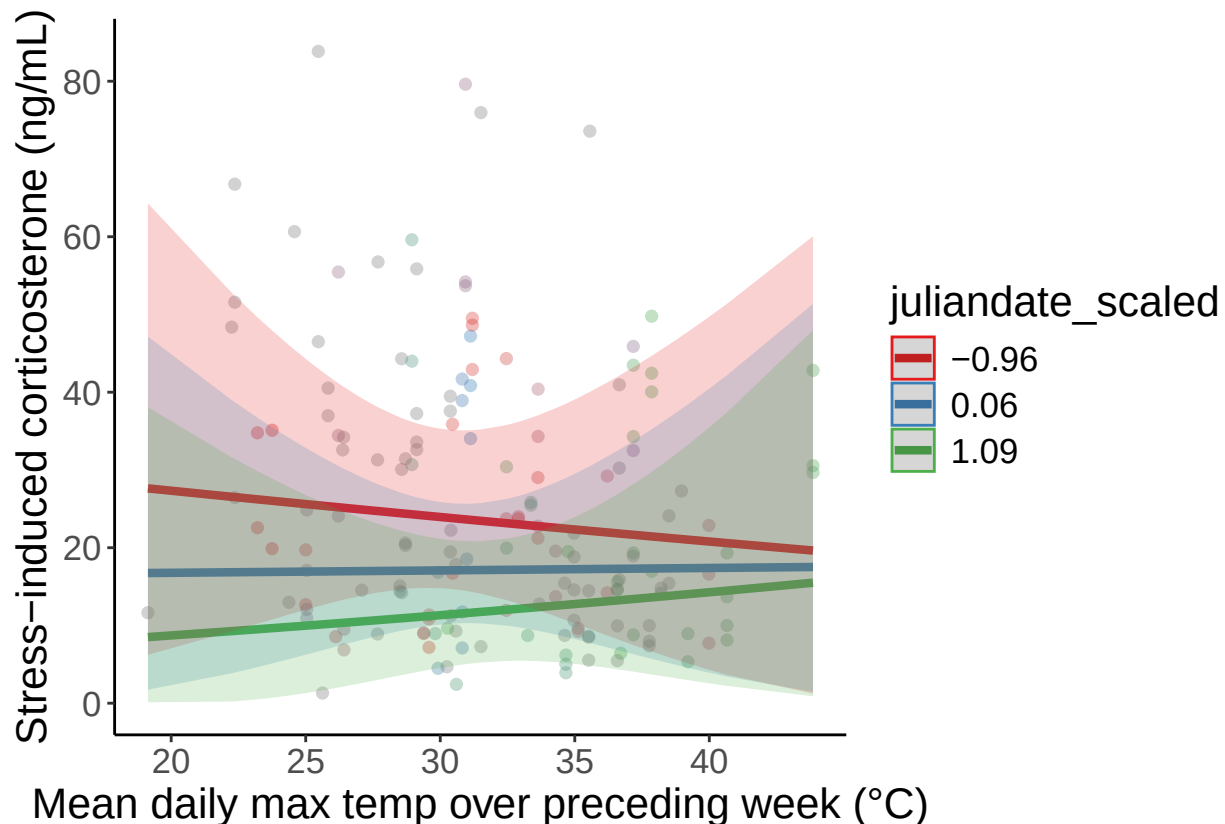
## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



No interaction

```

prior_model <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + age,
                               mutate(across(c(gweight,meanmaxtempI,meanmintempI,juliandate,brood

```

```

      ~ scale(.x)[,1],
      .names = "{.col}_scaled"),
      meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled

s2_lintemp <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat,
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

s2_lintemp_addmax <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled + meanmintempI_scaled * habitat,
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

s2_lintemp_addmin <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled * habitat,
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

s2_lintemp_noint <- lmerTest::lmer(sqrt(cort_s2) ~ meanmaxtempI_scaled + meanmintempI_scaled + habitat,
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

c1 <- anova(s2_lintemp, s2_lintemp_addmin, s2_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction", "single interaction", "both interacting"))

```

TRES

refitting model(s) with ML (instead of REML)

```
c2 <- anova(s2_lintemp, s2_lintemp_addmax, s2_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction", "single interaction", "both interacting"))
```

refitting model(s) with ML (instead of REML)

```

(int_tab_s2_tres <- bind_rows(c1, c2) %>%
  as_tibble() %>%
  mutate(across(where(is.numeric), ~round(.x, digits = 3)),
    P = `Pr(>Chisq)` %>%
    dplyr::select(Model, AIC, Chisq, P) %>%
  mutate(max_or_min = c(rep("Max temp", times = 3), rep("Min temp", times = 3)),
    across(c(AIC, Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001, "<0.001", as.character(P)) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	351.88	NA	NA
single interaction	347.64	10.24	0.017
both interacting	349.88	3.76	0.289

Min temp

no interaction	351.88	NA	NA
single interaction	356.22	1.66	0.646
both interacting	349.88	12.34	0.006

```
# gtsave(int_tab_s2_tres,"figures/int_tab_s2_tres.html")

s2_tres <- s2_lintemp_admin

summary(s2_lintemp_admin)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled +
##      age_scaled + meanmaxtempI_scaled * juliandate_scaled + year_fct +
##      (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
##      mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate,
##      brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##      meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)
##
## REML criterion at convergence: 318.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.98523 -0.52482  0.05419  0.48390  2.78941
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.4418   0.6646
## Residual              2.4431   1.5630
## Number of obs: 86, groups: attempt_id, 38
##
## Fixed effects:
##
##              Estimate Std. Error    df t value
## (Intercept)      5.78734    1.10137 24.86882    5.255
## meanmaxtempI_scaled -0.13856    0.79749 25.14618   -0.174
## habitatOrchard     -0.67812    1.35152 23.36877   -0.502
## habitatGrassland   -0.40767    1.11531 23.25685   -0.366
## habitatRow crop      0.68003    1.10839 21.39809    0.614
## meanmintempI_scaled -0.28743    0.34734 19.64917   -0.828
## age_scaled         0.58570    0.29573 39.72161    1.981
## juliandate_scaled  -0.07137    0.26435 20.42010   -0.270
## year_fct2023        0.39649    0.50323 17.74065    0.788
## meanmaxtempI_scaled:habitatOrchard -1.54121    1.34328 23.59248   -1.147
## meanmaxtempI_scaled:habitatGrassland  1.96599    1.02449 21.47017    1.919
## meanmaxtempI_scaled:habitatRow crop  0.90468    1.08505 22.09866    0.834
## meanmaxtempI_scaled:juliandate_scaled -0.47650    0.36444 24.82763   -1.307
##
##              Pr(>|t|)
## (Intercept)    1.96e-05 ***
## meanmaxtempI_scaled    0.8635
```

```
## habitatOrchard                0.6205
## habitatGrassland              0.7180
## habitatRow crop               0.5460
## meanmintempI_scaled           0.4179
## age_scaled                    0.0546 .
## juliandate_scaled             0.7899
## year_fct2023                  0.4412
## meanmaxtempI_scaled:habitatOrchard 0.2627
## meanmaxtempI_scaled:habitatGrassland 0.0684 .
## meanmaxtempI_scaled:habitatRow crop 0.4133
## meanmaxtempI_scaled:juliandate_scaled 0.2030
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it
anova(s2_lintemp_addmin,prior_model)

## refitting model(s) with ML (instead of REML)
## Data: dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>% ...
## Models:
## prior_model: sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + age_scaled + juli
## s2_lintemp_addmin: sqrt(cort_s2) ~ meanmaxtempI_scaled * habitat + meanmintempI_scaled + age_scaled +
##               npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model      14 348.00 382.37 -160.00   320.00
## s2_lintemp_addmin 15 347.64 384.46 -158.82   317.64 2.3636 1      0.1242

Interaction is not significant, does not increase fit, does not change most parsimonious model

(tress2trendmax <- emtrends(prior_model,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test() %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "
  gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.146	0.787	30	0.185	0.854
Orchard	-1.377	1.125	25	-1.223	0.233
Grassland	1.462	0.627	46	2.333	0.024
Row crop	0.239	0.563	31	0.425	0.674

```
(tress2trendmax <- emtrends(s2_lintemp_addmin,specs = ~ habitat, var = c("meanmaxtempI_scaled")) %>% test() %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "meanmaxtempI_scaled.trend",Df = "df", `T-ratio` = "
  gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.271	0.845	26	-0.321	0.751
Orchard	-1.812	1.165	23	-1.556	0.133
Grassland	1.695	0.643	34	2.636	0.013
Row crop	0.634	0.632	26	1.002	0.326

Same direction & significance

```
samp_s2_tres <- s2_lintemp_addmin@frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_tres_s2.html")

dat_text_s2_tres <- data.frame(
  label = paste("N =", samp_s2_tres$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_s2_tres = dplyr::filter(g, Species == "TRES", !is.na(meanmaxtempI), !is.na(meanmintempI)) %>%
  mutate(across(c(gweight, meanmaxtempI, meanmintempI, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    meanmaxtempI_scaled_sq = meanmaxtempI_scaled * meanmaxtempI_scaled)

mean_temp_s2_tres <- mean(data_s2_tres %>% pull(meanmaxtempI))
sd_temp_s2_tres <- sd(data_s2_tres %>% pull(meanmaxtempI))

temp_trans_s2_tres <- trans_new("temp_trans_s2_tres",
  transform = function(x){(x * sd_temp_s2_tres) + mean_temp_s2_tres},
  inverse = function(x){x})

(fig_tres_s2 <- ggpredict(prior_model, terms = c("meanmaxtempI_scaled [all]", "habitat"), bias_correction = FALSE) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (\u00b0C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_tres,
    breaks = c((20-mean_temp_s2_tres)/sd_temp_s2_tres,
      (25-mean_temp_s2_tres)/sd_temp_s2_tres,
      (30-mean_temp_s2_tres)/sd_temp_s2_tres,
      (35-mean_temp_s2_tres)/sd_temp_s2_tres,
      (40-mean_temp_s2_tres)/sd_temp_s2_tres),
    # breaks = c(20, 30, 40, 50),
```

```

# labels = c("20", "30", "40", "50"),
# limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
#           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
# limits = c(18,5)

) +
scale_linetype_manual(values = c("dotted","dotted","solid","dotted")) +
#ylim(-5,5) +
geom_text(data = dat_text_s2_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust
theme(legend.position = "none")
)

```

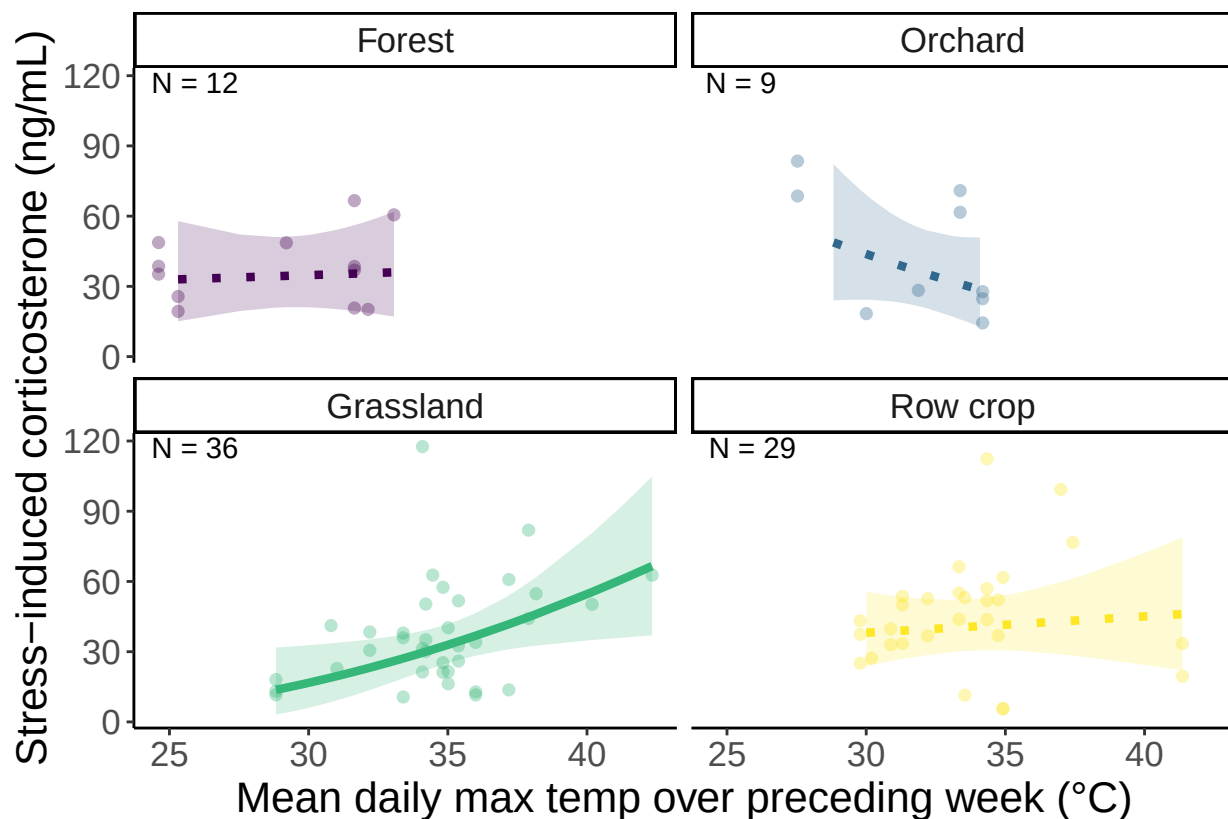
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(fig_tres_s2 <- ggpredict(s2_lintemp_addmin,terms = c("meanmaxtempI_scaled [all]","habitat"),bias_corre
plot(line_size = 1.5,alpha = .2,show_data = TRUE,limit_range = TRUE) +
aes(linetype = .data[["group"]]) +
theme_classic() +
facet_wrap(~ group, ncol = 2) +
xlab("Mean daily max temp over preceding week (°C)") +

```

```

ylab("Stress-induced corticosterone (ng/mL)") +
scale_fill_viridis(discrete = TRUE) +
scale_color_viridis(discrete = TRUE) +
theme(text = element_text(size = 16)) +
labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_s2_tres,
  breaks = c((20-mean_temp_s2_tres)/sd_temp_s2_tres,
    (25-mean_temp_s2_tres)/sd_temp_s2_tres,
    (30-mean_temp_s2_tres)/sd_temp_s2_tres,
    (35-mean_temp_s2_tres)/sd_temp_s2_tres,
    (40-mean_temp_s2_tres)/sd_temp_s2_tres),
  # breaks = c(20,30,40,50),
  # labels = c("20", "30", "40", "50"),
  # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # limits = c(18,5)
) +
scale_linetype_manual(values = c("dotted","dotted","solid","dotted")) +
#ylim(-5,5) +
geom_text(data = dat_text_s2_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust
theme(legend.position = "none")
)

```

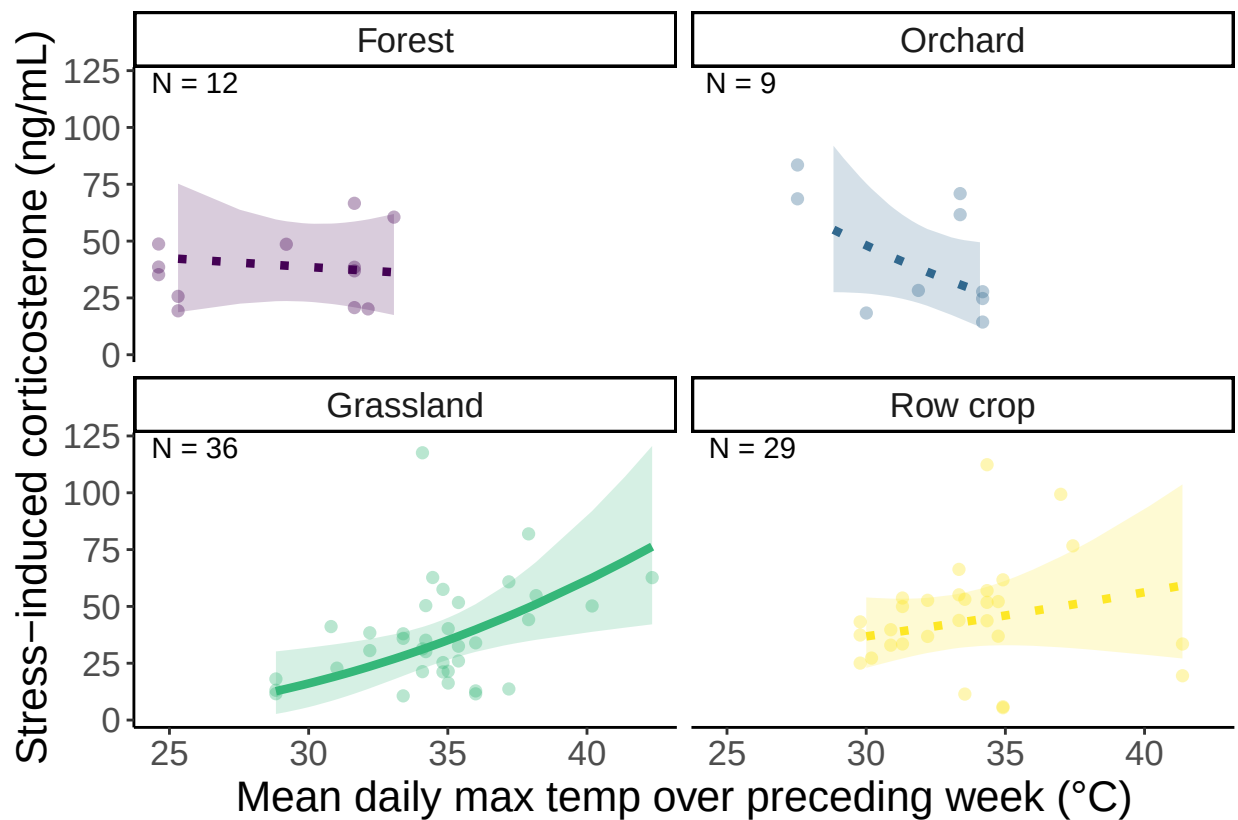
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
(fig_tres_s2 <- ggpredict(s2_lintemp_admin, terms = c("meanmaxtempI_scaled [all]", "juliandate_scaled"),
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean daily max temp over preceding week (°C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_tres,
    breaks = c((20-mean_temp_s2_tres)/sd_temp_s2_tres,
      (25-mean_temp_s2_tres)/sd_temp_s2_tres,
      (30-mean_temp_s2_tres)/sd_temp_s2_tres,
      (35-mean_temp_s2_tres)/sd_temp_s2_tres,
      (40-mean_temp_s2_tres)/sd_temp_s2_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18,5)
  ) # +
  # scale_linetype_manual(values = c("dotted", "dotted", "solid", "dotted")) +
  # ylim(-5,5) +
  # geom_text(data = dat_text_s2_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vj
```

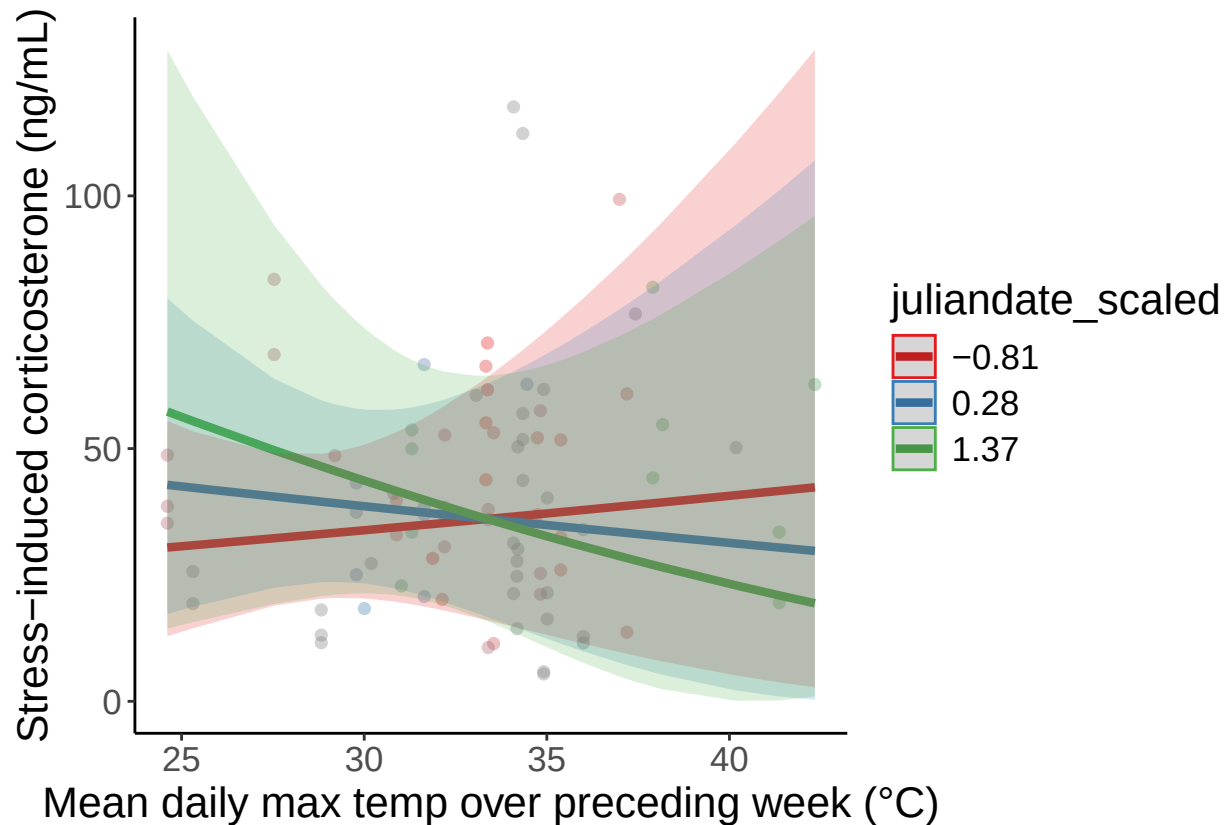
```

# theme(legend.position = "none")
)

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Interaction is not significant.

Baseline cort by prior day temp

```

prior_model <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_scaled +
  mutate(across(c(gweight, maxt_prior, mint_prior, juliandate, brood_size),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)

s1_lintemp <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat +
  mutate(across(c(gweight, maxt_prior, mint_prior, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

```

```

s1_lintemp_addmax <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_s1,
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size_s1),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)

s1_lintemp_addmin <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled * habitat + mint_prior_scaled + age_s1,
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size_s1),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)

s1_lintemp_noint <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled + habitat + age_s1,
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size_s1),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)

c1 <- anova(s1_lintemp,s1_lintemp_addmin,s1_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",

```

WEBL

```

## refitting model(s) with ML (instead of REML)
c2 <- anova(s1_lintemp,s1_lintemp_addmax,s1_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",
## refitting model(s) with ML (instead of REML)
(int_tab_s1_priordayt_webl <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
    P = `Pr(>Chisq)` ) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	279.73	NA	NA
single interaction	281.37	4.36	0.225
both interacting	282.57	4.80	0.187
Min temp			
no interaction	279.73	NA	NA
single interaction	278.44	7.29	0.063
both interacting	282.57	1.86	0.601

```

# gtsave(int_tab_s1_priordayt_webl,"figures/int_tab_s1_priordayt_webl.html")

s1_prior_day_webl <- s1_lintemp_addmax

```

```
summary(s1_lintemp_addmax)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat +
##      age_scaled + maxt_prior_scaled * juliandate_scaled + year_fct +
##      (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>%
##      mutate(across(c(gweight, maxt_prior, mint_prior, juliandate,
##      brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)
##
## REML criterion at convergence: 288.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.0129 -0.5704 -0.0164  0.5512  4.3501
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.08812  0.2969
## Residual              0.20387  0.4515
## Number of obs: 175, groups: attempt_id, 52
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)      2.22601    0.12439  36.02964   17.895
## maxt_prior_scaled    0.07803    0.07177  94.29648    1.087
## mint_prior_scaled    0.03307    0.09797 155.61091    0.338
## habitatOrchard     -0.10195    0.15115  36.13959   -0.674
## habitatGrassland   -0.23365    0.20911  42.86609   -1.117
## habitatRow crop      0.09300    0.17300  35.16866    0.538
## age_scaled          0.11270    0.04559 160.15909    2.472
## juliandate_scaled   -0.19849    0.07310  55.19382   -2.715
## year_fct2023         0.08222    0.13816  44.71265    0.595
## mint_prior_scaled:habitatOrchard  0.06630    0.11713 122.97420    0.566
## mint_prior_scaled:habitatGrassland -0.55730    0.24521 153.06224   -2.273
## mint_prior_scaled:habitatRow crop  -0.02963    0.14540  79.21061   -0.204
## maxt_prior_scaled:juliandate_scaled 0.03343    0.05458  99.34196    0.612
##
##              Pr(>|t|)
## (Intercept)    < 2e-16 ***
## maxt_prior_scaled  0.27969
## mint_prior_scaled  0.73613
## habitatOrchard    0.50429
## habitatGrassland  0.27006
## habitatRow crop    0.59425
## age_scaled        0.01448 *
## juliandate_scaled  0.00882 **
## year_fct2023      0.55478
## mint_prior_scaled:habitatOrchard  0.57241
## mint_prior_scaled:habitatGrassland 0.02443 *
## mint_prior_scaled:habitatRow crop  0.83906
```

```
## maxt_prior_scaled:juliandate_scaled 0.54165
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
anova(s1_lintemp_addmax,prior_model)

## refitting model(s) with ML (instead of REML)

## Data: dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>% ...
## Models:
## prior_model: sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_scaled + juliandate_scaled
## s1_lintemp_addmax: sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_scaled + juliandate_scaled
##               npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model      14 277.02 321.33 -124.51   249.02
## s1_lintemp_addmax 15 278.44 325.91 -124.22   248.44 0.5816  1    0.4457

Interaction not significant, does not add to fit, does not change most parsimonious model

(webbs1_priordayt_trendmax <- emtrends(prior_model, specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
    gt()))
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.062	0.068	101	0.912	0.364
Orchard	0.062	0.068	101	0.912	0.364
Grassland	0.062	0.068	101	0.912	0.364
Row crop	0.062	0.068	101	0.912	0.364

```
(webbs1_priordayt_trendmax <- emtrends(s1_lintemp_addmax, specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
    gt()))
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.079	0.074	103	1.076	0.284
Orchard	0.079	0.074	103	1.076	0.284
Grassland	0.079	0.074	103	1.076	0.284
Row crop	0.079	0.074	103	1.076	0.284

No change in trend or significance

```

samp_s1_priordayt_webl <- s1_lintemp_addmax@frame %>% group_by(habitat) %>% summarize(count = n())
# samp_s1_priordayt_webl %>% gt() %>% gtsave("figures/ss_webl_s1_priordayt_.html")

dat_text_s1_priordayt_webl <- data.frame(
  label = paste("N =", samp_s1_priordayt_webl$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_s1_priordayt_webl = dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>%
  mutate(across(c(gweight, maxt_prior, mint_prior, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)

mean_temp_s1_priordayt_webl <- mean(data_s1_priordayt_webl %>% pull(maxt_prior))
sd_temp_s1_priordayt_webl <- sd(data_s1_priordayt_webl %>% pull(maxt_prior))

temp_trans_s1_priordayt_webl <- trans_new("temp_trans_s1_priordayt_webl",
  transform = function(x){(x * sd_temp_s1_priordayt_webl) + mean_temp_s1_
  inverse = function(x){x}}

(figs2_priordayt_webl <- ggpredict(prior_model, terms = c("maxt_prior_scaled [all]", "habitat"), bias_corr
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s1_priordayt_webl,
    breaks = c((20-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl,
      (25-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl,
      (30-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl,
      (35-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl,
      (40-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("dotted", "dotted", "dotted", "dotted")) +
  #ylim(-5,5) +

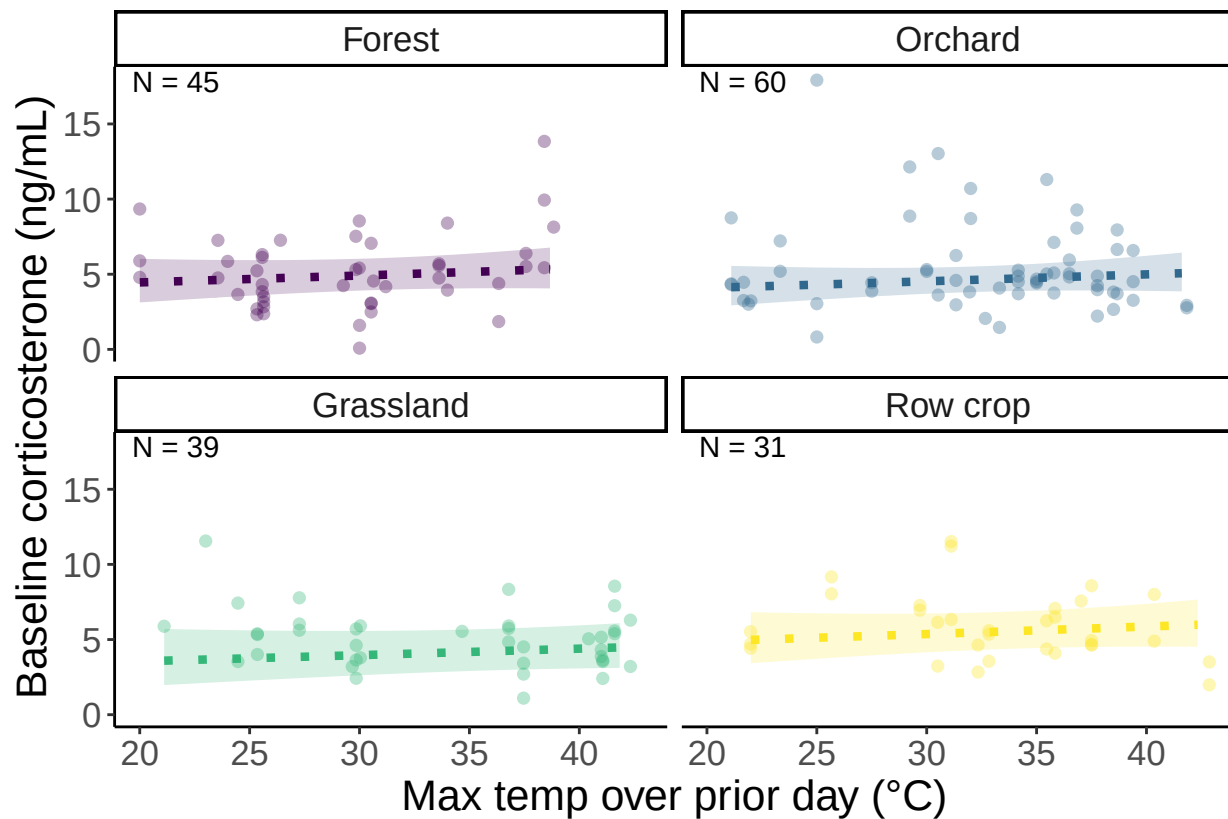
```

```
geom_text(data = dat_text_s1_priordayt_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust =
  theme(legend.position = "none")
)
```

```
## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
(figs2_priordayt_webl <- ggpredict(s1_lintemp_addmax, terms = c("maxt_prior_scaled [all]", "habitat"), bias
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s1_priordayt_webl,
    breaks = c((20 - mean_temp_s1_priordayt_webl) / sd_temp_s1_priordayt_webl,
```

```

      (25-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl,
      (30-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl,
      (35-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl,
      (40-mean_temp_s1_priordayt_webl)/sd_temp_s1_priordayt_webl),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #             (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("dotted","dotted","dotted","dotted")) +
  #ylim(-5,5) +
  geom_text(data = dat_text_s1_priordayt_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust =
  theme(legend.position = "none")
)

```

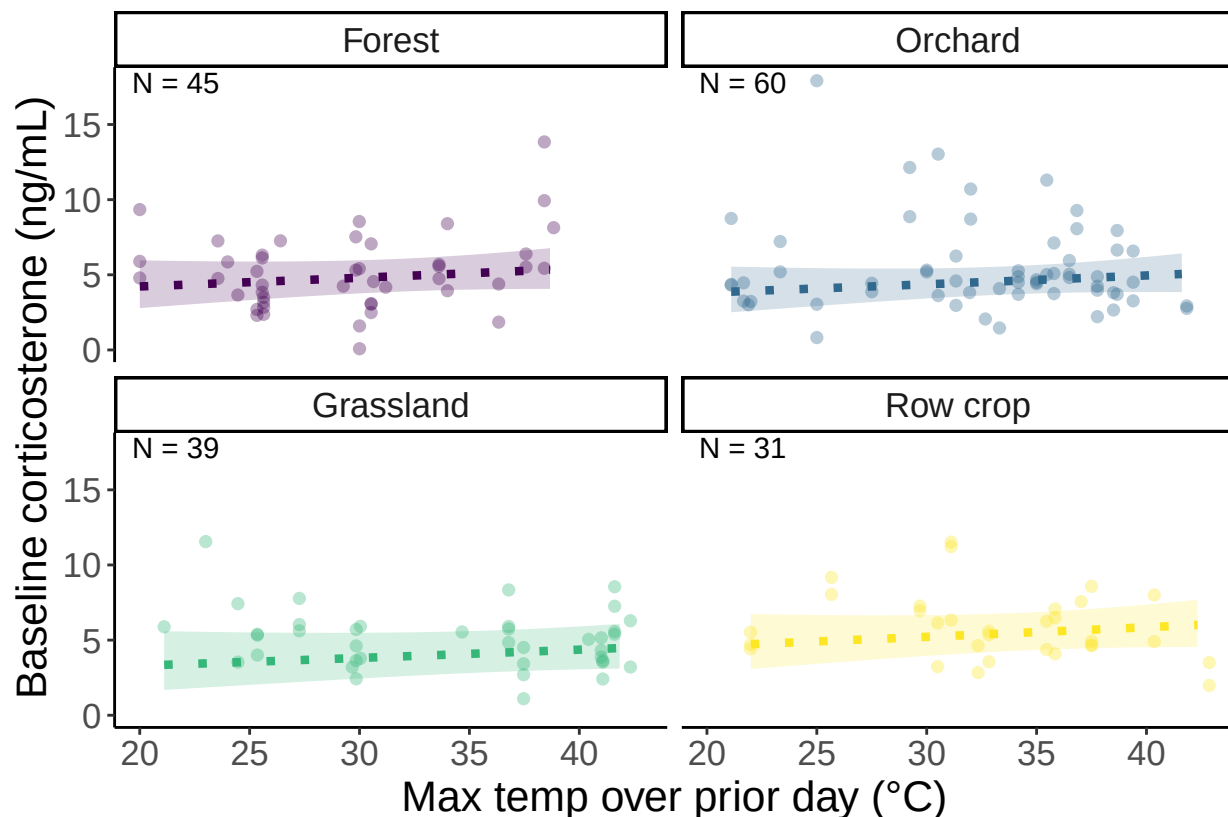
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(figs2_priordayt_webl <- ggpredict(s1_lintemp_addmax, terms = c("maxt_prior_scaled [all]", "juliandate_sc
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
# aes(linetype = .data[["group"]]) +
theme_classic() +
# facet_wrap(~ group, ncol = 2) +
xlab("Max temp over prior day (\u00b0C)") +
ylab("Baseline corticosterone (ng/mL)") +
# scale_fill_viridis(discrete = TRUE) +
# scale_color_viridis(discrete = TRUE) +
theme(text = element_text(size = 16)) +
labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_s1_priordayt_webl,
                    breaks = c((20 - mean_temp_s1_priordayt_webl) / sd_temp_s1_priordayt_webl,
                                (25 - mean_temp_s1_priordayt_webl) / sd_temp_s1_priordayt_webl,
                                (30 - mean_temp_s1_priordayt_webl) / sd_temp_s1_priordayt_webl,
                                (35 - mean_temp_s1_priordayt_webl) / sd_temp_s1_priordayt_webl,
                                (40 - mean_temp_s1_priordayt_webl) / sd_temp_s1_priordayt_webl),
                    # breaks = c(20, 30, 40, 50),
                    # labels = c("20", "30", "40", "50"),
                    # limits = c((15 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TR
                    #          (55 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TR
                    # limits = c(18, 5)
) # +
# scale_linetype_manual(values = c("dotted", "dotted", "dotted", "dotted")) +
# #ylim(-5, 5) +
# geom_text(data = dat_text_s1_priordayt_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust
# theme(legend.position = "none")
)

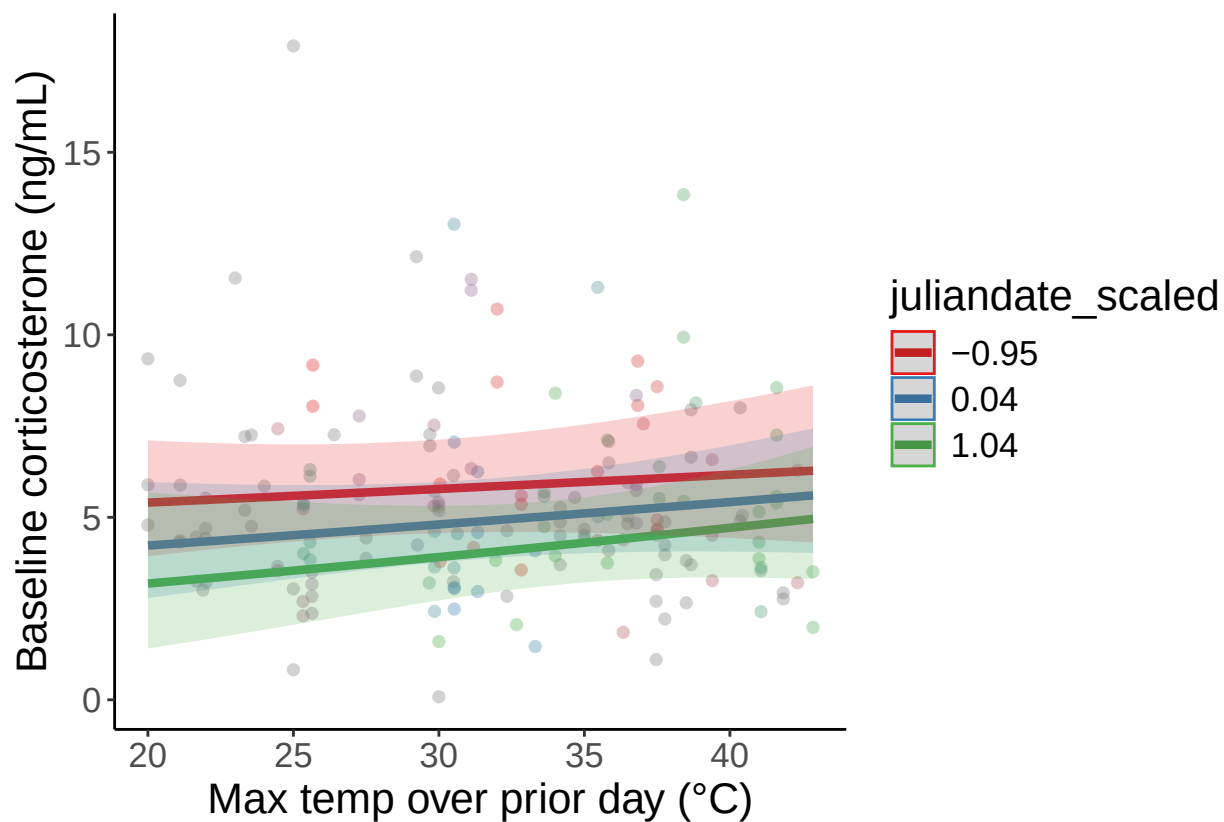
```

```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Interaction not significant

```
prior_model <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_scaled +
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))
```

TRES

```
## boundary (singular) fit: see help('isSingular')
```

```
s1_lintemp <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat + age_scaled +
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))
```

```
s1_lintemp_addmax <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_scaled +
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))
```

```
s1_lintemp_addmin <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat + age_scaled +
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
  maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))
```

```

~ scale(.x)[,1],
.names = "{.col}_scaled"),
maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled

s1_lintemp_noint <- lmerTest::lmer(sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled + habitat + age,
mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size),
~ scale(.x)[,1],
.names = "{.col}_scaled"),
maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled

c1 <- anova(s1_lintemp,s1_lintemp_addmin,s1_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",
"maxt prior scaled", "mint prior scaled", "habitat", "age", "brood size", "gweight", "juliandate"))

## refitting model(s) with ML (instead of REML)
c2 <- anova(s1_lintemp,s1_lintemp_addmax,s1_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction",
"maxt prior scaled", "mint prior scaled", "habitat", "age", "brood size", "gweight", "juliandate"))

## refitting model(s) with ML (instead of REML)
(int_tab_s1_priordayt_tres <- bind_rows(c1,c2) %>%
as.tibble() %>%
mutate(across(where(is.numeric),~round(.x,digits = 3)),
P = `Pr(>Chisq)` ) %>%
dplyr::select(Model,AIC,Chisq,P) %>%
mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
across(c(AIC,Chisq), ~ round(.x, digits = 2)),
P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	189.89	NA	NA
single interaction	190.17	5.72	0.126
both interacting	189.77	6.40	0.094
Min temp			
no interaction	189.89	NA	NA
single interaction	185.92	9.97	0.019
both interacting	189.77	2.15	0.542

```
# gtsave(int_tab_s1_priordayt_tres,"figures/int_tab_s1_priordayt_tres.html")
```

```
check_collinearity(s1_lintemp_noint)
```

```
## # Check for Multicollinearity
```

```
##
```

```
## Low Correlation
```

```
##
```

```
##
```

	Term	VIF	VIF 95% CI	Increased SE	Tolerance
##	maxt_prior_scaled	2.32	[1.83, 3.10]	1.52	0.43
##	mint_prior_scaled	1.90	[1.53, 2.52]	1.38	0.53
##	habitat	1.56	[1.29, 2.06]	1.25	0.64

```

##               age_scaled 1.23 [1.08, 1.67]           1.11      0.81
##               juliandate_scaled 1.59 [1.32, 2.10]       1.26      0.63
##               year_fct 1.62 [1.33, 2.14]           1.27      0.62
## maxt_prior_scaled:juliandate_scaled 1.15 [1.03, 1.65]   1.07      0.87
## Tolerance 95% CI
##   [0.32, 0.55]
##   [0.40, 0.65]
##   [0.49, 0.77]
##   [0.60, 0.93]
##   [0.48, 0.76]
##   [0.47, 0.75]
##   [0.61, 0.97]

s1_prior_day_tres <- s1_lintemp_admin

summary(s1_lintemp_admin)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: sqrt(cort_s1) ~ maxt_prior_scaled * habitat + mint_prior_scaled +
##   age_scaled + maxt_prior_scaled * juliandate_scaled + year_fct +
##   (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>%
##   mutate(across(c(gweight, maxt_prior, mint_prior, juliandate,
##     brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##     maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)
##
## REML criterion at convergence: 194.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.95603 -0.69922 -0.09299  0.70588  2.36506
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.03196  0.1788
## Residual              0.28304  0.5320
## Number of obs: 105, groups: attempt_id, 41
##
## Fixed effects:
##
##               Estimate Std. Error    df t value
## (Intercept)      2.48060    0.30066 30.44917   8.251
## maxt_prior_scaled    0.05863    0.24476 37.84554   0.240
## habitatOrchard    -0.23382    0.35398 30.80768  -0.661
## habitatGrassland  -0.25410    0.30456 29.21192  -0.834
## habitatRow crop   -0.01501    0.29775 27.70750  -0.050
## mint_prior_scaled  -0.19970    0.09690 37.72912  -2.061
## age_scaled        -0.01096    0.07646 73.69638  -0.143
## juliandate_scaled   0.05832    0.07459 30.28372   0.782
## year_fct2023       0.15856    0.16182 28.01251   0.980
## maxt_prior_scaled:habitatOrchard -0.46424    0.33822 21.16964  -1.373
## maxt_prior_scaled:habitatGrassland  0.09143    0.25821 32.90547   0.354
## maxt_prior_scaled:habitatRow crop   0.12844    0.25019 31.95274   0.513
## maxt_prior_scaled:juliandate_scaled  0.09507    0.08499 40.51060   1.119

```

```
##                                Pr(>|t|)
## (Intercept)                   2.93e-09 ***
## maxt_prior_scaled             0.8120
## habitatOrchard                0.5138
## habitatGrassland              0.4109
## habitatRow crop               0.9602
## mint_prior_scaled             0.0463 *
## age_scaled                    0.8864
## juliandate_scaled             0.4404
## year_fct2023                  0.3356
## maxt_prior_scaled:habitatOrchard 0.1842
## maxt_prior_scaled:habitatGrassland 0.7255
## maxt_prior_scaled:habitatRow crop 0.6112
## maxt_prior_scaled:juliandate_scaled 0.2699
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it
anova(s1_lintemp_addmax,prior_model)

## refitting model(s) with ML (instead of REML)

## Warning in optwrap(optimizer, devfun, x@theta, lower = x@lower, calc.derivs =
## TRUE, : convergence code 3 from bobyqa: bobyqa -- a trust region step failed to
## reduce q

## Data: dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>% ...
## Models:
## prior_model: sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_scaled + juliandate_scaled
## s1_lintemp_addmax: sqrt(cort_s1) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_scaled + juliandate_scaled
##               npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model      14 184.90 222.05 -78.448   156.90
## s1_lintemp_addmax 15 185.92 225.72 -77.958   155.92 0.981  1    0.3219
```

So this more clearly favors addmax. I'm going to change the manuscript to reflect this.

Interaction is not significant, does not increase fit.

```
(tress1_priordayt_trendmax <- emtrends(prior_model, specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
         gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.196	0.094	50	2.08	0.043
Orchard	0.196	0.094	50	2.08	0.043
Grassland	0.196	0.094	50	2.08	0.043
Row crop	0.196	0.094	50	2.08	0.043

```
(tress1_priordayt_trendmax <- emtrends(s1_lintemp_addmax, specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
         df = round(df),
         p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
         gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	0.195	0.095	50	2.049	0.046
Orchard	0.195	0.095	50	2.049	0.046
Grassland	0.195	0.095	50	2.049	0.046
Row crop	0.195	0.095	50	2.049	0.046

```
samp_s1_priordayt_tres <- s1_lintemp_addmax@frame %>% group_by(habitat) %>% summarize(count = n())
# samp_s1_priordayt_tres %>% gt() %>% gtsave("figures/ss_tres_s1_priordayt_.html")

dat_text_s1_priordayt_tres <- data.frame(
  label = paste("N =", samp_s1_priordayt_tres$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_s1_priordayt_tres = dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>%
  mutate(across(c(gweight, maxt_prior, mint_prior, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)

mean_temp_s1_priordayt_tres <- mean(data_s1_priordayt_tres %>% pull(maxt_prior))
sd_temp_s1_priordayt_tres <- sd(data_s1_priordayt_tres %>% pull(maxt_prior))

temp_trans_s1_priordayt_tres <- trans_new("temp_trans_s1_priordayt_tres",
  transform = function(x){(x * sd_temp_s1_priordayt_tres) + mean_temp_s1_priordayt_tres},
  inverse = function(x){x})

(figs2_priordayt_tres <- ggpredict(prior_model, terms = c("maxt_prior_scaled [all]", "habitat"), bias_correction = "none",
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
```

```

labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_s1_priordayt_tres,
  breaks = c((25-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
    (35-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
    (45-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
    (55-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
    (65-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres),
  # breaks = c(20,30,40,50),
  # labels = c("20","30","40","50"),
  # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # limits = c(18,5)
) +
scale_linetype_manual(values = c("solid","solid","solid","solid")) +
#ylim(-5,5) +
geom_text(data = dat_text_s1_priordayt_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust =
theme(legend.position = "none")
)

```

```

## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Solution: Respecify random structure! You may also decrease the
## `tolerance` level to enforce the calculation of random effect variances.

```

```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

```

```

## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Solution: Respecify random structure! You may also decrease the
## `tolerance` level to enforce the calculation of random effect variances.

```

```

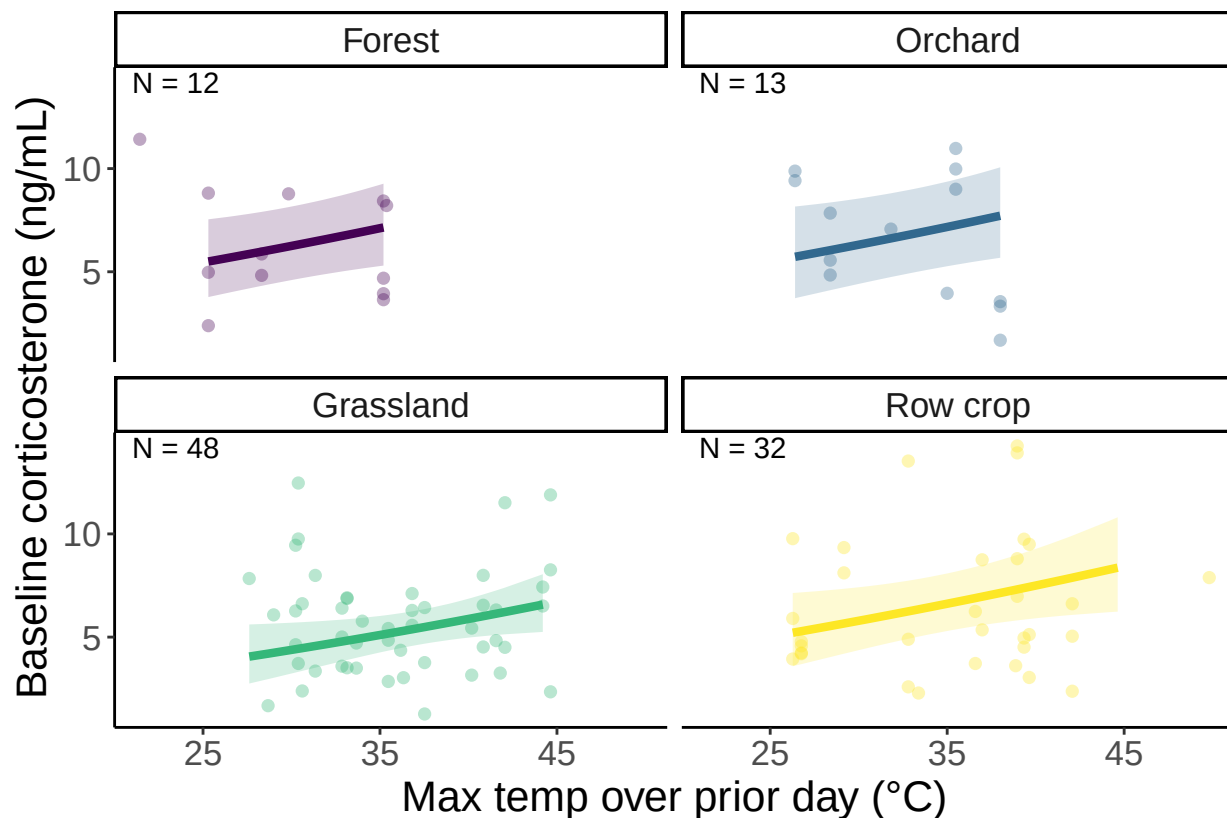
## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```

```

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
(figs2_priordayt_tres <- ggpredict(s1_lintemp_addmax, terms = c("maxt_prior_scaled [all]", "habitat"), bias = TRUE,
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (°C)") +
  ylab("Baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s1_priordayt_tres,
    breaks = c((25-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
      (35-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
      (45-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
      (55-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
      (65-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("solid", "solid", "solid", "solid")) +
  #ylim(-5,5) +
  geom_text(data = dat_text_s1_priordayt_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust =
```

```

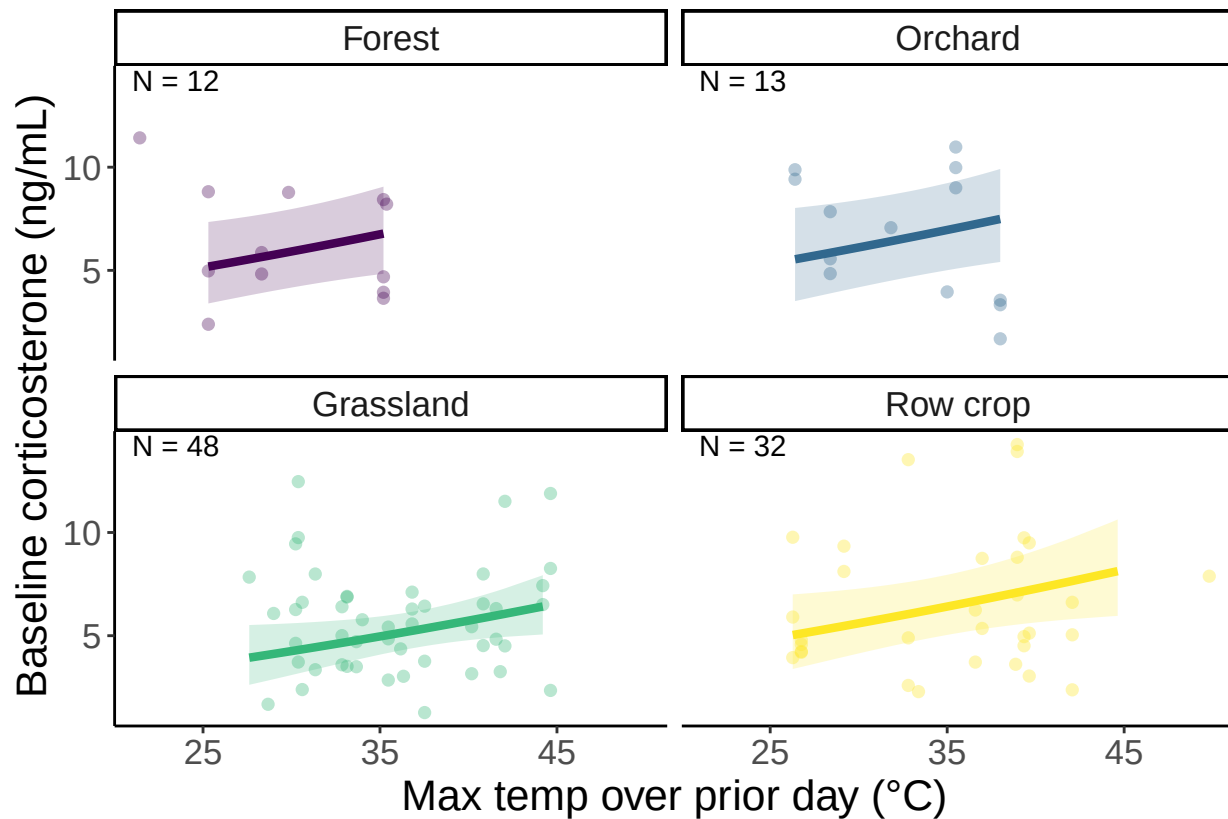
theme(legend.position = "none")
)

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(figs2_priordayt_tres <- ggpredict(s1_lintemp_addmax, terms = c("maxt_prior_scaled [all]", "juliandate_sc
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
# aes(linetype = .data[["group"]]) +
theme_classic() +
# facet_wrap(~ group, ncol = 2) +
xlab("Max temp over prior day (\u00b0C)") +
ylab("Baseline corticosterone (ng/mL)") +
# scale_fill_viridis(discrete = TRUE) +
# scale_color_viridis(discrete = TRUE) +
theme(text = element_text(size = 16)) +
labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_s1_priordayt_tres,
breaks = c((25-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
(35-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,

```

```

      (45-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
      (55-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres,
      (65-mean_temp_s1_priordayt_tres)/sd_temp_s1_priordayt_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) # +
#   scale_linetype_manual(values = c("dotted","dashed","dotted","dotted")) +
#   ylim(-5,5) +
#   geom_text(data = dat_text_s1_priordayt_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust
#   theme(legend.position = "none")
)

```

```

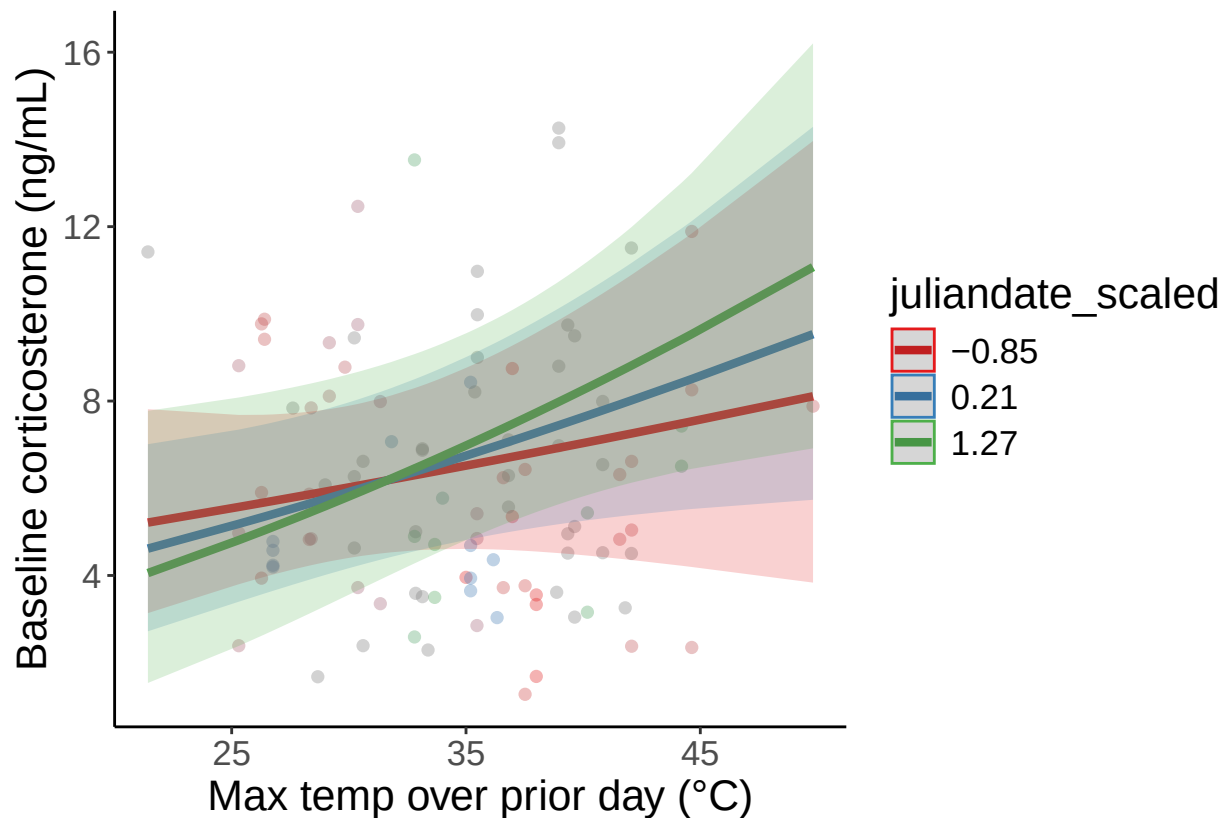
## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Interaction not significant

Abs difference by prior day temp

```
prior_model <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled + mint_prior_scaled + habitat +
                             mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size),
                             ~ scale(.x)[,1],
                             .names = "{.col}_scaled"),
                             maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))
```

WEBL

```
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

abs_lintemp <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled * habitat + mint_prior_scaled *
                             mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age),
                             ~ scale(.x)[,1],
                             .names = "{.col}_scaled"),
                             maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))
```

```
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

abs_lintemp_addmax <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled + mint_prior_scaled * habitat +
                             mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age),
                             ~ scale(.x)[,1],
                             .names = "{.col}_scaled"),
                             maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))
```

```
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

abs_lintemp_addmin <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled * habitat + mint_prior_scaled *
                             mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age),
                             ~ scale(.x)[,1],
                             .names = "{.col}_scaled"),
                             maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))
```

```
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced

abs_lintemp_noint <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled + mint_prior_scaled + habitat +
                             mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age),
```

```

~ scale(.x)[,1],
.names = "{.col}_scaled"),
maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled

## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
## Warning in sqrt(abs_change_cort): NaNs produced
c1 <- anova(abs_lintemp,abs_lintemp_addmin,abs_lintemp_noint) %>% tibble() %>% mutate(Model = c("no int", "min temp", "max temp", "both"))

## refitting model(s) with ML (instead of REML)
c2 <- anova(abs_lintemp,abs_lintemp_addmax,abs_lintemp_noint) %>% tibble() %>% mutate(Model = c("no int", "min temp", "max temp", "both"))

## refitting model(s) with ML (instead of REML)
(int_tab_abs_priordayt_webl <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
    P = `Pr(>Chisq)` ) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	561.97	NA	NA
single interaction	566.13	1.84	0.605
both interacting	568.05	4.08	0.253
Min temp			
no interaction	561.97	NA	NA
single interaction	564.96	3.01	0.39
both interacting	568.05	2.91	0.405

```

# gtsave(int_tab_abs_priordayt_webl,"figures/int_tab_abs_priordayt_webl.html")

summary(abs_lintemp_noint)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: sqrt(abs_change_cort) ~ maxt_prior_scaled + mint_prior_scaled +
##       habitat + age_scaled + maxt_prior_scaled * juliandate_scaled +
##       year_fct + (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>%
##       mutate(across(c(gweight, maxt_prior, mint_prior, juliandate,

```

```

##      brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)
##
## REML criterion at convergence: 543.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.92326 -0.61599 -0.00913  0.59051  2.20929
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 2.112      1.453
## Residual              1.704      1.306
## Number of obs: 142, groups: attempt_id, 49
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)      3.66143    0.53928  36.55653    6.789
## maxt_prior_scaled -0.38905    0.25701 121.48550   -1.514
## mint_prior_scaled  0.10511    0.24088 107.12421    0.436
## habitatOrchard     0.51292    0.65740  36.46864    0.780
## habitatGrassland   0.06861    0.78999  36.98537    0.087
## habitatRow crop    0.54319    0.74121  35.60347    0.733
## age_scaled         0.29551    0.15617 129.40807    1.892
## juliandate_scaled  -0.29924    0.28678  49.71214   -1.043
## year_fct2023       -0.17510    0.58673  41.28524   -0.298
## maxt_prior_scaled:juliandate_scaled -0.16970    0.19886 112.39477   -0.853
##
##              Pr(>|t|)
## (Intercept)      5.75e-08 ***
## maxt_prior_scaled  0.1327
## mint_prior_scaled  0.6634
## habitatOrchard     0.4403
## habitatGrassland   0.9313
## habitatRow crop    0.4685
## age_scaled         0.0607 .
## juliandate_scaled  0.3018
## year_fct2023       0.7669
## maxt_prior_scaled:juliandate_scaled  0.3953
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) mxt_p_ mnt_p_ hbttOr hbttGr hbttRc ag_scl jlndt_ y_2023
## mxt_prr_scl  0.023
## mnt_prr_scl  0.003 -0.387
## habttOrchrd -0.720 -0.087 -0.144
## hbttGrsslnd -0.563 -0.245  0.248  0.467
## habittRwcrp -0.621 -0.163  0.027  0.531  0.447
## age_scaled  0.000  0.120  0.200 -0.135  0.016 -0.106
## jlndt_scl  0.024 -0.355 -0.233  0.094 -0.052  0.132 -0.288
## yer_fct2023 -0.340  0.336 -0.231  0.073 -0.114 -0.047  0.108 -0.184
## mxt_prr_s:_ -0.160  0.375  0.075 -0.133 -0.123 -0.071  0.104 -0.199  0.226

```

```
abs_webl_priordayt <- abs_lintemp_noint
anova(abs_lintemp_noint,prior_model)

## refitting model(s) with ML (instead of REML)

## Data: dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>% ...
## Models:
## prior_model: sqrt(abs_change_cort) ~ maxt_prior_scaled + mint_prior_scaled + habitat + age_scaled + ...
## abs_lintemp_noint: sqrt(abs_change_cort) ~ maxt_prior_scaled + mint_prior_scaled + habitat + age_scaled + ...
##               npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model      11 560.69 593.20 -269.35   538.69
## abs_lintemp_noint 12 561.97 597.44 -268.99   537.97 0.7193  1    0.3964

Interaction not significant, does not add to fit, does not change parsimony

(weblabs_priordayt_trendmax <- emtrends(prior_model,specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend",Df = "df", `T-ratio` = "t.ratio",
    gt()))
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.307	0.242	126	-1.27	0.206
Orchard	-0.307	0.242	126	-1.27	0.206
Grassland	-0.307	0.242	126	-1.27	0.206
Row crop	-0.307	0.242	126	-1.27	0.206

```
(weblabs_priordayt_trendmax <- emtrends(abs_lintemp_noint,specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend",Df = "df", `T-ratio` = "t.ratio",
    gt()))
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.396	0.265	123	-1.497	0.137
Orchard	-0.396	0.265	123	-1.497	0.137
Grassland	-0.396	0.265	123	-1.497	0.137
Row crop	-0.396	0.265	123	-1.497	0.137

Same trend and significance.

```
samp <- abs_lintemp_noint@frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_webl_abs_priordayt.html")

dat_text_webl <- data.frame(
  label = paste("N =",samp$count),
  group = factor(c("Forest","Orchard","Grassland","Row crop"))
)
```

```

data_webl_priordayt = dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>%
  mutate(across(c(gweight, maxt_prior, mint_prior, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)

mean_temp_webl_priordayt <- mean(data_webl_priordayt %>% pull(maxt_prior))
sd_temp_webl_priordayt <- sd(data_webl_priordayt %>% pull(maxt_prior))

temp_trans_webl_priordayt <- trans_new("temp_trans_webl_priordayt",
  transform = function(x){(x * sd_temp_webl_priordayt) + mean_temp_webl_priordayt},
  inverse = function(x){x})

(fig4_webl_priordayt <- ggpredict(prior_model, terms = c("maxt_prior_scaled [all]", "habitat"), bias_correction = TRUE,
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Stress-induced - Baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "WEBL cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_webl_priordayt,
    breaks = c((20-mean_temp_webl_priordayt)/sd_temp_webl_priordayt,
      (25-mean_temp_webl_priordayt)/sd_temp_webl_priordayt,
      (30-mean_temp_webl_priordayt)/sd_temp_webl_priordayt,
      (35-mean_temp_webl_priordayt)/sd_temp_webl_priordayt,
      (40-mean_temp_webl_priordayt)/sd_temp_webl_priordayt),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "dotted", "Rangeland" = "dotted"),
  # ylim(0,60) +
  geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1) +
  theme(legend.position = "none")
)

```

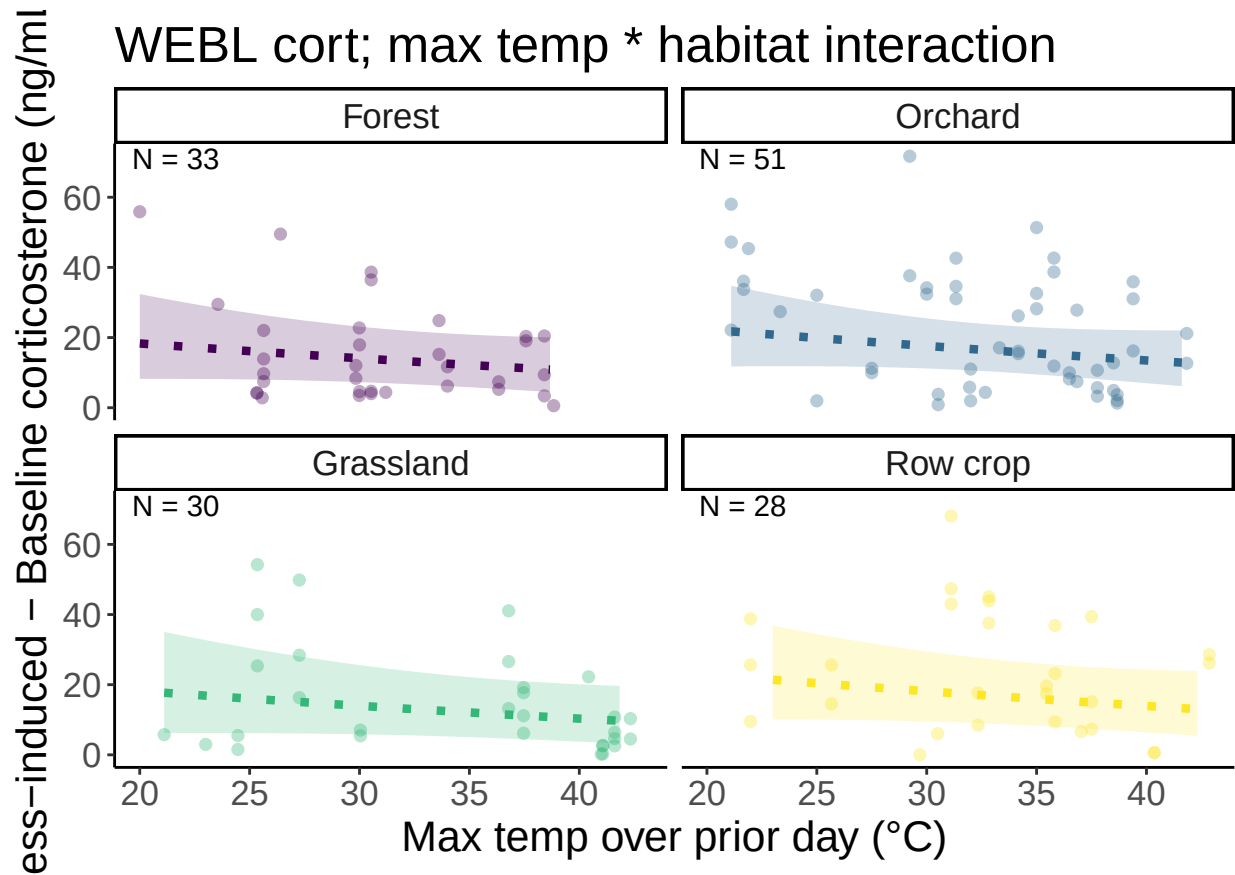
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
(fig4_webl_priordayt <- ggpredict(abs_lintemp_noint, terms = c("maxt_prior_scaled [all]", "habitat"), bias = FALSE,
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (°C)") +
  ylab("Stress-induced - Baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "WEBL cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_webl_priordayt,
    breaks = c((20 - mean_temp_webl_priordayt) / sd_temp_webl_priordayt,
      (25 - mean_temp_webl_priordayt) / sd_temp_webl_priordayt,
      (30 - mean_temp_webl_priordayt) / sd_temp_webl_priordayt,
      (35 - mean_temp_webl_priordayt) / sd_temp_webl_priordayt,
      (40 - mean_temp_webl_priordayt) / sd_temp_webl_priordayt),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18, 5)
  ) +
  scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "dotted", "Row crop" = "dotted")) +
  # ylim(0, 60) +
  geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1.5))
```

```

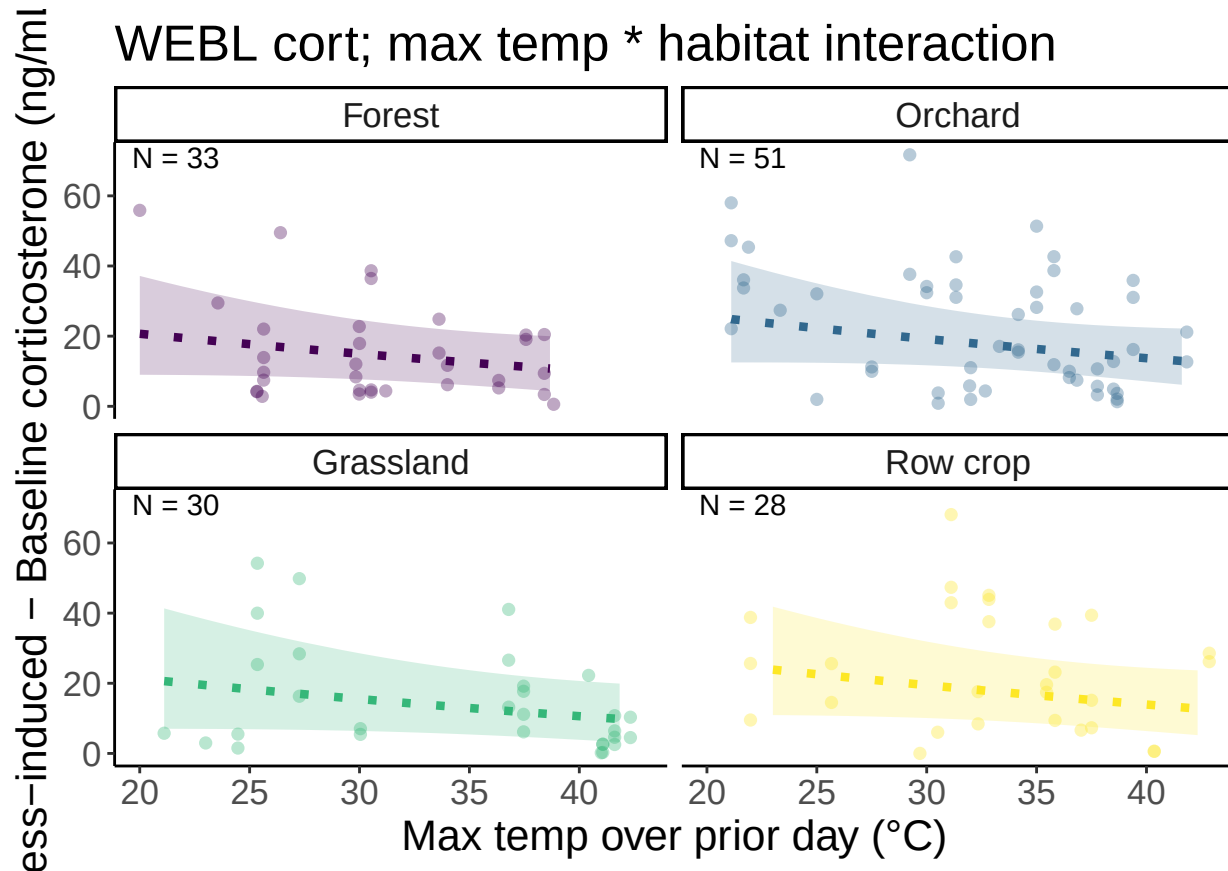
theme(legend.position = "none")
)

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(fig4_webl_priordayt <- ggpredict(abs_lintemp_noint, terms = c("maxt_prior_scaled [all]", "juliandate_scaled [all]"),
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (°C)") +
  ylab("Stress-induced - Baseline corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "WEBL cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_webl_priordayt,
    breaks = c((20-mean_temp_webl_priordayt)/sd_temp_webl_priordayt,
      (25-mean_temp_webl_priordayt)/sd_temp_webl_priordayt,

```

```

      (30-mean_temp_webl_priordayt)/sd_temp_webl_priordayt,
      (35-mean_temp_webl_priordayt)/sd_temp_webl_priordayt,
      (40-mean_temp_webl_priordayt)/sd_temp_webl_priordayt),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) # +
  # scale_linetype_manual(values = c("Forest" = "dotted","Orchard" = "dotted","Grassland" = "dotted",
  # # ylim(0,60) +
  # geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust
  # theme(legend.position = "none")
)

```

```

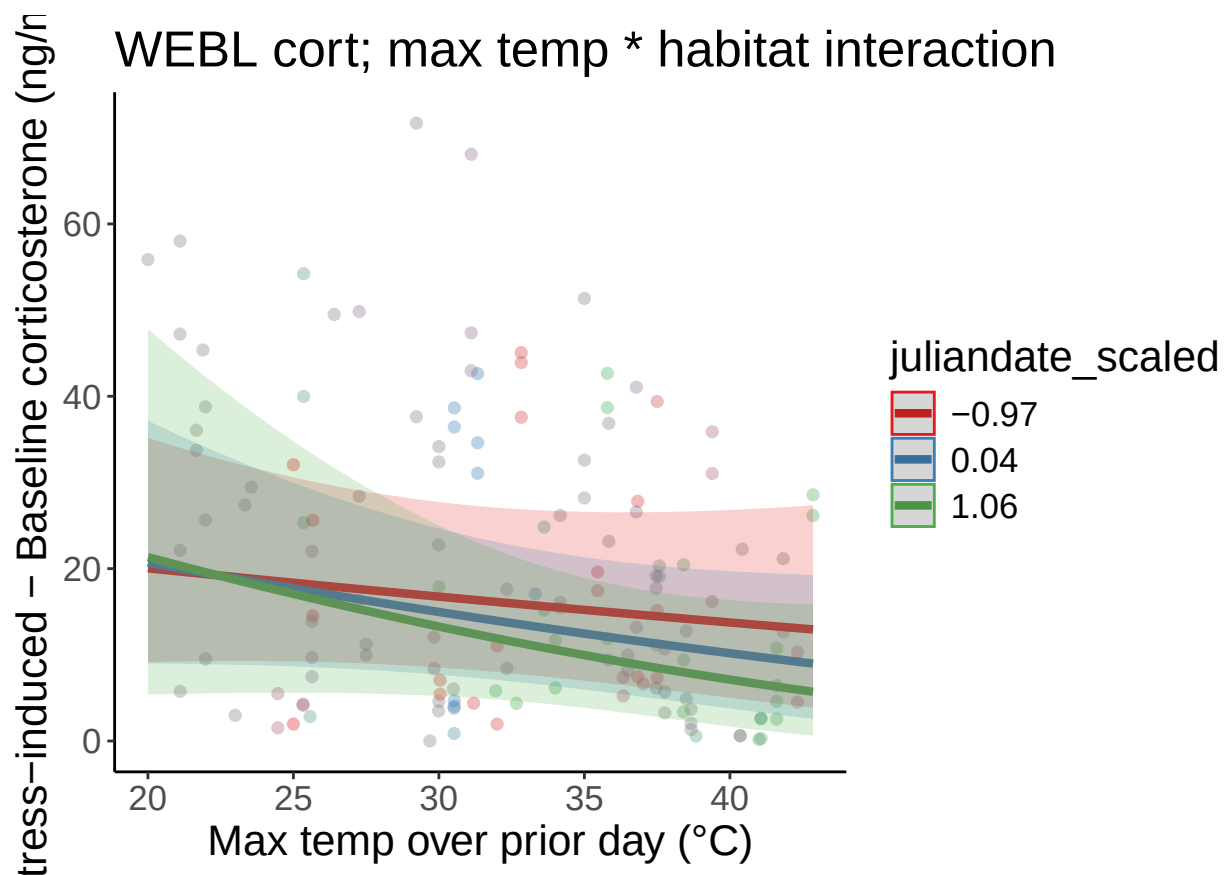
## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Interaction not significant

```

prior_model <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled * habitat + mint_prior_scaled *
mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age)

```

```

      ~ scale(.x)[,1],
      .names = "{.col}_scaled"),
      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

abs_lintemp <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled * habitat + mint_prior_scaled *
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

abs_lintemp_addmax <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled + mint_prior_scaled * ha
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_s
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_sca

abs_lintemp_addmin <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled * habitat + mint_prior_s
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_s
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_sca

abs_lintemp_noint <- lmerTest::lmer(sqrt(abs_change_cort) ~ maxt_prior_scaled + mint_prior_scaled + hab
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_si
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_sca

c1 <- anova(abs_lintemp,abs_lintemp_addmin,abs_lintemp_noint) %>% tibble() %>% mutate(Model = c("no int

```

TRES

```
## refitting model(s) with ML (instead of REML)
```

```
c2 <- anova(abs_lintemp,abs_lintemp_addmax,abs_lintemp_noint) %>% tibble() %>% mutate(Model = c("no int

```

```
## refitting model(s) with ML (instead of REML)
```

```

(int_tab_abs_priordayt_tres <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
    P = `Pr(>Chisq)` %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	349.25	NA	NA
single interaction	353.87	1.38	0.711
both interacting	352.45	7.42	0.06

Min temp

no interaction	349.25	NA	NA
single interaction	352.29	2.96	0.398
both interacting	352.45	5.84	0.12

```
# gtsave(int_tab_abs_priordayt_tres,"figures/int_tab_abs_priordayt_tres.html")
```

```
check_collinearity(abs_lintemp_noint)
```

```
## # Check for Multicollinearity
```

```
##
```

```
## Low Correlation
```

```
##
```

```
##               Term  VIF   VIF 95% CI Increased SE Tolerance
##      maxt_prior_scaled 2.62 [2.00, 3.64]      1.62      0.38
##      mint_prior_scaled 2.00 [1.57, 2.74]      1.41      0.50
##      habitat          1.55 [1.27, 2.11]      1.24      0.65
##      age_scaled       1.36 [1.15, 1.87]      1.16      0.74
##      juliandate_scaled 1.90 [1.50, 2.60]      1.38      0.53
##      year_fct         1.68 [1.36, 2.29]      1.30      0.60
##      maxt_prior_scaled:juliandate_scaled 1.19 [1.05, 1.73]      1.09      0.84
```

```
## Tolerance 95% CI
```

```
##      [0.27, 0.50]
```

```
##      [0.36, 0.64]
```

```
##      [0.47, 0.79]
```

```
##      [0.54, 0.87]
```

```
##      [0.38, 0.67]
```

```
##      [0.44, 0.74]
```

```
##      [0.58, 0.95]
```

```
summary(abs_lintemp)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```
## lmerModLmerTest]
```

```
## Formula:
```

```
## sqrt(abs_change_cort) ~ maxt_prior_scaled * habitat + mint_prior_scaled *
```

```
##      habitat + age_scaled + maxt_prior_scaled * juliandate_scaled +
```

```
##      year_fct + (1 | attempt_id)
```

```
## Data:
```

```
## dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>%
```

```
##      mutate(across(c(gweight, maxt_prior, mint_prior, juliandate,
```

```
##      brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
```

```
##      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)
```

```
##
```

```
## REML criterion at convergence: 304.6
```

```
##
```

```
## Scaled residuals:
```

```
##      Min      1Q   Median      3Q      Max
```

```
## -2.07743 -0.43764 -0.00556  0.53431  1.99129
```

```
##
```

```
## Random effects:
```

```
## Groups      Name      Variance Std.Dev.
```

```
## attempt_id (Intercept) 1.215    1.102
```

```

## Residual                2.994    1.730
## Number of obs: 80, groups: attempt_id, 38
##
## Fixed effects:
##
##               Estimate Std. Error    df t value
## (Intercept)      5.16450    2.02600 21.92854   2.549
## maxt_prior_scaled -0.07147    1.62636 17.57261  -0.044
## habitatOrchard    -0.29006    2.14818 23.17499  -0.135
## habitatGrassland   0.13940    1.96667 21.86326   0.071
## habitatRow crop    0.93403    1.99773 21.93510   0.468
## mint_prior_scaled  1.25545    1.69406 23.45415   0.741
## age_scaled         0.19805    0.47777 29.16734   0.415
## juliandate_scaled -0.14298    0.37381 19.77735  -0.382
## year_fct2023       0.33738    0.78188 19.90464   0.431
## maxt_prior_scaled:habitatOrchard -3.37972    2.70386 23.13891  -1.250
## maxt_prior_scaled:habitatGrassland  0.53060    1.75470 17.95375   0.302
## maxt_prior_scaled:habitatRow crop  0.82721    1.63217 17.08809   0.507
## habitatOrchard:mint_prior_scaled -5.42777    3.16203 34.39238  -1.717
## habitatGrassland:mint_prior_scaled -0.86422    1.72879 22.17413  -0.500
## habitatRow crop:mint_prior_scaled -1.94181    1.84017 21.79424  -1.055
## maxt_prior_scaled:juliandate_scaled -0.03498    0.41319 26.72488  -0.085
##
##               Pr(>|t|)
## (Intercept)      0.0183 *
## maxt_prior_scaled  0.9654
## habitatOrchard    0.8938
## habitatGrassland  0.9441
## habitatRow crop    0.6447
## mint_prior_scaled  0.4660
## age_scaled         0.6815
## juliandate_scaled  0.7062
## year_fct2023       0.6707
## maxt_prior_scaled:habitatOrchard  0.2238
## maxt_prior_scaled:habitatGrassland  0.7658
## maxt_prior_scaled:habitatRow crop  0.6188
## habitatOrchard:mint_prior_scaled  0.0951 .
## habitatGrassland:mint_prior_scaled  0.6221
## habitatRow crop:mint_prior_scaled  0.3029
## maxt_prior_scaled:juliandate_scaled  0.9332
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 16 > 12.
## Use print(x, correlation=TRUE) or
##       vcov(x)           if you need it

abs_tres_priordayt <- abs_lintemp

anova(abs_lintemp,prior_model)

## refitting model(s) with ML (instead of REML)

## Data: dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>% ...
## Models:
## prior_model: sqrt(abs_change_cort) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat + age
## abs_lintemp: sqrt(abs_change_cort) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat + age

```

```
##          npar    AIC    BIC logLik deviance  Chisq Df Pr(>Chisq)
## prior_model    17 350.45 390.95 -158.23   316.45
## abs_lintemp    18 352.45 395.33 -158.23   316.45 0.0017  1    0.9667
```

Interaction not significant, does not improve fit, does not change parsimony

```
(tresabs_priordayt_trendmax <- emtrends(prior_model, specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
    gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.055	1.587	19	-0.035	0.973
Orchard	-3.431	2.218	28	-1.546	0.133
Grassland	0.441	0.719	25	0.613	0.546
Row crop	0.767	0.617	21	1.245	0.227

```
(tresabs_priordayt_trendmax <- emtrends(abs_lintemp, specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
    gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.082	1.654	18	-0.049	0.961
Orchard	-3.461	2.320	27	-1.492	0.148
Grassland	0.449	0.740	25	0.607	0.55
Row crop	0.746	0.653	20	1.142	0.267

Same trend and non significance

```
samp <- abs_lintemp@frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_tres_abs_priordayt.html")
```

```
dat_text_tres <- data.frame(
  label = paste("N =", samp$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)
```

```
data_tres_priordayt = dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>%
  mutate(across(c(gweight, maxt_prior, mint_prior, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)
```

```

mean_temp_tres_priordayt <- mean(data_tres_priordayt %>% pull(maxt_prior))
sd_temp_tres_priordayt <- sd(data_tres_priordayt %>% pull(maxt_prior))

temp_trans_tres_priordayt <- trans_new("temp_trans_tres_priordayt",
                                       transform = function(x){(x * sd_temp_tres_priordayt) + mean_temp_tres_priordayt},
                                       inverse = function(x){x})

(fig4_tres_priordayt <- ggpredict(prior_model, terms = c("maxt_prior_scaled [all]", "habitat"), bias_correction = TRUE,
                                plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
                                aes(linetype = .data[["group"]]) +
                                theme_classic() +
                                facet_wrap(~ group, ncol = 2) +
                                xlab("Max temp over prior day (\u00b0C)") +
                                ylab("Stress-induced - baseline corticosterone (ng/mL)") +
                                scale_fill_viridis(discrete = TRUE) +
                                scale_color_viridis(discrete = TRUE) +
                                theme(text = element_text(size = 16)) +
                                labs(title = "TRES cort; max temp * habitat interaction") +
                                scale_x_continuous(trans = temp_trans_tres_priordayt,
                                                    breaks = c((20-mean_temp_tres_priordayt)/sd_temp_tres_priordayt,
                                                                (25-mean_temp_tres_priordayt)/sd_temp_tres_priordayt,
                                                                (30-mean_temp_tres_priordayt)/sd_temp_tres_priordayt,
                                                                (35-mean_temp_tres_priordayt)/sd_temp_tres_priordayt,
                                                                (40-mean_temp_tres_priordayt)/sd_temp_tres_priordayt),
                                                    # breaks = c(20,30,40,50),
                                                    # labels = c("20", "30", "40", "50"),
                                                    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE),
                                                                (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TRUE)),
                                                    # limits = c(18,5)
                                ) +
                                scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "dotted", "Rangeland" = "dotted"),
                                                       # ylim(0,60) +
                                geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1) +
                                theme(legend.position = "none")
                                )

```

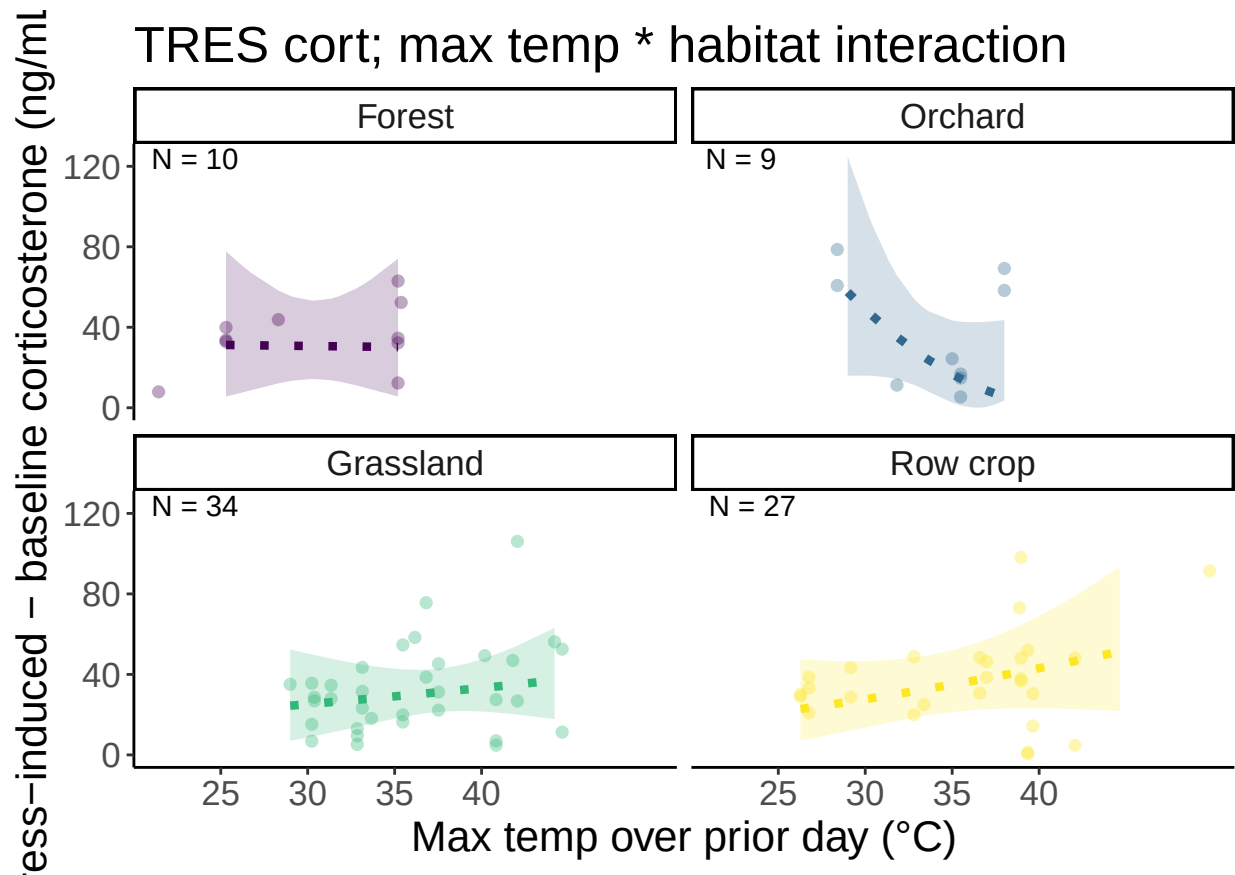
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
(fig4_tres_priordayt <- ggpredict(abs_lintemp, terms = c("maxt_prior_scaled [all]", "habitat"), bias_correction = "none",
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (°C)") +
  ylab("Stress-induced - baseline corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "TRES cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_tres_priordayt,
    breaks = c((20 - mean_temp_tres_priordayt) / sd_temp_tres_priordayt,
      (25 - mean_temp_tres_priordayt) / sd_temp_tres_priordayt,
      (30 - mean_temp_tres_priordayt) / sd_temp_tres_priordayt,
      (35 - mean_temp_tres_priordayt) / sd_temp_tres_priordayt,
      (40 - mean_temp_tres_priordayt) / sd_temp_tres_priordayt),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18, 5)
  ) +
  scale_linetype_manual(values = c("Forest" = "dotted", "Orchard" = "dotted", "Grassland" = "dotted", "Row crop" = "dotted")) +
  # ylim(0, 60) +
  geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1.5))
```

```

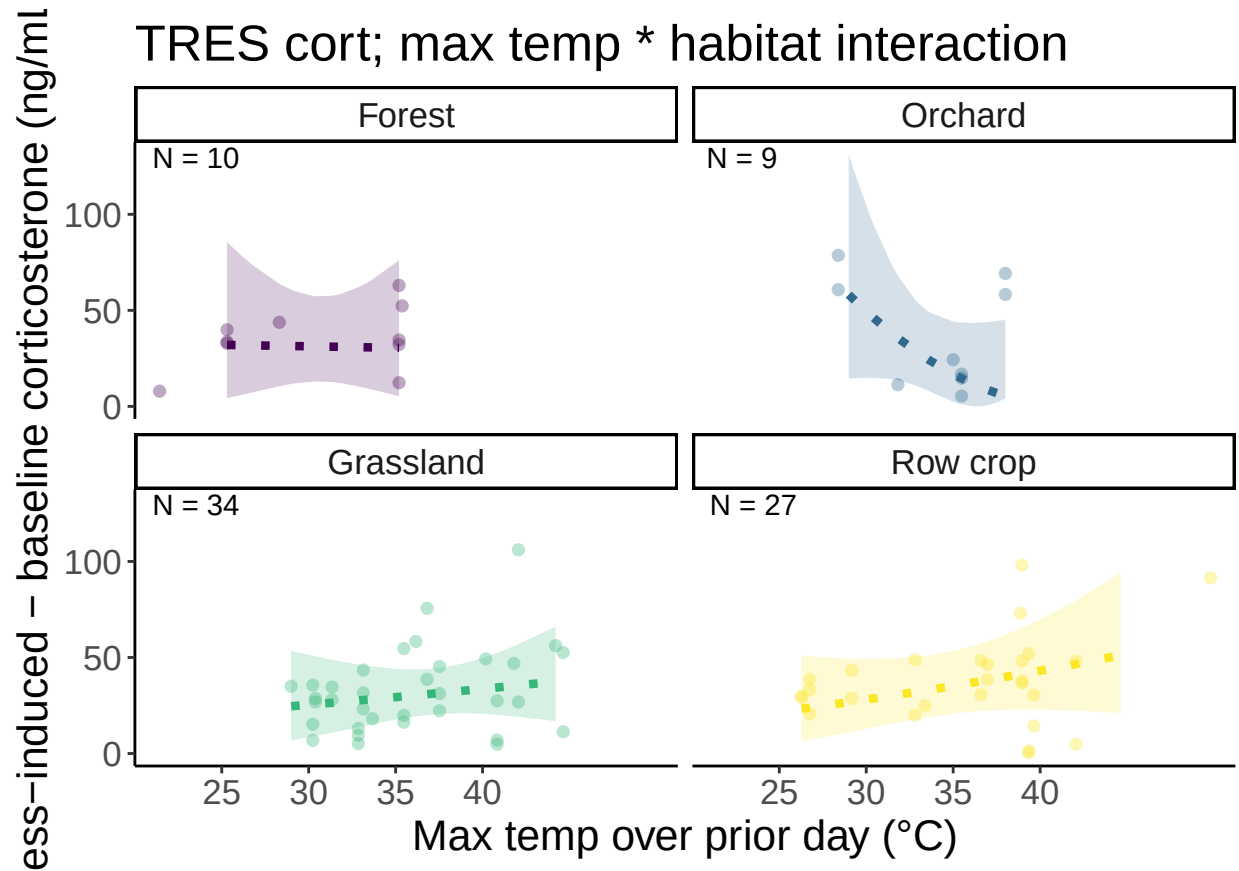
theme(legend.position = "none")
)

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(fig4_tres_priordayt <- ggpredict(abs_lintemp, terms = c("maxt_prior_scaled [all]", "juliandate_scaled"),
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Stress-induced - baseline corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "TRES cort; max temp * habitat interaction") +
  scale_x_continuous(trans = temp_trans_tres_priordayt,
    breaks = c((20-mean_temp_tres_priordayt)/sd_temp_tres_priordayt,
      (25-mean_temp_tres_priordayt)/sd_temp_tres_priordayt,

```

```

      (30-mean_temp_tres_priordayt)/sd_temp_tres_priordayt,
      (35-mean_temp_tres_priordayt)/sd_temp_tres_priordayt,
      (40-mean_temp_tres_priordayt)/sd_temp_tres_priordayt),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) # +
  # scale_linetype_manual(values = c("Forest" = "dotted","Orchard" = "solid","Grassland" = "dotted"),
  # # ylim(0,60) +
  # geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust
  # theme(legend.position = "none")
)

```

```

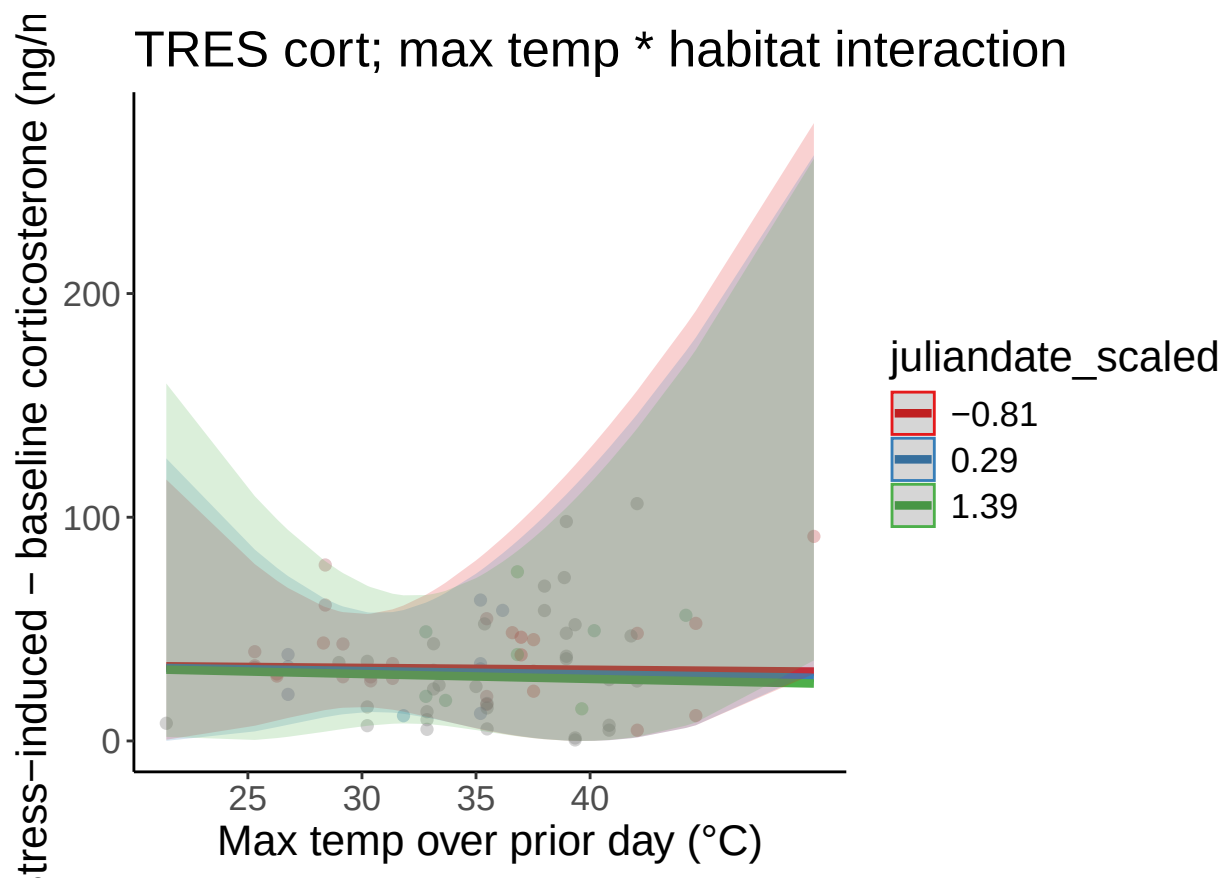
## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Interaction is not significant

s2 cort by prior day temp

```

prior_model <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled + mint_prior_scaled + habitat + age_scaled,
                             mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size),
                                             ~ scale(.x)[,1],
                                             .names = "{.col}_scaled"),
                             maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

s2_lintemp <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat + age_scaled,
                             mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age_scaled),
                                             ~ scale(.x)[,1],
                                             .names = "{.col}_scaled"),
                             maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

s2_lintemp_addmax <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age_scaled,
                                   mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age_scaled),
                                                   ~ scale(.x)[,1],
                                                   .names = "{.col}_scaled"),
                                   maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

s2_lintemp_addmin <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled * habitat + mint_prior_scaled + age_scaled,
                                   mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age_scaled),
                                                   ~ scale(.x)[,1],
                                                   .names = "{.col}_scaled"),
                                   maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

s2_lintemp_noint <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled + mint_prior_scaled + habitat + age_scaled,
                                   mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age_scaled),
                                                   ~ scale(.x)[,1],
                                                   .names = "{.col}_scaled"),
                                   maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

c1 <- anova(s2_lintemp,s2_lintemp_addmin,s2_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction", "max temp", "min temp"))

```

WEBL

```

## refitting model(s) with ML (instead of REML)

c2 <- anova(s2_lintemp,s2_lintemp_addmax,s2_lintemp_noint) %>% tibble() %>% mutate(Model = c("no interaction", "max temp", "min temp"))

## refitting model(s) with ML (instead of REML)

(int_tab_s2_priordayt_webl <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
         P = `Pr(>Chisq)` ) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
         across(c(AIC,Chisq), ~ round(.x, digits = 2)),
         P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	597.75	NA	NA

single interaction	603.60	0.15	0.985
both interacting	607.68	1.92	0.589
Min temp			
no interaction	597.75	NA	NA
single interaction	602.76	0.99	0.803
both interacting	607.68	1.08	0.783

```
# gtsave(int_tab_s2_priordayt_webl, "figures/int_tab_s2_priordayt_webl.html")

s2_prior_day_webl <- s2_lintemp_noint

summary(s2_lintemp_noint)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: sqrt(cort_s2) ~ maxt_prior_scaled + mint_prior_scaled + habitat +
##      age_scaled + maxt_prior_scaled * juliandate_scaled + year_fct +
##      (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>%
##      mutate(across(c(gweight, maxt_prior, mint_prior, juliandate,
##      brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)
##
## REML criterion at convergence: 582.8
##
## Scaled residuals:
##      Min      1Q   Median      3Q      Max
## -2.54090 -0.53812 -0.03287  0.53650  2.58382
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 1.515    1.231
## Residual              1.379    1.174
## Number of obs: 162, groups: attempt_id, 53
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)      4.16591    0.45077  41.86016    9.242
## maxt_prior_scaled -0.03670    0.21003 131.48324   -0.175
## mint_prior_scaled -0.05280    0.20077 122.85650   -0.263
## habitatOrchard      0.59054    0.55254  40.33891    1.069
## habitatGrassland    0.22096    0.64588  42.67041    0.342
## habitatRow crop     0.84404    0.61049  39.20813    1.383
## age_scaled          0.19688    0.12714 148.83841    1.549
## juliandate_scaled   -0.35971    0.23905  52.93454   -1.505
## year_fct2023        0.07127    0.47759  43.74986    0.149
## maxt_prior_scaled:juliandate_scaled 0.04792    0.16340 123.11599    0.293
##
##              Pr(>|t|)
## (Intercept)      1.16e-11 ***
## maxt_prior_scaled      0.862
```

```
## mint_prior_scaled 0.793
## habitatOrchard 0.292
## habitatGrassland 0.734
## habitatRow crop 0.175
## age_scaled 0.124
## juliandate_scaled 0.138
## year_fct2023 0.882
## maxt_prior_scaled:juliandate_scaled 0.770
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) mxt_p_ mnt_p_ hbttOr hbttGr hbttRc ag_scl jlndt_ y_2023
## mxt_prr_scl 0.050
## mnt_prr_scl 0.011 -0.366
## hbttOrchrd -0.723 -0.105 -0.137
## hbttGrsslnd -0.565 -0.267 0.253 0.476
## hbttRwcrp -0.626 -0.195 0.027 0.541 0.479
## age_scaled 0.034 0.163 0.138 -0.137 -0.024 -0.145
## jlndt_scl 0.036 -0.371 -0.249 0.090 -0.068 0.111 -0.294
## yer_fct2023 -0.321 0.319 -0.223 0.065 -0.168 -0.084 0.097 -0.215
## mxt_prr_s:_ -0.181 0.338 0.094 -0.105 -0.105 -0.074 0.124 -0.217 0.217
```

```
anova(s2_lintemp_noint,prior_model)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>% ...
## Models:
## prior_model: sqrt(cort_s2) ~ maxt_prior_scaled + mint_prior_scaled + habitat + age_scaled + julianda
## s2_lintemp_noint: sqrt(cort_s2) ~ maxt_prior_scaled + mint_prior_scaled + habitat + age_scaled + max
##      npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## prior_model      11 595.85 629.82 -286.93 573.85
## s2_lintemp_noint  12 597.75 634.80 -286.88 573.75 0.1013 1 0.7503
```

Not significant, does not improve fit, does not change parsimony

```
(webls2_priordayt_trendmax <- emtrends(prior_model,specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend",Df = "df", `T-ratio` = "t-
  gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.058	0.2	139	-0.288	0.774
Orchard	-0.058	0.2	139	-0.288	0.774
Grassland	-0.058	0.2	139	-0.288	0.774
Row crop	-0.058	0.2	139	-0.288	0.774

```
(webls2_priordayt_trendmax <- emtrends(s2_lintemp_noint,specs = ~ habitat, var = c("maxt_prior_scaled"))
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
```

```
rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.034	0.217	134	-0.155	0.877
Orchard	-0.034	0.217	134	-0.155	0.877
Grassland	-0.034	0.217	134	-0.155	0.877
Row crop	-0.034	0.217	134	-0.155	0.877

Changes nothing

```
samp_s2_priordayt_webl <- s2_lintemp_noint@frame %>% group_by(habitat) %>% summarize(count = n())
# samp_s2_priordayt_webl %>% gt() %>% gtsave("figures/ss_webl_s2_priordayt_.html")

dat_text_s2_priordayt_webl <- data.frame(
  label = paste("N =", samp_s2_priordayt_webl$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_s2_priordayt_webl = dplyr::filter(g, Species == "WEBL", !is.na(maxt_prior), !is.na(mint_prior)) %>%
  mutate(across(c(gweight, maxt_prior, mint_prior, juliandate, brood_size, age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)

mean_temp_s2_priordayt_webl <- mean(data_s2_priordayt_webl %>% pull(maxt_prior))
sd_temp_s2_priordayt_webl <- sd(data_s2_priordayt_webl %>% pull(maxt_prior))

temp_trans_s2_priordayt_webl <- trans_new("temp_trans_s2_priordayt_webl",
  transform = function(x){(x * sd_temp_s2_priordayt_webl) + mean_temp_s2_priordayt_webl},
  inverse = function(x){x})

(fig_priordayt_webl_s2 <- ggpredict(prior_model, terms = c("maxt_prior_scaled [all]", "habitat"), bias_corr = FALSE) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_priordayt_webl,
    breaks = c((20 - mean_temp_s2_priordayt_webl) / sd_temp_s2_priordayt_webl,
```

```

      (25-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (30-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (35-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (40-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("dotted","dotted","dotted","dotted")) +
  #ylim(-5,5) +
  geom_text(data = dat_text_s2_priordayt_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust =
  theme(legend.position = "none")
)

```

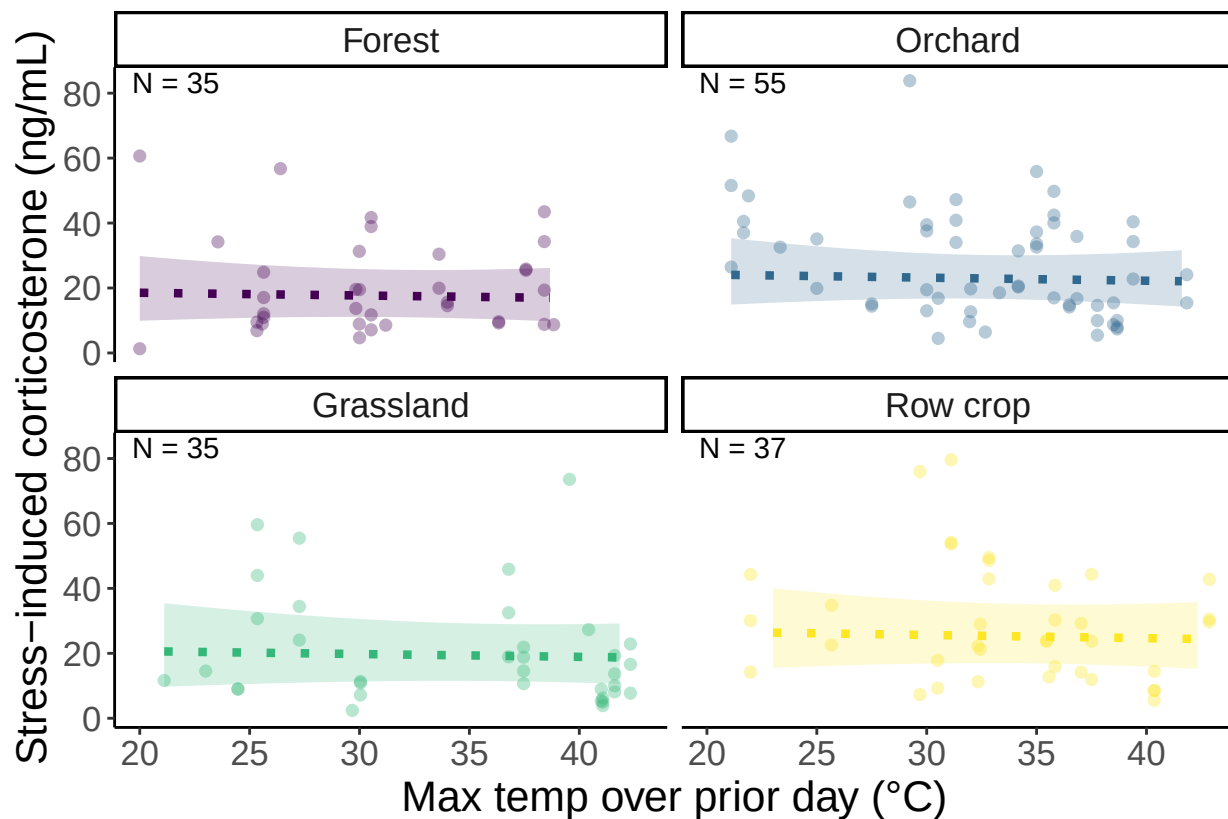
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(fig_priordayt_webl_s2 <- ggpredict(s2_lintemp_addmax, terms = c("maxt_prior_scaled [all]", "habitat"), bi
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_priordayt_webl,
    breaks = c((20-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (25-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (30-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (35-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (40-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("dotted", "dotted", "dotted", "dotted")) +
  #ylim(-5,5) +
  geom_text(data = dat_text_s2_priordayt_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust =
  theme(legend.position = "none")
)

```

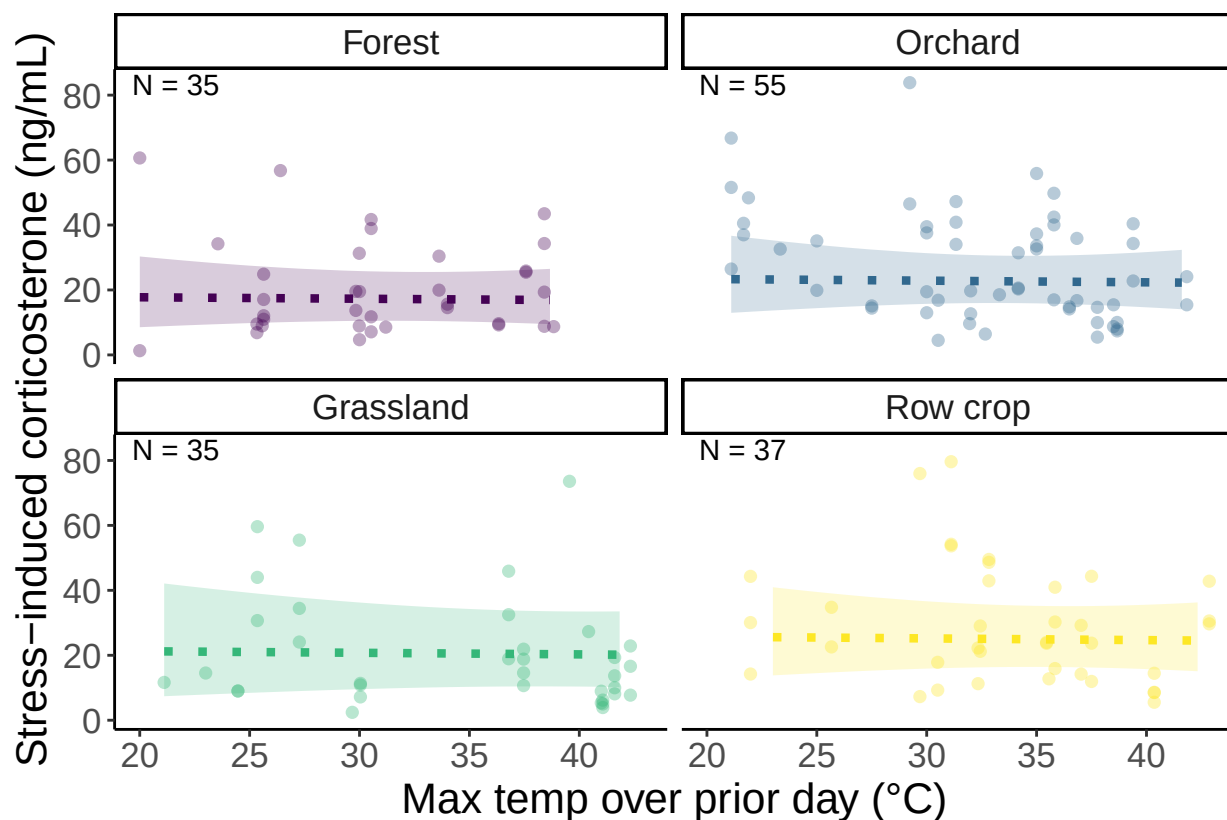
```

## Model has sqrt-transformed response. Back-transforming predictions to
##   original response scale. Standard errors are still on the transformed
##   scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
##   random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
(fig_priordayt_webl_s2 <- ggpredict(s2_lintemp_addmax, terms = c("maxt_prior_scaled [all]", "juliandate_s2"),
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_priordayt_webl,
    breaks = c((20-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (25-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (30-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (35-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl,
      (40-mean_temp_s2_priordayt_webl)/sd_temp_s2_priordayt_webl),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18,5)
  ) # +
  # scale_linetype_manual(values = c("dotted", "dotted", "dotted", "dotted")) +
  # #ylim(-5,5) +
  # geom_text(data = dat_text_s2_priordayt_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust
```

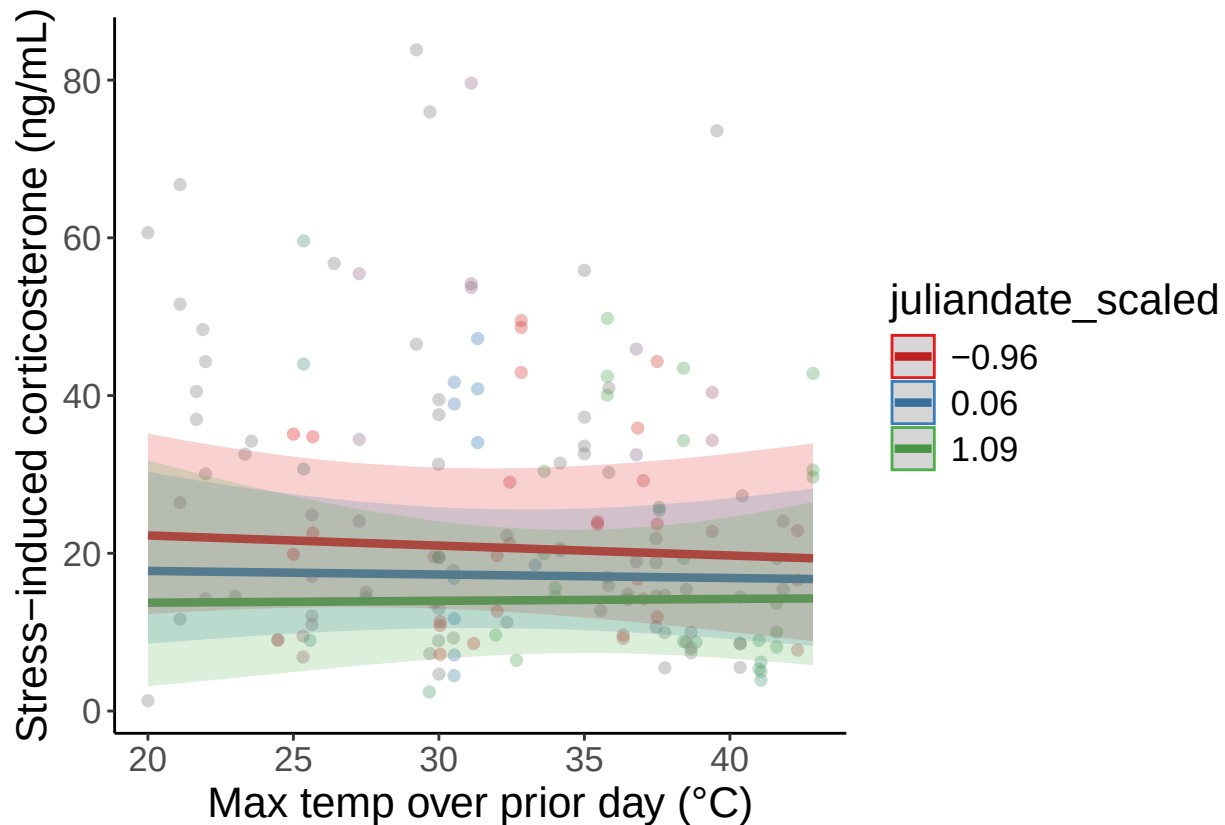
```

# theme(legend.position = "none")
)

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



No interaction

```

prior_model <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat +
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age)
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

s2_lintemp <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat +
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age)
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

s2_lintemp_addmax <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled + mint_prior_scaled * habitat + age +
  mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size,age)
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"),
    maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled))

```

```

      ~ scale(.x)[,1],
      .names = "{.col}_scaled"),
      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled

s2_lintemp_addmin <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled * habitat + mint_prior_scaled + ag
      mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size),
      ~ scale(.x)[,1],
      .names = "{.col}_scaled"),
      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled

s2_lintemp_noint <- lmerTest::lmer(sqrt(cort_s2) ~ maxt_prior_scaled + mint_prior_scaled + habitat + ag
      mutate(across(c(gweight,maxt_prior,mint_prior,juliandate,brood_size),
      ~ scale(.x)[,1],
      .names = "{.col}_scaled"),
      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled

c1 <- anova(s2_lintemp,s2_lintemp_addmin,s2_lintemp_noint) %>% tibble() %>% mutate(Model = c("no intera

```

TRES

```

## refitting model(s) with ML (instead of REML)
c2 <- anova(s2_lintemp,s2_lintemp_addmax,s2_lintemp_noint) %>% tibble() %>% mutate(Model = c("no intera

## refitting model(s) with ML (instead of REML)
(int_tab_s2_priordayt_tres <- bind_rows(c1,c2) %>%
  as.tibble() %>%
  mutate(across(where(is.numeric),~round(.x,digits = 3)),
    P = `Pr(>Chisq)` ) %>%
  dplyr::select(Model,AIC,Chisq,P) %>%
  mutate(max_or_min = c(rep("Max temp",times = 3),rep("Min temp",times = 3)),
    across(c(AIC,Chisq), ~ round(.x, digits = 2)),
    P = if_else(P < 0.001,"<0.001",as.character(P))) %>%
  group_by(max_or_min) %>% gt())

```

Model	AIC	Chisq	P
Max temp			
no interaction	352.78	NA	NA
single interaction	357.15	1.63	0.653
both interacting	352.15	10.99	0.012
Min temp			
no interaction	352.78	NA	NA
single interaction	354.47	4.30	0.23
both interacting	352.15	8.32	0.04

```

# gtsave(int_tab_s2_priordayt_tres,"figures/int_tab_s2_priordayt_tres.html")

```

```

check_collinearity(s2_lintemp_noint)

```

```

## # Check for Multicollinearity

```

```
##
## Low Correlation
##
##               Term   VIF   VIF 95% CI Increased SE Tolerance
##      maxt_prior_scaled 2.38 [1.84, 3.25]      1.54      0.42
##      mint_prior_scaled 1.91 [1.52, 2.60]      1.38      0.52
##      habitat          1.51 [1.25, 2.05]      1.23      0.66
##      age_scaled        1.26 [1.09, 1.75]      1.12      0.80
##      juliandate_scaled 1.83 [1.46, 2.48]      1.35      0.55
##      year_fct          1.63 [1.33, 2.21]      1.28      0.61
##      maxt_prior_scaled:juliandate_scaled 1.21 [1.06, 1.71]      1.10      0.83
## Tolerance 95% CI
##      [0.31, 0.54]
##      [0.39, 0.66]
##      [0.49, 0.80]
##      [0.57, 0.92]
##      [0.40, 0.68]
##      [0.45, 0.75]
##      [0.58, 0.94]

s2_prior_day_tres <- s2_lintemp

summary(s2_lintemp)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: sqrt(cort_s2) ~ maxt_prior_scaled * habitat + mint_prior_scaled *
##      habitat + age_scaled + maxt_prior_scaled * juliandate_scaled +
##      year_fct + (1 | attempt_id)
## Data:
## dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>%
##      mutate(across(c(gweight, maxt_prior, mint_prior, juliandate,
##      brood_size, age), ~scale(.x)[, 1], .names = "{.col}_scaled"),
##      maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)
##
## REML criterion at convergence: 312.8
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -1.8989 -0.4610 -0.0794  0.6523  2.7448
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
##      attempt_id (Intercept) 0.4928   0.702
##      Residual              2.5097   1.584
## Number of obs: 86, groups: attempt_id, 38
##
## Fixed effects:
##
##               Estimate Std. Error    df t value
## (Intercept)      5.468092   1.113122 24.185829   4.912
## maxt_prior_scaled -0.467088   0.924403 24.845881  -0.505
## habitatOrchard     0.286043   1.302504 22.808810   0.220
## habitatGrassland   0.313396   1.120265 21.548622   0.280
## habitatRow crop    1.452465   1.183465 19.567147   1.227
## mint_prior_scaled  1.124999   1.040246 18.329276   1.081
```

```
## age_scaled 0.196457 0.348216 35.572380 0.564
## juliandate_scaled -0.065404 0.278860 19.033146 -0.235
## year_fct2023 0.067736 0.593186 16.005897 0.114
## maxt_prior_scaled:habitatOrchard -2.572828 2.040486 27.424447 -1.261
## maxt_prior_scaled:habitatGrassland 0.579063 1.107919 24.712832 0.523
## maxt_prior_scaled:habitatRow crop 1.315326 1.016108 19.688182 1.294
## habitatOrchard:mint_prior_scaled -4.492488 2.470465 40.881204 -1.818
## habitatGrassland:mint_prior_scaled -0.572247 1.076563 18.126738 -0.532
## habitatRow crop:mint_prior_scaled -2.039762 1.237614 16.052456 -1.648
## maxt_prior_scaled:juliandate_scaled -0.002027 0.330463 23.628668 -0.006
## Pr(>|t|)
## (Intercept) 5.09e-05 ***
## maxt_prior_scaled 0.6178
## habitatOrchard 0.8281
## habitatGrassland 0.7823
## habitatRow crop 0.2343
## mint_prior_scaled 0.2935
## age_scaled 0.5762
## juliandate_scaled 0.8171
## year_fct2023 0.9105
## maxt_prior_scaled:habitatOrchard 0.2180
## maxt_prior_scaled:habitatGrassland 0.6059
## maxt_prior_scaled:habitatRow crop 0.2105
## habitatOrchard:mint_prior_scaled 0.0763 .
## habitatGrassland:mint_prior_scaled 0.6015
## habitatRow crop:mint_prior_scaled 0.1188
## maxt_prior_scaled:juliandate_scaled 0.9952
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 16 > 12.
## Use print(x, correlation=TRUE) or
## vcov(x) if you need it
```

```
anova(s2_lintemp,prior_model)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>% ...
```

```
## Models:
```

```
## prior_model: sqrt(cort_s2) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat + age_scaled +
```

```
## s2_lintemp: sqrt(cort_s2) ~ maxt_prior_scaled * habitat + mint_prior_scaled * habitat + age_scaled +
```

```
## npar AIC BIC logLik deviance Chisq Df Pr(>Chisq)
```

```
## prior_model 17 350.17 391.89 -158.08 316.17
```

```
## s2_lintemp 18 352.16 396.33 -158.08 316.15 0.0101 1 0.9198
```

Not significant, does not improve fit, does not change parsimony

```
(tress2_priordayt_trendmax <- emtrends(prior_model, specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
    gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.455	0.914	28	-0.497	0.623
Orchard	-3.050	1.808	34	-1.687	0.101
Grassland	0.112	0.563	24	0.198	0.845
Row crop	0.853	0.474	21	1.798	0.087

```
(tress2_priordayt_trendmax <- emtrends(s2_lintemp, specs = ~ habitat, var = c("maxt_prior_scaled")) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "maxt_prior_scaled.trend", Df = "df", `T-ratio` = "t.ratio",
    gt())
```

Habitat	Max temp trend	SE	Df	T-ratio	P
Forest	-0.468	0.966	27	-0.484	0.632
Orchard	-3.040	1.886	31	-1.612	0.117
Grassland	0.111	0.587	25	0.190	0.851
Row crop	0.848	0.504	20	1.682	0.108

No change in trends or significance

```
samp_s2_priordayt_tres <- s2_lintemp@frame %>% group_by(habitat) %>% summarize(count = n())
# samp_s2_priordayt_tres %>% gt() %>% gtsave("figures/ss_tres_s2_priordayt_.html")
```

```
dat_text_s2_priordayt_tres <- data.frame(
  label = paste("N =", samp_s2_priordayt_tres$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)
```

```
data_s2_priordayt_tres = dplyr::filter(g, Species == "TRES", !is.na(maxt_prior), !is.na(mint_prior)) %>%
  mutate(across(c(gweight, maxt_prior, mint_prior, juliandate, brood_size, age),
    ~ scale(.x)[, 1],
    .names = "{.col}_scaled"),
  maxt_prior_scaled_sq = maxt_prior_scaled * maxt_prior_scaled)
```

```
mean_temp_s2_priordayt_tres <- mean(data_s2_priordayt_tres %>% pull(maxt_prior))
sd_temp_s2_priordayt_tres <- sd(data_s2_priordayt_tres %>% pull(maxt_prior))
```

```
temp_trans_s2_priordayt_tres <- trans_new("temp_trans_s2_priordayt_tres",
  transform = function(x){(x * sd_temp_s2_priordayt_tres) + mean_temp_s2_priordayt_tres},
  inverse = function(x){x})
```

```

(fig_priordayt_tres_s2 <- ggpredict(prior_model, terms = c("maxt_prior_scaled [all]", "habitat"), bias_cor:
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_priordayt_tres,
    breaks = c((25-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres,
      (35-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres,
      (45-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres,
      (55-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres,
      (65-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("dotted", "dotted", "dotted", "dashed")) +
  #ylim(-5,5) +
  geom_text(data = dat_text_s2_priordayt_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust =
  theme(legend.position = "none")
)

```

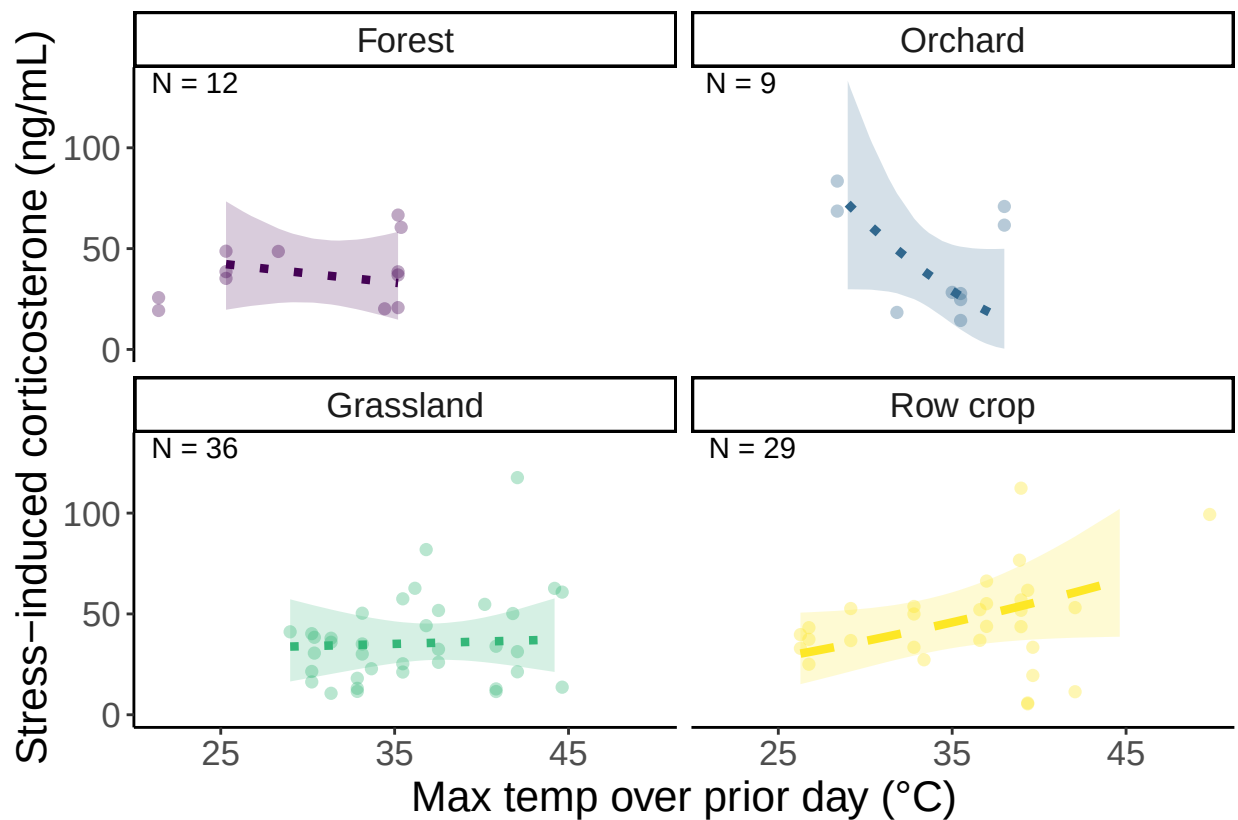
```

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
(fig_priordayt_tres_s2 <- ggpredict(s2_lintemp, terms = c("maxt_prior_scaled [all]", "habitat"), bias_corr = TRUE) +
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (°C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_priordayt_tres,
    breaks = c((25 - mean_temp_s2_priordayt_tres) / sd_temp_s2_priordayt_tres,
      (35 - mean_temp_s2_priordayt_tres) / sd_temp_s2_priordayt_tres,
      (45 - mean_temp_s2_priordayt_tres) / sd_temp_s2_priordayt_tres,
      (55 - mean_temp_s2_priordayt_tres) / sd_temp_s2_priordayt_tres,
      (65 - mean_temp_s2_priordayt_tres) / sd_temp_s2_priordayt_tres),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE),
    #   (55 - mean(g$meanmaxtempI, na.rm = TRUE)) / sd(g$meanmaxtempI, na.rm = TRUE)),
    # limits = c(18, 5)
  ) +
  scale_linetype_manual(values = c("dotted", "dotted", "dotted", "dotted")) +
  #ylim(-5, 5) +
  geom_text(data = dat_text_s2_priordayt_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust =
```

```

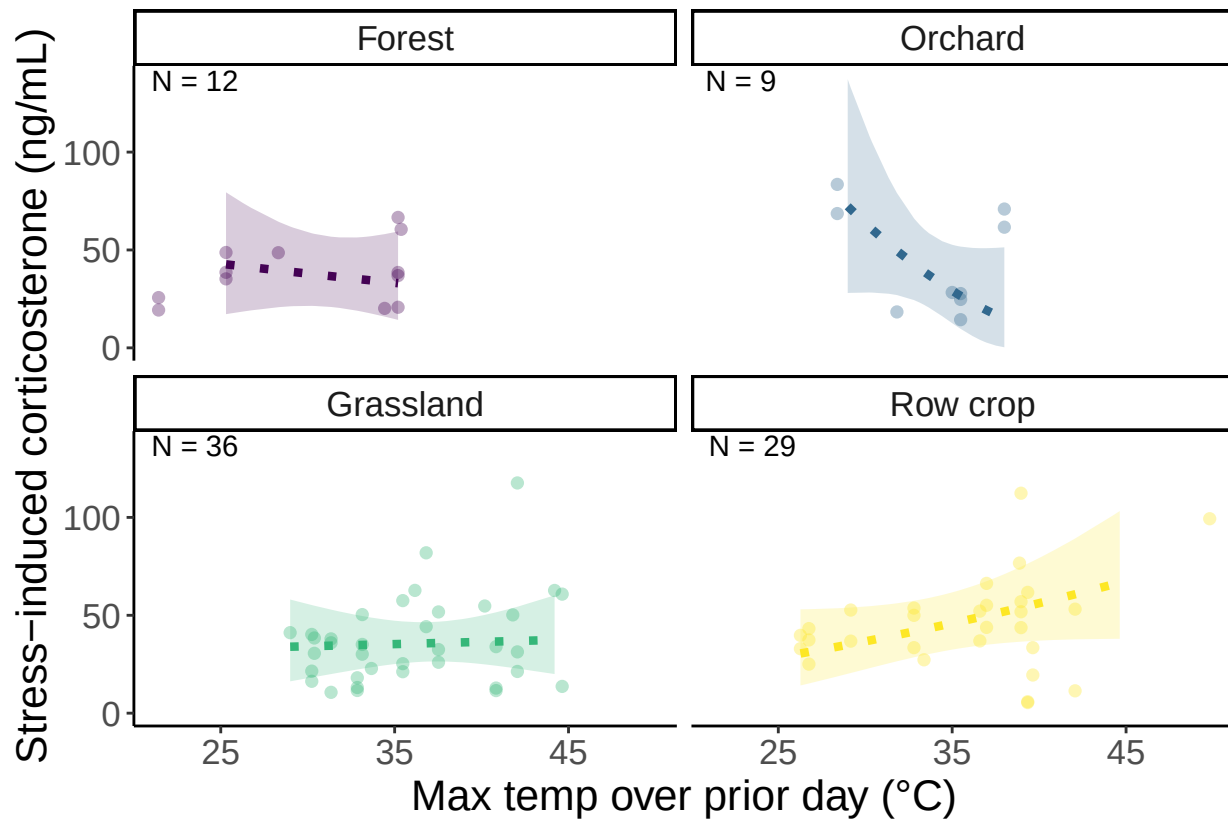
theme(legend.position = "none")
)

## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(fig_priordayt_tres_s2 <- ggpredict(s2_lintemp, terms = c("maxt_prior_scaled [all]", "juliandate_scaled")
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Max temp over prior day (\u00b0C)") +
  ylab("Stress-induced corticosterone (ng/mL)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_s2_priordayt_tres,
    breaks = c((25-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres,
      (35-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres,

```

```

      (45-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres,
      (55-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres,
      (65-mean_temp_s2_priordayt_tres)/sd_temp_s2_priordayt_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) # +
  # scale_linetype_manual(values = c("dotted","dotted","dotted","dashed")) +
  # ylim(-5,5) +
  # geom_text(data = dat_text_s2_priordayt_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust
  # theme(legend.position = "none")
)

```

```

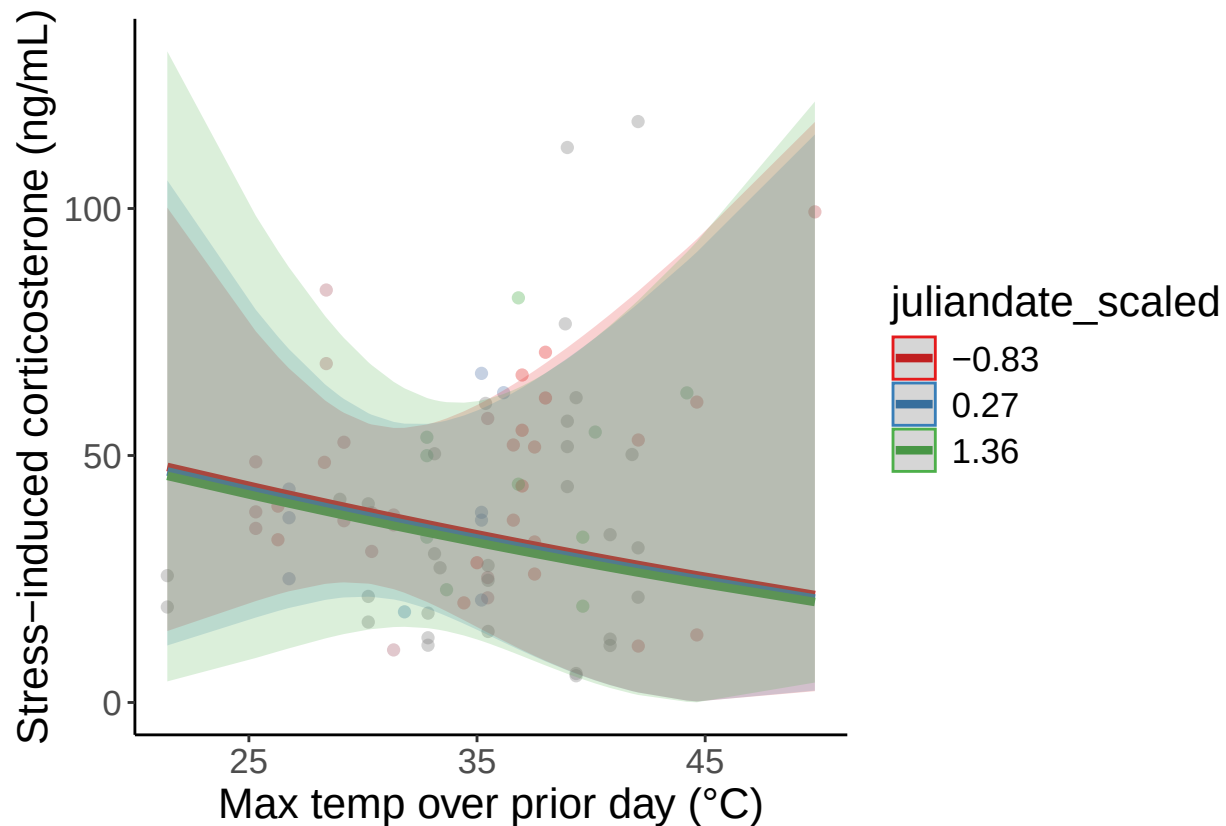
## Model has sqrt-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.

```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```



Interaction is not significant

Provis

```
prior_model <- glmmTMB(valid_detections ~ mean_temp_scaled * habitat + poly(mean_temp_scaled,2) + julian
  ziformula = ~ 1,
  family = nbinom2(),
  data = dplyr::filter(p,!is.na(mean_temp),
    !is.na(julian_date),
    !is.na(mean_nestling_age),
    !is.na(tod_h),
    !is.na(site),
    !is.na(attempt_id),
    !is.na(year),
    species == "WEBL"# ,
    # tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
    # tod_end <= hms::as.hms('17:00:00',tz = "America/Los_Angeles"),
    # tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
  ) %>%
  mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled")))
```

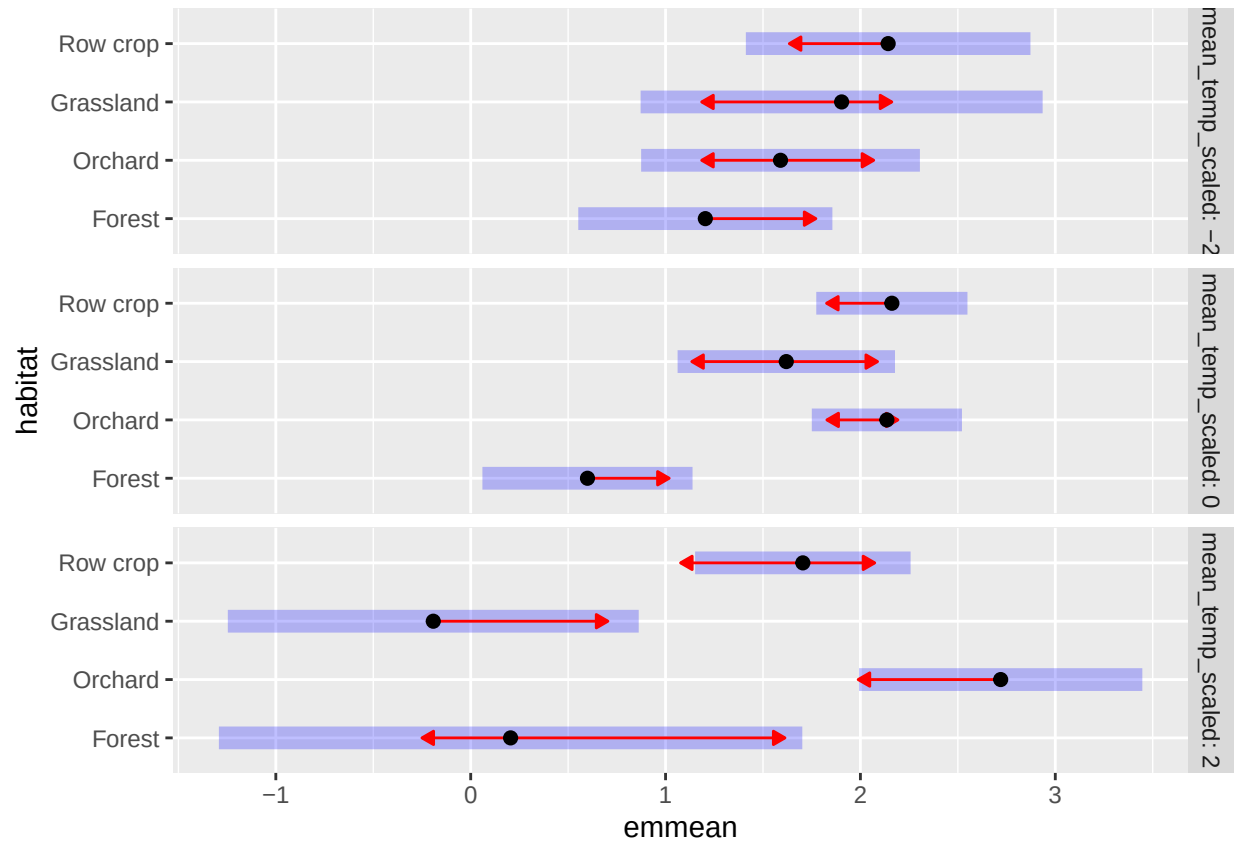
WEBL

dropping columns from rank-deficient conditional model: poly(mean_temp_scaled, 2)1

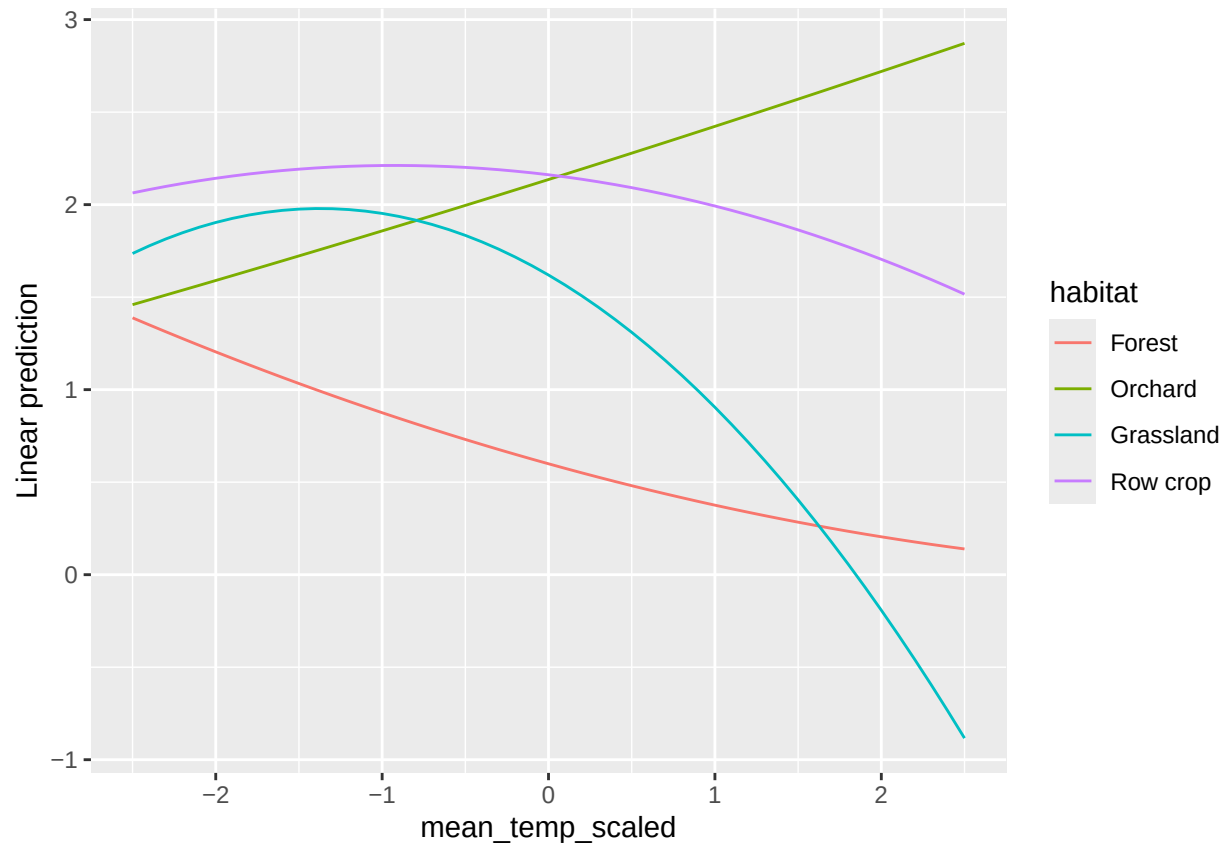
```
m <- glmmTMB(valid_detections ~ poly(mean_temp_scaled,2) * habitat + mean_temp_scaled * julian_date_scaled
  ziformula = ~ 1,
  family = nbinom2(),
  data = dplyr::filter(p,!is.na(mean_temp),
    !is.na(julian_date),
    !is.na(mean_nestling_age),
    !is.na(tod_h),
    !is.na(site),
    !is.na(attempt_id),
    !is.na(year),
    species == "WEBL"# ,
    # tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
    # tod_end <= hms::as.hms('17:00:00',tz = "America/Los_Angeles"),
    # tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
  ) %>%
  mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled")))
```

dropping columns from rank-deficient conditional model: mean_temp_scaled

```
emmeans(m,specs = pairwise ~ habitat,by = c("mean_temp_scaled"), at = list(mean_temp_scaled = c(-2,0,2)))
```



```
emmip(m, formula = habitat ~ mean_temp_scaled, at = list(mean_temp_scaled = seq(from = -2.5, to = 2.5, by = 2)))
```



```
m_linint <- glmmTMB(valid_detections ~ mean_temp_scaled * habitat + poly(mean_temp_scaled,2) + mean_temp_scaled,
  ziformula = ~ 1,
  family = nbinom2(),
  data = dplyr::filter(p,!is.na(mean_temp),
    !is.na(julian_date),
    !is.na(mean_nestling_age),
    !is.na(tod_h),
    !is.na(site),
    !is.na(attempt_id),
    !is.na(year),
    species == "WEBL"# ,
    # tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
    # tod_end <= hms::as.hms('17:00:00',tz = "America/Los_Angeles"),
    # tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
  ) %>%
  mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"))))

## dropping columns from rank-deficient conditional model: poly(mean_temp_scaled, 2)1
m_noint <- glmmTMB(valid_detections ~ poly(mean_temp_scaled,2) + habitat + mean_temp_scaled * julian_date,
  ziformula = ~ 1,
  family = nbinom2(),
  data = dplyr::filter(p,!is.na(mean_temp),
    !is.na(julian_date),
    !is.na(mean_nestling_age),
```

```

!is.na(tod_h),
!is.na(site),
!is.na(attempt_id),
!is.na(year),
species == "WEBL"# ,
# tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
# tod_end <= hms::as.hms('17:00:00', tz = "America/Los_Angeles")
# tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
) %>%
mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
~ scale(.x)[,1],
.names = "{.col}_scaled")))

## dropping columns from rank-deficient conditional model: mean_temp_scaled
c <- anova(m,m_linint,m_noint) %>% tibble() %>% mutate(Model = c("No temp * LC interaction","linear temp
(int_tab_provis_webl <- c %>%
as.tibble() %>%
mutate(across(where(is.numeric),~round(.x,digits = 4)),
P = `Pr(>Chisq)` ) %>%
dplyr::select(Model,AIC,Chisq,P) %>%
mutate(across(c(AIC,Chisq), ~ round(.x, digits = 2)),
P = if_else(P < 0.001,"<0.001",as.character(P))) %>% gt())

```

Model	AIC	Chisq	P
No temp * LC interaction	1880.31	NA	NA
linear temp * LC interaction	1858.93	27.38	<0.001
quadratic temp * LC interaction	1862.25	2.68	0.4431

```

# gtsave(int_tab_provis_webl,"figures/int_tab_provis_webl.html")

summary(m_linint)

## Family: nbinom2 ( log )
## Formula:
## valid_detections ~ mean_temp_scaled * habitat + poly(mean_temp_scaled,
## 2) + mean_temp_scaled * julian_date_scaled + poly(tod_h_scaled,
## 2) + mean_nestling_age_scaled + (1 | attempt_id) + year_fct
## Zero inflation: ~1
## Data:
## dplyr::filter(p, !is.na(mean_temp), !is.na(julian_date), !is.na(mean_nestling_age),
## !is.na(tod_h), !is.na(site), !is.na(attempt_id), !is.na(year),
## species == "WEBL") %>% mutate(across(c(tod_h, mean_temp,
## julian_date, mean_nestling_age), ~scale(.x)[, 1], .names = "{.col}_scaled"))
##
## AIC BIC logLik deviance df.resid
## 1858.9 1930.4 -910.5 1820.9 298
##
## Random effects:
##
## Conditional model:

```

```

## Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.1371  0.3703
## Number of obs: 317, groups: attempt_id, 29
##
## Dispersion parameter for nbinom2 family (): 4.3
##
## Conditional model:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.64322    0.41710   1.542 0.123049
## mean_temp_scaled -0.34385    0.13361  -2.574 0.010066 *
## habitatOrchard    1.58431    0.28147   5.629 1.82e-08 ***
## habitatGrassland  0.87435    0.33760   2.590 0.009600 **
## habitatRow crop   1.54070    0.29246   5.268 1.38e-07 ***
## poly(mean_temp_scaled, 2)1      NA         NA      NA      NA
## poly(mean_temp_scaled, 2)2    -0.94560    0.91585  -1.032 0.301845
## julian_date_scaled -0.01230    0.09889  -0.124 0.901032
## poly(tod_h_scaled, 2)1    -0.80704    0.72094  -1.119 0.262955
## poly(tod_h_scaled, 2)2     1.60436    1.19560   1.342 0.179633
## mean_nestling_age_scaled  0.16730    0.05725   2.923 0.003472 **
## year_fct2022         0.12313    0.33844   0.364 0.716002
## year_fct2023        -0.16125    0.33236  -0.485 0.627545
## mean_temp_scaled:habitatOrchard  0.61479    0.16400   3.749 0.000178 ***
## mean_temp_scaled:habitatGrassland -0.16461    0.19031  -0.865 0.387067
## mean_temp_scaled:habitatRow crop  0.24530    0.16900   1.451 0.146647
## mean_temp_scaled:julian_date_scaled 0.13959    0.05707   2.446 0.014453 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.8859    0.4866  -5.931 3.01e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

check_collinearity(m_noint)

```

```

## Model matrix is rank deficient. VIFs may not be sensible.
## # Check for Multicollinearity
##
## * conditional component:
##
## Low Correlation
##
##              Term  VIF   VIF 95% CI Increased SE Tolerance
## poly(mean_temp_scaled, 2) 3.16 [2.67, 3.79]      1.78      0.32
## habitat 1.61 [1.42, 1.89]      1.27      0.62
## julian_date_scaled 1.76 [1.54, 2.07]      1.33      0.57
## poly(tod_h_scaled, 2) 2.42 [2.07, 2.88]      1.55      0.41
## mean_nestling_age_scaled 1.52 [1.35, 1.79]      1.23      0.66
## year_fct 1.39 [1.24, 1.63]      1.18      0.72
## mean_temp_scaled:julian_date_scaled 1.55 [1.37, 1.81]      1.24      0.65
## Tolerance 95% CI
## [0.26, 0.37]
## [0.53, 0.70]

```

```
##      [0.48, 0.65]
##      [0.35, 0.48]
##      [0.56, 0.74]
##      [0.61, 0.80]
##      [0.55, 0.73]

provis_webl <- m_linint

summary(m_linint)

## Family: nbinom2 ( log )
## Formula:
## valid_detections ~ mean_temp_scaled * habitat + poly(mean_temp_scaled,
##      2) + mean_temp_scaled * julian_date_scaled + poly(tod_h_scaled,
##      2) + mean_nestling_age_scaled + (1 | attempt_id) + year_fct
## Zero inflation: ~1
## Data:
## dplyr::filter(p, !is.na(mean_temp), !is.na(julian_date), !is.na(mean_nestling_age),
##      !is.na(tod_h), !is.na(site), !is.na(attempt_id), !is.na(year),
##      species == "WEBL") %>% mutate(across(c(tod_h, mean_temp,
##      julian_date, mean_nestling_age), ~scale(.x)[, 1], .names = "{.col}_scaled"))
##
##      AIC      BIC   logLik deviance df.resid
##  1858.9   1930.4   -910.5   1820.9      298
##
## Random effects:
##
## Conditional model:
##      Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.1371   0.3703
## Number of obs: 317, groups: attempt_id, 29
##
## Dispersion parameter for nbinom2 family (): 4.3
##
## Conditional model:
##
##      Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.64322    0.41710   1.542 0.123049
## mean_temp_scaled -0.34385    0.13361  -2.574 0.010066 *
## habitatOrchard    1.58431    0.28147   5.629 1.82e-08 ***
## habitatGrassland  0.87435    0.33760   2.590 0.009600 **
## habitatRow crop    1.54070    0.29246   5.268 1.38e-07 ***
## poly(mean_temp_scaled, 2)1      NA         NA      NA      NA
## poly(mean_temp_scaled, 2)2    -0.94560    0.91585  -1.032 0.301845
## julian_date_scaled -0.01230    0.09889  -0.124 0.901032
## poly(tod_h_scaled, 2)1    -0.80704    0.72094  -1.119 0.262955
## poly(tod_h_scaled, 2)2     1.60436    1.19560   1.342 0.179633
## mean_nestling_age_scaled  0.16730    0.05725   2.923 0.003472 **
## year_fct2022      0.12313    0.33844   0.364 0.716002
## year_fct2023    -0.16125    0.33236  -0.485 0.627545
## mean_temp_scaled:habitatOrchard  0.61479    0.16400   3.749 0.000178 ***
## mean_temp_scaled:habitatGrassland -0.16461    0.19031  -0.865 0.387067
## mean_temp_scaled:habitatRow crop  0.24530    0.16900   1.451 0.146647
## mean_temp_scaled:julian_date_scaled 0.13959    0.05707   2.446 0.014453 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Zero-inflation model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.8859      0.4866  -5.931 3.01e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(m_linint,prior_model)

## Data: dplyr::filter(p, !is.na(mean_temp), !is.na(julian_date), !is.na(mean_nestling_age), ...
## Models:
## prior_model: valid_detections ~ mean_temp_scaled * habitat + poly(mean_temp_scaled, , zi=~1, disp=~1
## prior_model:      2) + julian_date_scaled + poly(tod_h_scaled, 2) + mean_nestling_age_scaled + , zi=~1
## prior_model:      (1 | attempt_id) + year_fct, zi=~1, disp=~1
## m_linint: valid_detections ~ mean_temp_scaled * habitat + poly(mean_temp_scaled, , zi=~1, disp=~1
## m_linint:      2) + mean_temp_scaled * julian_date_scaled + poly(tod_h_scaled, , zi=~1, disp=~1
## m_linint:      2) + mean_nestling_age_scaled + (1 | attempt_id) + year_fct, zi=~1, disp=~1
##           Df      AIC      BIC logLik deviance Chisq Chi Df Pr(>Chisq)
## prior_model 18 1862.8 1930.5 -913.42   1826.8
## m_linint    19 1858.9 1930.3 -910.47   1820.9 5.8986      1   0.01515 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interaction is significant, improves fit, does not change parsimony

```
(webplprovistrend <- emtrends(prior_model,specs = ~ habitat, var = c("mean_temp_scaled"),max.degree = 1) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "mean_temp_scaled.trend",Df = "df", `Z-ratio` = "z.ratio",
    #group_by(degree) %>%
  gt())
```

Habitat	Max temp trend	SE	Df	Z-ratio	P
Forest	-0.403	0.139	Inf	-2.901	0.004
Orchard	0.222	0.119	Inf	1.868	0.062
Grassland	-0.528	0.145	Inf	-3.634	<0.001
Row crop	-0.064	0.100	Inf	-0.640	0.522

```
(webplprovistrend <- emtrends(m_linint,specs = ~ habitat, var = c("mean_temp_scaled"),max.degree = 1) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000,"<0.001",as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "mean_temp_scaled.trend",Df = "df", `Z-ratio` = "z.ratio",
    #group_by(degree) %>%
  gt())
```

Habitat	Max temp trend	SE	Df	Z-ratio	P
Forest	-0.356	0.137	Inf	-2.601	0.009
Orchard	0.258	0.117	Inf	2.215	0.027
Grassland	-0.521	0.145	Inf	-3.591	<0.001

Row crop	-0.111	0.101	Inf	-1.105	0.269
----------	--------	-------	-----	--------	-------

Same trends, but makes orchard trend slightly more significant

```
samp <- m_linint$frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_webl_provis.html")

dat_text_webl <- data.frame(
  label = paste("N =", samp$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)

data_webl = dplyr::filter(p, !is.na(mean_temp),
  !is.na(julian_date),
  !is.na(mean_nestling_age),
  !is.na(tod_h),
  !is.na(site),
  !is.na(attempt_id),
  !is.na(year),
  species == "WEBL" # ,
  # tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
  # tod_end <= hms::as.hms('17:00:00', tz = "America/Los_Angeles"),
  # tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
) %>%
  mutate(across(c(tod_h, mean_temp, julian_date, mean_nestling_age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"))

mean_temp_webl <- mean(data_webl %>% pull(mean_temp))
sd_temp_webl <- sd(data_webl %>% pull(mean_temp))

temp_trans_webl <- trans_new("temp_trans_webl",
  transform = function(x){(x * sd_temp_webl) + mean_temp_webl},
  inverse = function(x){x})

(fig5_webl <- predict_response(prior_model, terms = c("mean_temp_scaled [all]", "habitat"), bias_correction = FALSE) %>%
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean temperature (\u00b0C)") +
  ylab("Provisioning (count/hour)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = element_blank()) +
  scale_x_continuous(trans = temp_trans_webl,
    breaks = c((20-mean_temp_webl)/sd_temp_webl,
      (25-mean_temp_webl)/sd_temp_webl,
      (30-mean_temp_webl)/sd_temp_webl,
```

```

      (35-mean_temp_webl)/sd_temp_webl,
      (40-mean_temp_webl)/sd_temp_webl),
  # breaks = c(20,30,40,50),
  # labels = c("20", "30", "40", "50"),
  # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  #           (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # limits = c(18,5)
) +
scale_linetype_manual(values = c("Forest" = "solid","Orchard" = "dashed","Grassland" = "solid","Row
#ylim(-5,5) +
geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust =
theme(legend.position = "none")
)

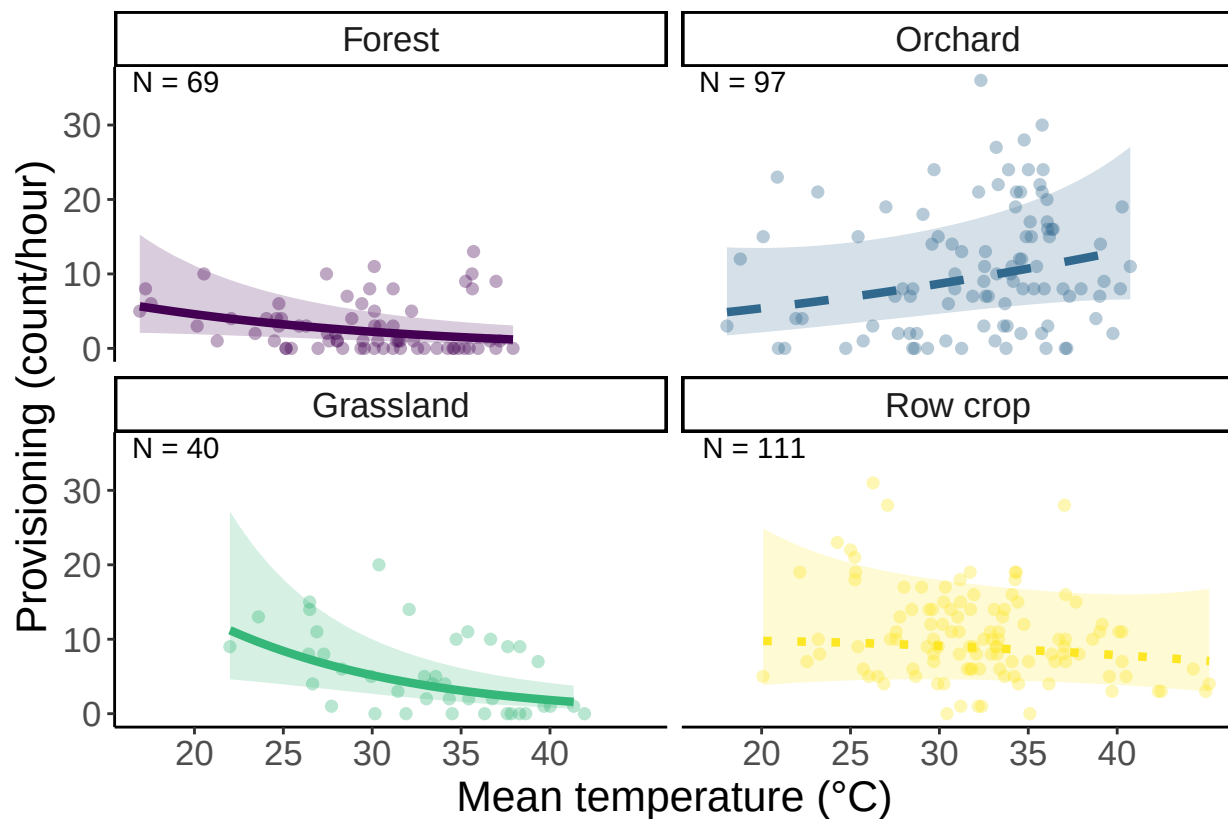
```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```

(fig5_webl <- predict_response(m_linint,terms = c("mean_temp_scaled [all]","habitat"),bias_correction =
plot(line_size = 1.5,alpha = .2,show_data = TRUE,limit_range = TRUE) +
aes(linetype = .data[["group"]]) +
theme_classic() +
facet_wrap(~ group, ncol = 2) +
xlab("Mean temperature (\u00b0C)") +
ylab("Provisioning (count/hour)") +

```

```

scale_fill_viridis(discrete = TRUE) +
scale_color_viridis(discrete = TRUE) +
theme(text = element_text(size = 16)) +
labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_webl,
  breaks = c((20-mean_temp_webl)/sd_temp_webl,
    (25-mean_temp_webl)/sd_temp_webl,
    (30-mean_temp_webl)/sd_temp_webl,
    (35-mean_temp_webl)/sd_temp_webl,
    (40-mean_temp_webl)/sd_temp_webl),
  # breaks = c(20,30,40,50),
  # labels = c("20", "30", "40", "50"),
  # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # limits = c(18,5)
) +
scale_linetype_manual(values = c("Forest" = "solid","Orchard" = "solid","Grassland" = "solid","Row
#ylim(-5,5) +
geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust =
theme(legend.position = "none")
)

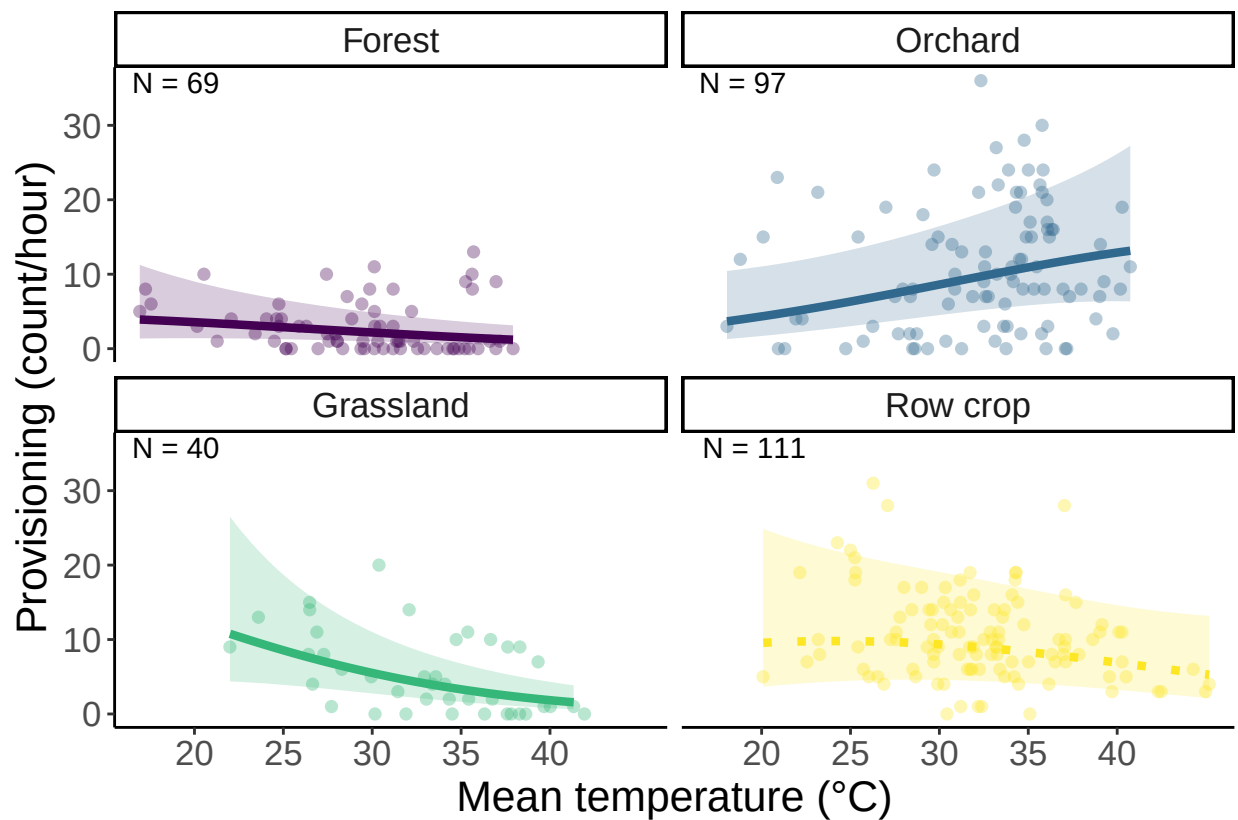
```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
##   random variation to the location of data points and avoid overplotting.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

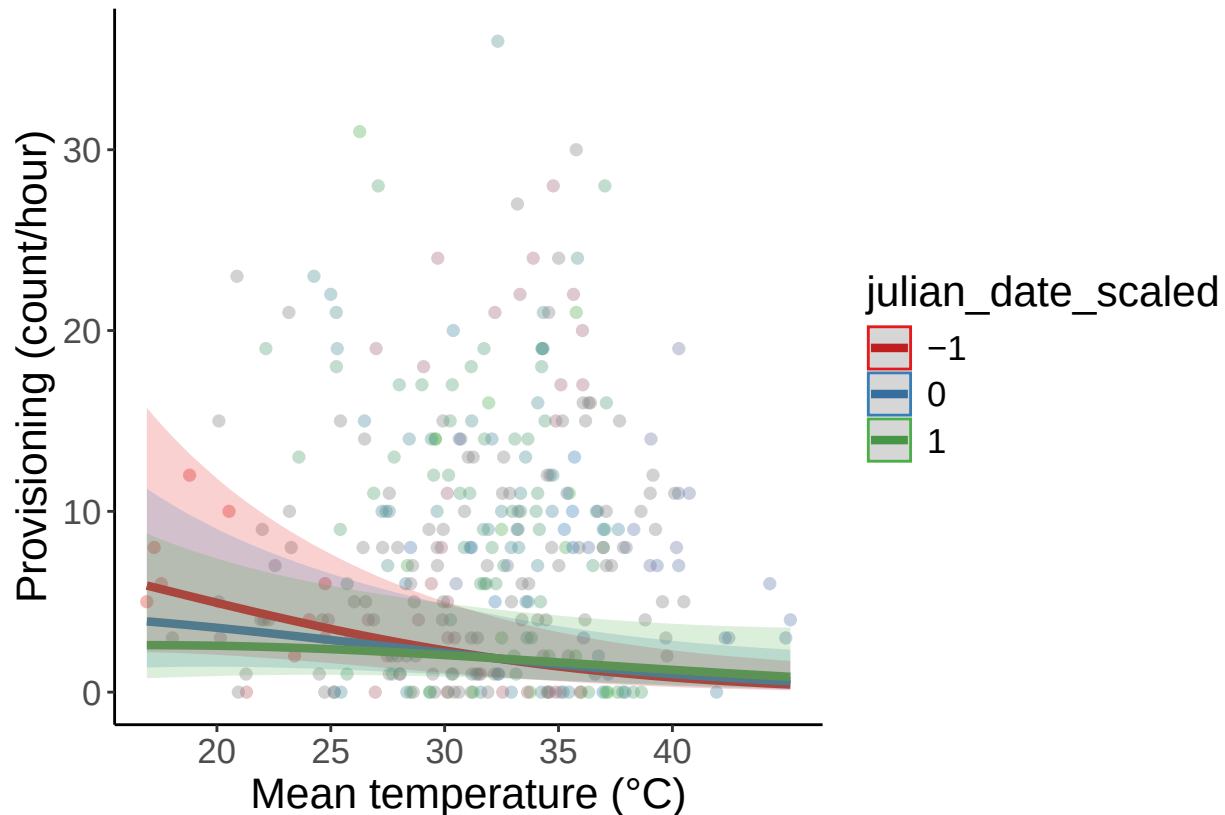
```



```
(fig5_webl <- predict_response(m_linint, terms = c("mean_temp_scaled [all]", "julian_date_scaled"), bias_c
plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
# aes(linetype = .data[["group"]]) +
theme_classic() +
# facet_wrap(~ group, ncol = 2) +
xlab("Mean temperature (\u00b0C)") +
ylab("Provisioning (count/hour)") +
# scale_fill_viridis(discrete = TRUE) +
# scale_color_viridis(discrete = TRUE) +
theme(text = element_text(size = 16)) +
labs(title = element_blank()) +
scale_x_continuous(trans = temp_trans_webl,
breaks = c((20-mean_temp_webl)/sd_temp_webl,
(25-mean_temp_webl)/sd_temp_webl,
(30-mean_temp_webl)/sd_temp_webl,
(35-mean_temp_webl)/sd_temp_webl,
(40-mean_temp_webl)/sd_temp_webl),
# breaks = c(20,30,40,50),
# labels = c("20", "30", "40", "50"),
# limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
# (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
# limits = c(18,5)
) # +
# scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "dashed", "Grassland" = "solid", "R
# #ylim(-5,5) +
# geom_text(data = dat_text_webl, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust
```

```
# theme(legend.position = "none")
)
```

```
## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.
```



Higher mean temp is worse for provisioning in the early season relative to late season.

```
prior_model <- glmmTMB(valid_detections ~ poly(mean_temp_scaled,2) * habitat + julian_date_scaled + poly(
  ziformula = ~ 1,
  family = nbinom2(),
  data = dplyr::filter(p,!is.na(mean_temp),
    !is.na(julian_date),
    !is.na(mean_nestling_age),
    !is.na(tod_h),
    !is.na(site),
    !is.na(attempt_id),
    !is.na(year),
    valid_detections < 40,
    species == "TRES"# ,
    # tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
    # tod_end <= hms::as.hms('17:00:00',tz = "America/Los_Angeles"),
    # tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
  ) %>%
  mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
```

```

      ~ scale(.x)[,1],
      .names = "{.col}_scaled"))))

m <- glmmTMB(valid_detections ~ poly(mean_temp_scaled,2) * habitat + mean_temp_scaled * julian_date_scaled,
  ziformula = ~ 1,
  family = nbinom2(),
  data = dplyr::filter(p,!is.na(mean_temp),
    !is.na(julian_date),
    !is.na(mean_nestling_age),
    !is.na(tod_h),
    !is.na(site),
    !is.na(attempt_id),
    !is.na(year),
    valid_detections < 40,
    species == "TRES"# ,
    # tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
    # tod_end <= hms::as.hms('17:00:00',tz = "America/Los_Angeles"),
    # tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
  ) %>%
  mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"))))

```

TRES

dropping columns from rank-deficient conditional model: mean_temp_scaled

```

m_linint <- glmmTMB(valid_detections ~ mean_temp_scaled * habitat + poly(mean_temp_scaled,2) + mean_temp_scaled,
  ziformula = ~ 1,
  family = nbinom2(),
  data = dplyr::filter(p,!is.na(mean_temp),
    !is.na(julian_date),
    !is.na(mean_nestling_age),
    !is.na(tod_h),
    !is.na(site),
    !is.na(attempt_id),
    !is.na(year),
    valid_detections < 40,
    species == "TRES"# ,
    # tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
    # tod_end <= hms::as.hms('17:00:00',tz = "America/Los_Angeles"),
    # tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
  ) %>%
  mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
    ~ scale(.x)[,1],
    .names = "{.col}_scaled"))))

```

dropping columns from rank-deficient conditional model: poly(mean_temp_scaled, 2)1

```

m_noint <- glmmTMB(valid_detections ~ poly(mean_temp_scaled,2) + habitat + mean_temp_scaled * julian_date_scaled,
  ziformula = ~ 1,
  family = nbinom2(),
  data = dplyr::filter(p,!is.na(mean_temp),
    !is.na(julian_date),
    !is.na(mean_nestling_age),

```

```

!is.na(tod_h),
!is.na(site),
!is.na(attempt_id),
!is.na(year),
valid_detections < 40,
species == "TRES"# ,
# tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
# tod_end <= hms::as.hms('17:00:00',tz = "America/Los_Angeles")
# tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
) %>%
mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
~ scale(.x)[,1],
.names = "{.col}_scaled")))

## dropping columns from rank-deficient conditional model: mean_temp_scaled
c <- anova(m,m_linint,m_noint) %>% tibble() %>% mutate(Model = c("No temp * LC interaction","Linear temp
(int_tab_provis_tres <- c %>%
as.tibble() %>%
mutate(across(where(is.numeric),~round(.x,digits = 4)),
P = `Pr(>Chisq)` ) %>%
dplyr::select(Model,AIC,Chisq,P) %>%
mutate(across(c(AIC,Chisq), ~ round(.x, digits = 2)),
P = if_else(P < 0.001,"<0.001",as.character(P))) %>% gt())

```

Model	AIC	Chisq	P
No temp * LC interaction	1362.39	NA	NA
Linear temp * LC interaction	1367.50	0.88	0.8296
Squared temp * LC interaction	1361.07	12.43	0.006

```

# gtsave(int_tab_provis_tres,"figures/int_tab_provis_tres.html")

summary(m)

## Family: nbinom2 ( log )
## Formula:
## valid_detections ~ poly(mean_temp_scaled, 2) * habitat + mean_temp_scaled *
## julian_date_scaled + poly(tod_h_scaled, 2) + mean_nestling_age_scaled +
## (1 | attempt_id) + year_fct
## Zero inflation: ~1
## Data:
## dplyr::filter(p, !is.na(mean_temp), !is.na(julian_date), !is.na(mean_nestling_age),
## !is.na(tod_h), !is.na(site), !is.na(attempt_id), !is.na(year),
## valid_detections < 40, species == "TRES") %>% mutate(across(c(tod_h,
## mean_temp, julian_date, mean_nestling_age), ~scale(.x)[,
## 1], .names = "{.col}_scaled"))
##
## AIC BIC logLik deviance df.resid
## 1361.1 1435.3 -658.5 1317.1 194
##
## Random effects:

```

```

##
## Conditional model:
##   Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.03126 0.1768
## Number of obs: 216, groups: attempt_id, 23
##
## Dispersion parameter for nbinom2 family (): 3.85
##
## Conditional model:
##                                     Estimate Std. Error z value
## (Intercept)                       -0.70414    0.72850  -0.967
## poly(mean_temp_scaled, 2)1         -43.38985   13.98517  -3.103
## poly(mean_temp_scaled, 2)2        -28.05817    8.08114  -3.472
## habitatOrchard                     3.35871    2.69749   1.245
## habitatGrassland                   3.45503    0.73791   4.682
## habitatRow crop                    3.33027    0.72829   4.573
## mean_temp_scaled                   NA          NA      NA
## julian_date_scaled                 -0.08337    0.09882  -0.844
## poly(tod_h_scaled, 2)1              1.17103    0.86096   1.360
## poly(tod_h_scaled, 2)2              1.29523    1.26673   1.022
## mean_nestling_age_scaled            -0.03208    0.06136  -0.523
## year_fct2022                       -0.09700    0.22055  -0.440
## year_fct2023                       0.18150    0.22146   0.820
## poly(mean_temp_scaled, 2)1:habitatOrchard 76.89333   45.33198   1.696
## poly(mean_temp_scaled, 2)2:habitatOrchard 41.76934   19.22585   2.173
## poly(mean_temp_scaled, 2)1:habitatGrassland 46.22564   14.26549   3.240
## poly(mean_temp_scaled, 2)2:habitatGrassland 27.33728    8.10412   3.373
## poly(mean_temp_scaled, 2)1:habitatRow crop 45.33226   14.23214   3.185
## poly(mean_temp_scaled, 2)2:habitatRow crop 25.55780    8.28334   3.085
## mean_temp_scaled:julian_date_scaled -0.04591    0.07306  -0.628
##                                     Pr(>|z|)
## (Intercept)                       0.333758
## poly(mean_temp_scaled, 2)1          0.001919 **
## poly(mean_temp_scaled, 2)2          0.000516 ***
## habitatOrchard                     0.213085
## habitatGrassland                   2.84e-06 ***
## habitatRow crop                    4.81e-06 ***
## mean_temp_scaled                   NA
## julian_date_scaled                 0.398826
## poly(tod_h_scaled, 2)1              0.173783
## poly(tod_h_scaled, 2)2              0.306547
## mean_nestling_age_scaled            0.601097
## year_fct2022                       0.660059
## year_fct2023                       0.412462
## poly(mean_temp_scaled, 2)1:habitatOrchard 0.089843 .
## poly(mean_temp_scaled, 2)2:habitatOrchard 0.029813 *
## poly(mean_temp_scaled, 2)1:habitatGrassland 0.001194 **
## poly(mean_temp_scaled, 2)2:habitatGrassland 0.000743 ***
## poly(mean_temp_scaled, 2)1:habitatRow crop 0.001447 **
## poly(mean_temp_scaled, 2)2:habitatRow crop 0.002032 **
## mean_temp_scaled:julian_date_scaled 0.529805
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Zero-inflation model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.3111      0.1795  -7.303 2.82e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

check_collinearity(m)

## Model matrix is rank deficient. VIFs may not be sensible.
## Model has interaction terms. VIFs might be inflated.
## Try to center the variables used for the interaction, or check
## multicollinearity among predictors of a model without interaction terms.

## # Check for Multicollinearity
##
## * conditional component:
##
## Low Correlation
##
##           Term VIF           VIF 95% CI Increased SE
##           julian_date_scaled 2.36 [ 1.98, 2.88] 1.54
##           poly(tod_h_scaled, 2) 4.34 [ 3.54, 5.38] 2.08
##           mean_nestling_age_scaled 1.41 [ 1.24, 1.69] 1.19
##           year_fct 3.44 [ 2.83, 4.24] 1.85
## mean_temp_scaled:julian_date_scaled 1.45 [ 1.28, 1.75] 1.21
## Tolerance Tolerance 95% CI
##           0.42 [0.35, 0.50]
##           0.23 [0.19, 0.28]
##           0.71 [0.59, 0.81]
##           0.29 [0.24, 0.35]
##           0.69 [0.57, 0.78]
##
## High Correlation
##
##           Term VIF           VIF 95% CI Increased SE
##           poly(mean_temp_scaled, 2) 5515.82 [ 4353.44, 6988.63] 74.27
##           habitat 1328.06 [ 1048.28, 1682.58] 36.44
##           poly(mean_temp_scaled, 2):habitat 4.85e+05 [3.83e+05, 6.14e+05] 696.24
## Tolerance Tolerance 95% CI
##           1.81e-04 [0.00, 0.00]
##           7.53e-04 [0.00, 0.00]
##           2.06e-06 [0.00, 0.00]

provis_tres <- m

summary(m)

## Family: nbinom2 ( log )
## Formula:
## valid_detections ~ poly(mean_temp_scaled, 2) * habitat + mean_temp_scaled *
##           julian_date_scaled + poly(tod_h_scaled, 2) + mean_nestling_age_scaled +
##           (1 | attempt_id) + year_fct
## Zero inflation: ~1
## Data:
## dplyr::filter(p, !is.na(mean_temp), !is.na(julian_date), !is.na(mean_nestling_age),
```

```

##      !is.na(tod_h), !is.na(site), !is.na(attempt_id), !is.na(year),
##      valid_detections < 40, species == "TRES") %>% mutate(across(c(tod_h,
##      mean_temp, julian_date, mean_nestling_age), ~scale(.x)[,
##      1], .names = "{.col}_scaled"))
##
##      AIC      BIC    logLik deviance df.resid
##    1361.1    1435.3   -658.5   1317.1      194
##
## Random effects:
##
## Conditional model:
##   Groups      Name      Variance Std.Dev.
## attempt_id (Intercept) 0.03126  0.1768
## Number of obs: 216, groups:  attempt_id, 23
##
## Dispersion parameter for nbinom2 family (): 3.85
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)      -0.70414    0.72850  -0.967
## poly(mean_temp_scaled, 2)1      -43.38985    13.98517  -3.103
## poly(mean_temp_scaled, 2)2      -28.05817     8.08114  -3.472
## habitatOrchard         3.35871     2.69749   1.245
## habitatGrassland        3.45503     0.73791   4.682
## habitatRow crop        3.33027     0.72829   4.573
## mean_temp_scaled              NA          NA      NA
## julian_date_scaled      -0.08337     0.09882  -0.844
## poly(tod_h_scaled, 2)1         1.17103     0.86096   1.360
## poly(tod_h_scaled, 2)2         1.29523     1.26673   1.022
## mean_nestling_age_scaled     -0.03208     0.06136  -0.523
## year_fct2022              -0.09700     0.22055  -0.440
## year_fct2023               0.18150     0.22146   0.820
## poly(mean_temp_scaled, 2)1:habitatOrchard  76.89333    45.33198   1.696
## poly(mean_temp_scaled, 2)2:habitatOrchard  41.76934    19.22585   2.173
## poly(mean_temp_scaled, 2)1:habitatGrassland 46.22564    14.26549   3.240
## poly(mean_temp_scaled, 2)2:habitatGrassland 27.33728     8.10412   3.373
## poly(mean_temp_scaled, 2)1:habitatRow crop 45.33226    14.23214   3.185
## poly(mean_temp_scaled, 2)2:habitatRow crop 25.55780     8.28334   3.085
## mean_temp_scaled:julian_date_scaled     -0.04591     0.07306  -0.628
##
##              Pr(>|z|)
## (Intercept)      0.333758
## poly(mean_temp_scaled, 2)1      0.001919 **
## poly(mean_temp_scaled, 2)2      0.000516 ***
## habitatOrchard      0.213085
## habitatGrassland    2.84e-06 ***
## habitatRow crop     4.81e-06 ***
## mean_temp_scaled              NA
## julian_date_scaled      0.398826
## poly(tod_h_scaled, 2)1      0.173783
## poly(tod_h_scaled, 2)2      0.306547
## mean_nestling_age_scaled    0.601097
## year_fct2022              0.660059
## year_fct2023              0.412462
## poly(mean_temp_scaled, 2)1:habitatOrchard 0.089843 .

```

```
## poly(mean_temp_scaled, 2)2:habitatOrchard 0.029813 *
## poly(mean_temp_scaled, 2)1:habitatGrassland 0.001194 **
## poly(mean_temp_scaled, 2)2:habitatGrassland 0.000743 ***
## poly(mean_temp_scaled, 2)1:habitatRow crop 0.001447 **
## poly(mean_temp_scaled, 2)2:habitatRow crop 0.002032 **
## mean_temp_scaled:julian_date_scaled 0.529805
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
## Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.3111 0.1795 -7.303 2.82e-13 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
anova(m,prior_model)

## Data: dplyr::filter(p, !is.na(mean_temp), !is.na(julian_date), !is.na(mean_nestling_age), ...
## Models:
## prior_model: valid_detections ~ poly(mean_temp_scaled, 2) * habitat + julian_date_scaled + , zi=~1, disp=
## prior_model: poly(tod_h_scaled, 2) + mean_nestling_age_scaled + (1 | attempt_id) + , zi=~1, disp=
## prior_model: year_fct, zi=~1, disp=~1
## m: valid_detections ~ poly(mean_temp_scaled, 2) * habitat + mean_temp_scaled * , zi=~1, disp=~1
## m: julian_date_scaled + poly(tod_h_scaled, 2) + mean_nestling_age_scaled + , zi=~1, disp=~1
## m: (1 | attempt_id) + year_fct, zi=~1, disp=~1
## Df AIC BIC logLik deviance Chisq Chi Df Pr(>Chisq)
## prior_model 21 1359.5 1430.3 -658.73 1317.5
## m 22 1361.1 1435.3 -658.54 1317.1 0.3927 1 0.5309

Interaction is not significant, does not improve fit, does not change parsimony

(tresprovistrend <- emtrends(prior_model, specs = ~ degree | habitat, var = "mean_temp_scaled", max.degree = 2) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "mean_temp_scaled.trend", Df = "df", `Z-ratio` = "z.ratio", `P` = "p.value") %>%
  group_by(Habitat) %>%
  gt())
```

degree	Max temp trend	SE	Df	Z-ratio	P
Forest					
linear	-2.645	0.870	Inf	-3.041	0.002
quadratic	-1.700	0.484	Inf	-3.514	<0.001
Orchard					
linear	1.936	2.973	Inf	0.651	0.515
quadratic	0.719	1.138	Inf	0.632	0.528
Grassland					
linear	0.233	0.168	Inf	1.389	0.165
quadratic	-0.052	0.086	Inf	-0.609	0.543

Row crop

linear	0.172	0.136	Inf	1.262	0.207
quadratic	-0.172	0.108	Inf	-1.593	0.111

```
(tresprovistrend <- emtrends(m, specs = ~ degree | habitat, var = "mean_temp_scaled", max.degree = 2) %>%
  mutate(across(where(is.numeric), ~ round(.x, digits = 3)),
    df = round(df),
    p.value = if_else(p.value == 0.000, "<0.001", as.character(p.value))) %>%
  rename(Habitat = "habitat", `Max temp trend` = "mean_temp_scaled.trend", Df = "df", `Z-ratio` = "z.ratio")
  group_by(Habitat) %>%
  gt())
```

	degree	Max temp trend	SE	Df	Z-ratio	P
Forest						
linear		-2.673	0.879	Inf	-3.040	0.002
quadratic		-1.680	0.484	Inf	-3.472	0.001
Orchard						
linear		2.145	2.819	Inf	0.761	0.447
quadratic		0.821	1.088	Inf	0.754	0.451
Grassland						
linear		0.201	0.175	Inf	1.150	0.25
quadratic		-0.043	0.086	Inf	-0.500	0.617
Row crop						
linear		0.158	0.137	Inf	1.153	0.249
quadratic		-0.150	0.113	Inf	-1.322	0.186

Same trends and significance

```
samp <- m$frame %>% group_by(habitat) %>% summarize(count = n())
# samp %>% gt() %>% gtsave("figures/ss_tres_provis.html")
```

```
dat_text_tres <- data.frame(
  label = paste("N =", samp$count),
  group = factor(c("Forest", "Orchard", "Grassland", "Row crop"))
)
```

```
data_tres = dplyr::filter(p, !is.na(mean_temp),
  !is.na(julian_date),
  !is.na(mean_nestling_age),
  !is.na(tod_h),
```

```

      !is.na(site),
      !is.na(attempt_id),
      !is.na(year),
      valid_detections < 40,
      species == "TRES"# ,
      # tod >= hms::as.hms('11:00:00', tz = "America/Los_Angeles"),
      # tod_end <= hms::as.hms('17:00:00',tz = "America/Los_Angeles"),
      # tod_end >= hms::as.hms('11:00:00', tz = "America/Los_Angeles")
    ) %>%
      mutate(across(c(tod_h,mean_temp,julian_date,mean_nestling_age),
        ~ scale(.x)[,1],
        .names = "{.col}_scaled"))

mean_temp_tres <- mean(data_tres %>% pull(mean_temp))
sd_temp_tres <- sd(data_tres %>% pull(mean_temp))

temp_trans_tres <- trans_new("temp_trans_tres",
  transform = function(x){(x * sd_temp_tres) + mean_temp_tres},
  inverse = function(x){x})

(fig5_tres <- ggpredict(prior_model,terms = c("mean_temp_scaled [all]","habitat"),bias_correction = TRUE,
  plot(line_size = 1.5,alpha = .2,show_data = TRUE,limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean temperature (\u00b0C)") +
  ylab("Provisioning (count/hour)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "TRES provis: maxsq interaction with LC, but not with linear max") +
  scale_x_continuous(trans = temp_trans_tres,
    breaks = c((20-mean_temp_tres)/sd_temp_tres,
      (25-mean_temp_tres)/sd_temp_tres,
      (30-mean_temp_tres)/sd_temp_tres,
      (35-mean_temp_tres)/sd_temp_tres,
      (40-mean_temp_tres)/sd_temp_tres),
    # breaks = c(20,30,40,50),
    # labels = c("20","30","40","50"),
    # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("Forest" = "solid","Orchard" = "dotted","Grassland" = "dotted","Row
  ylim(0,50) +
  geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf,label = label),hjust = -.2, vjust =
  theme(legend.position = "none")
)

```

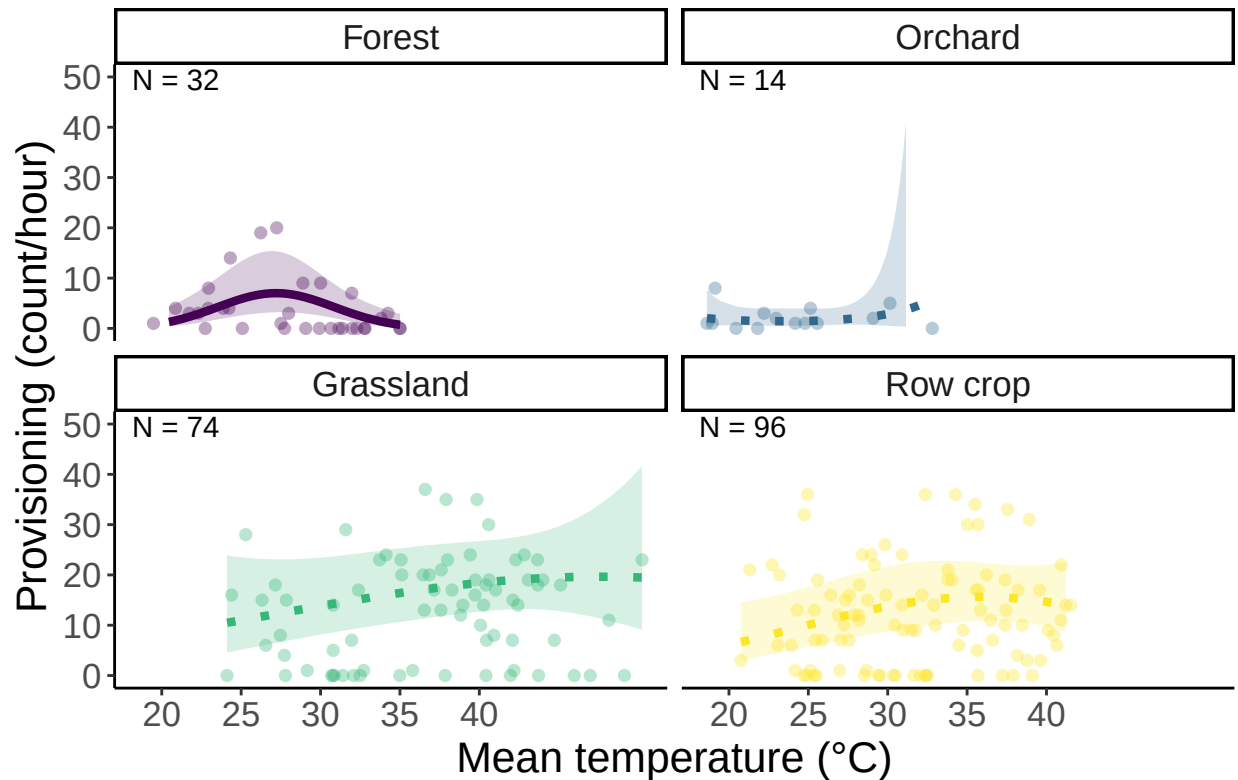
```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.
```

TRES provis: maxsq interaction with LC, but not wi



```
(fig5_tres <- ggpredict(m, terms = c("mean_temp_scaled [all]", "habitat"), bias_correction = TRUE) %>%
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  aes(linetype = .data[["group"]]) +
  theme_classic() +
  facet_wrap(~ group, ncol = 2) +
  xlab("Mean temperature (°C)") +
  ylab("Provisioning (count/hour)") +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "TRES provis: maxsq interaction with LC, but not with linear max") +
  scale_x_continuous(trans = temp_trans_tres,
    breaks = c((20-mean_temp_tres)/sd_temp_tres,
               (25-mean_temp_tres)/sd_temp_tres,
               (30-mean_temp_tres)/sd_temp_tres,
               (35-mean_temp_tres)/sd_temp_tres,
               (40-mean_temp_tres)/sd_temp_tres),
    # breaks = c(20, 30, 40, 50),
    # labels = c("20", "30", "40", "50"),
    # limits = c((15-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
    #           (55-mean(g$meanmaxtempI, na.rm = TRUE))/sd(g$meanmaxtempI, na.rm = TR
```

```

    # limits = c(18,5)
  ) +
  scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "dotted", "Grassland" = "dotted", "Row crop" = "dotted")) +
  ylim(0,50) +
  geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust = 1) +
  theme(legend.position = "none")
)

```

```

## Data points may overlap. Use the `jitter` argument to add some amount of
## random variation to the location of data points and avoid overplotting.

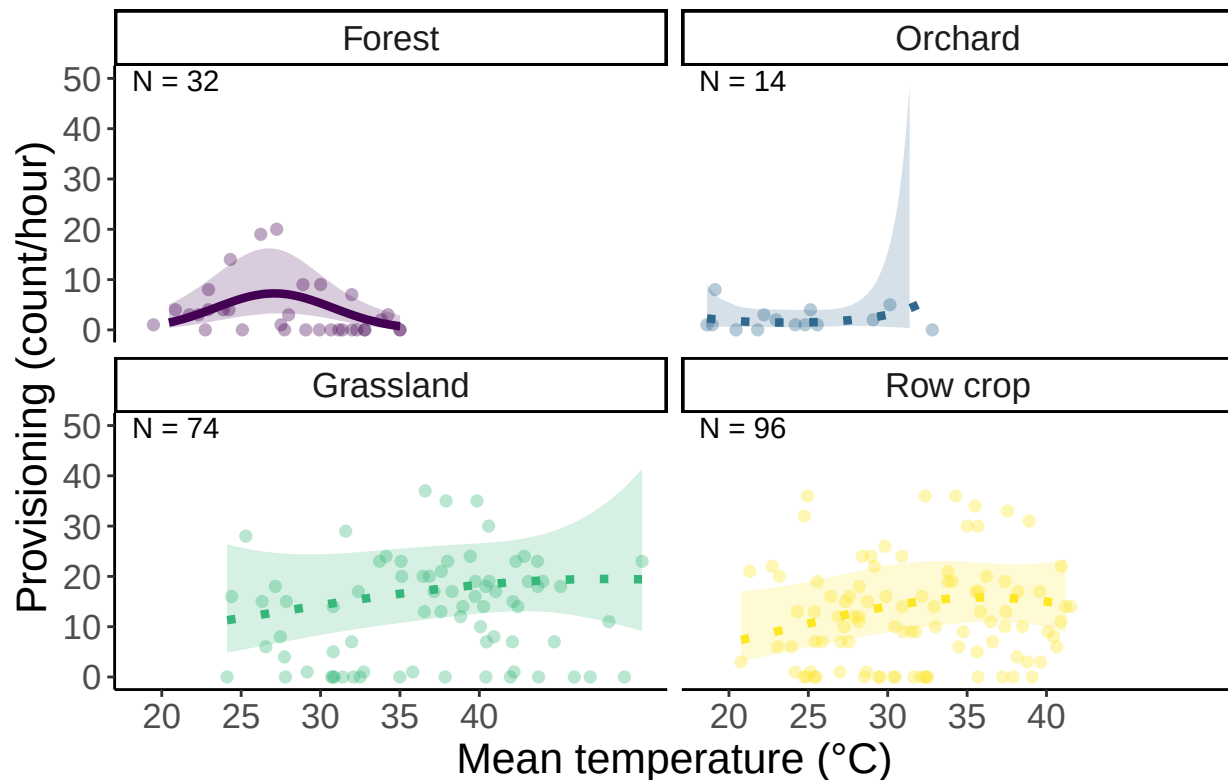
```

```

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.

```

TRES provis: maxsq interaction with LC, but not wi



```

(fig5_tres <- ggpredict(prior_model, terms = c("mean_temp_scaled [all]", "julian_date_scaled"), bias_correction = "none",
  plot(line_size = 1.5, alpha = .2, show_data = TRUE, limit_range = TRUE) +
  # aes(linetype = .data[["group"]]) +
  theme_classic() +
  # facet_wrap(~ group, ncol = 2) +
  xlab("Mean temperature (°C)") +
  ylab("Provisioning (count/hour)") +
  # scale_fill_viridis(discrete = TRUE) +
  # scale_color_viridis(discrete = TRUE) +
  theme(text = element_text(size = 16)) +
  labs(title = "TRES provis: maxsq interaction with LC, but not with linear max") +

```

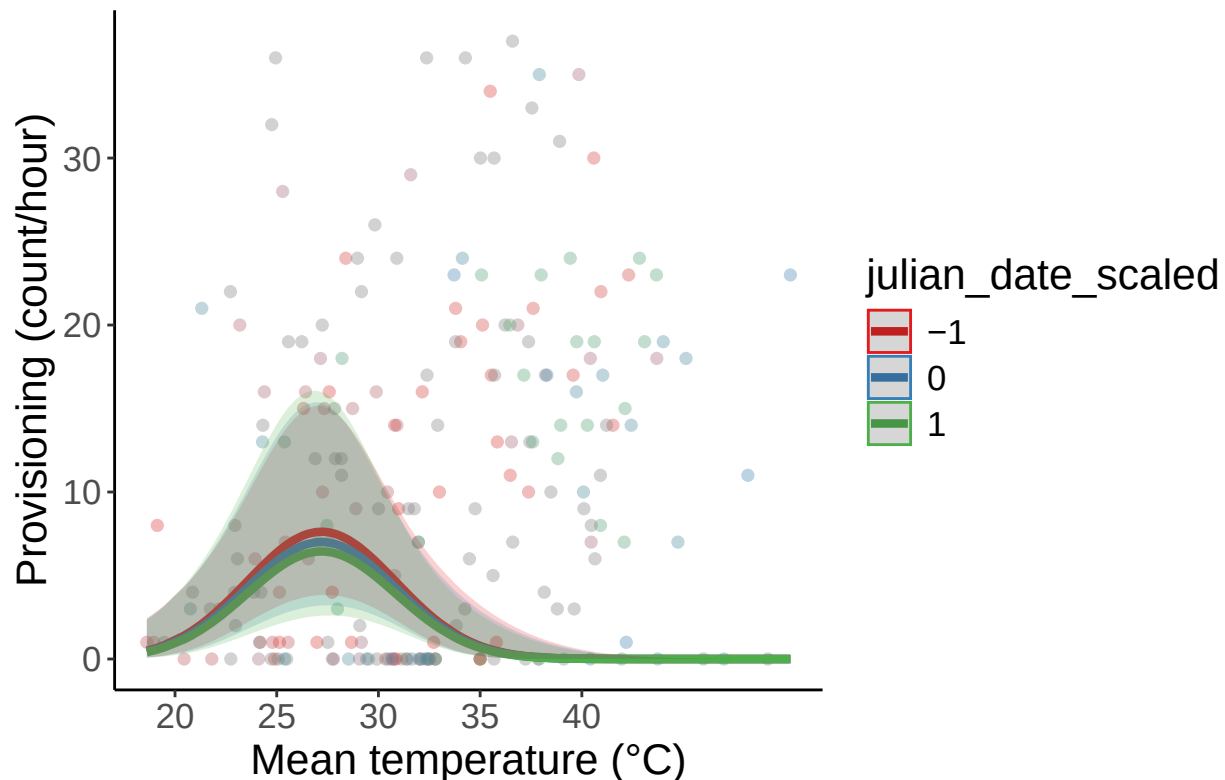
```

scale_x_continuous(trans = temp_trans_tres,
  breaks = c((20-mean_temp_tres)/sd_temp_tres,
    (25-mean_temp_tres)/sd_temp_tres,
    (30-mean_temp_tres)/sd_temp_tres,
    (35-mean_temp_tres)/sd_temp_tres,
    (40-mean_temp_tres)/sd_temp_tres),
  # breaks = c(20,30,40,50),
  # labels = c("20", "30", "40", "50"),
  # limits = c((15-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
    (55-mean(g$meanmaxtempI,na.rm = TRUE))/sd(g$meanmaxtempI,na.rm = TR
  # limits = c(18,5)
) # +
# scale_linetype_manual(values = c("Forest" = "solid", "Orchard" = "dotted", "Grassland" = "dotted", ".
# ylim(0,50) +
# geom_text(data = dat_text_tres, mapping = aes(x = -Inf, y = Inf, label = label), hjust = -.2, vjust
# theme(legend.position = "none")
)

```

Data points may overlap. Use the `jitter` argument to add some amount of
 ## random variation to the location of data points and avoid overplotting.

TRES provis: maxsq interaction with LC, but not wi



Interaction is not significant